Pinned group summary: SU-n5-q2-eps-minus1

Pinning-Sympy automated output

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Contents

1 Basic group attributes

Group	SU-n5-q2-eps-minus1
Matrix size	5×5
Rank	2
Defining equations	$X^* H X = H$ and $\det(X) = 1$

1.1 Skew-hermitian form

Dimension	5
Witt index	2
Primitive element	$\sqrt{d}$
Epsilon	-1

$$\text{Matrix: } \begin{bmatrix} 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ -1 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & c_0\sqrt{d} \end{bmatrix}$$

1.2 Lie algebra

The Lie algebra is denoted $\mathfrak{g}$. The defining equations for the $\mathfrak{g}$ are: $X^*H + HX = 0$ and $\operatorname{tr}(X) = 0$
A generic element of $\mathfrak{g}$ is:

$$\begin{bmatrix} \sqrt{d}x_{i00} + x_{r00} & \sqrt{d}x_{i01} + x_{r01} & -x_{r02} & -\sqrt{d}x_{i12} + x_{r12} & c_0\sqrt{d}\left(-\sqrt{d}x_{i42} + x_{r42}\right) \\ \sqrt{d}x_{i10} + x_{r10} & -\sqrt{d}x_{i00} - \frac{\sqrt{d}x_{i44}}{2} + x_{r11} & \sqrt{d}x_{i12} + x_{r12} & -x_{r13} & c_0\sqrt{d}\left(-\sqrt{d}x_{i43} + x_{r43}\right) \\ -x_{r20} & -\sqrt{d}x_{i30} + x_{r30} & \sqrt{d}x_{i00} - x_{r00} & \sqrt{d}x_{i10} - x_{r10} & -c_0\sqrt{d}\left(-\sqrt{d}x_{i40} + x_{r40}\right) \\ \sqrt{d}x_{i30} + x_{r30} & -x_{r31} & \sqrt{d}x_{i01} - x_{r01} & -\sqrt{d}x_{i00} - \frac{\sqrt{d}x_{i44}}{2} - x_{r11} & -c_0\sqrt{d}\left(-\sqrt{d}x_{i41} + x_{r41}\right) \\ \sqrt{d}x_{i40} + x_{r40} & \sqrt{d}x_{i41} + x_{r41} & \sqrt{d}x_{i42} + x_{r42} & \sqrt{d}x_{i43} + x_{r43} & \sqrt{d}x_{i44} \end{bmatrix}$$

1.3 Torus

The torus is denoted T. It has rank 2. A generic element of the T is

$$\begin{bmatrix} t_0 & 0 & 0 & 0 & 0 \\ 0 & t_1 & 0 & 0 & 0 \\ 0 & 0 & \frac{1}{t_0} & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{t_1} & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

A **character** of T is a group homomorphism from T to the multiplicative group of the ground field. The set of characters is denoted $X_*(T)$. In this document, elements of $X_*(T)$ are written as row vectors $\alpha = (a_1, a_2, \ldots, a_n)$ and such a vector is viewed as a function on diagonal matrices in the following way: for $\alpha = (a_1, a_2, \ldots, a_n) \in X_*(T)$ and $t = \operatorname{diag}(t_1, t_2, \ldots, t_n) \in T$,

$$\alpha(t) = \prod_{i=1}^{n} t_i^{a_i} = t_1^{a_1} t_2^{a_2} \cdots t_n^{a_n}$$

The rows of the matrix below are trivial characters, i.e. elements $\alpha \in X_*(T)$ which are the identity function on T. It would be more accurate to describe $X_*(T)$ as a quotient of a set of group homomorphisms modulo the lattice generated by trivial characters, but this distinction is not critical for our purposes.

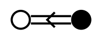
$$\begin{bmatrix} 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 1 & 1 & 1 & 1 & 1 \end{bmatrix}$$

2 Root system

Dynkin type	BC_2
Irreducible	True
Reduced	False
Simply laced	False
Number of roots	12

2.1 Dynkin diagram

The root system is irreducible and has Dynkin diagram:



2.2 Cartan matrix

The root system is irreducible and has Cartan matrix:

$$\begin{bmatrix} 2 & -1 \\ -2 & 2 \end{bmatrix}$$

2.3 Roots

Definition 2.1. A root $\alpha \in \Phi$ is **multipliable** if $2\alpha \in \Phi$.

Root	**Short Form**	**Norm^2**	**Simple**	**Sign**	**Multipliable**
$(0,\ 1,\ 0,\ 0,\ 0)$	$(0,\ 1)$	1	Yes	$+$	Yes
$(1,\ 0,\ 0,\ 0,\ 0)$	$(1,\ 0)$	1	No	$+$	Yes
$(-1,\ 0,\ 0,\ 0,\ 0)$	$(-1,\ 0)$	1	No	$-$	Yes
$(0,\ -1,\ 0,\ 0,\ 0)$	$(0,\ -1)$	1	No	$-$	Yes
$(1,\ -1,\ 0,\ 0,\ 0)$	$(1,\ -1)$	2	Yes	$+$	No
$(1,\ 1,\ 0,\ 0,\ 0)$	$(1,\ 1)$	2	No	$+$	No
$(-1,\ -1,\ 0,\ 0,\ 0)$	$(-1,\ -1)$	2	No	$-$	No
$(-1,\ 1,\ 0,\ 0,\ 0)$	$(-1,\ 1)$	2	No	$-$	No
$(0,\ 2,\ 0,\ 0,\ 0)$	$(0,\ 2)$	4	No	$+$	No
$(2,\ 0,\ 0,\ 0,\ 0)$	$(2,\ 0)$	4	No	$+$	No
$(-2,\ 0,\ 0,\ 0,\ 0)$	$(-2,\ 0)$	4	No	$-$	No
$(0,\ -2,\ 0,\ 0,\ 0)$	$(0,\ -2)$	4	No	$-$	No

2.4 Coroots

Given a root system Φ, for each root $\alpha \in \Phi$, we find a coroot $\alpha^\vee$ (an integer vector of the same length) so that $\langle \alpha, \alpha^\vee \rangle = 2$. There is not necessarily a unique choice of $\alpha^\vee$, but for the purposes of this document, we choose $\alpha^\vee$ so that it is the integer vector of minimal Euclidean norm among those satisfying $\langle \alpha, \alpha^\vee \rangle = 2$.

Root	**Coroot**
$(-2,\ 0,\ 0,\ 0,\ 0)$	$(-1,\ 0,\ 1,\ 0,\ 0)$
$(-1,\ -1,\ 0,\ 0,\ 0)$	$(-1,\ -1,\ 1,\ 1,\ 0)$
$(-1,\ 0,\ 0,\ 0,\ 0)$	$(-2,\ 0,\ 2,\ 0,\ 0)$
$(-1,\ 1,\ 0,\ 0,\ 0)$	$(-1,\ 1,\ 1,\ -1,\ 0)$
$(0,\ -2,\ 0,\ 0,\ 0)$	$(0,\ -1,\ 0,\ 1,\ 0)$
$(0,\ -1,\ 0,\ 0,\ 0)$	$(0,\ -2,\ 0,\ 2,\ 0)$
$(0,\ 1,\ 0,\ 0,\ 0)$	$(0,\ 2,\ 0,\ -2,\ 0)$
$(0,\ 2,\ 0,\ 0,\ 0)$	$(0,\ 1,\ 0,\ -1,\ 0)$
$(1,\ -1,\ 0,\ 0,\ 0)$	$(1,\ -1,\ -1,\ 1,\ 0)$
$(1,\ 0,\ 0,\ 0,\ 0)$	$(2,\ 0,\ -2,\ 0,\ 0)$
$(1,\ 1,\ 0,\ 0,\ 0)$	$(1,\ 1,\ -1,\ -1,\ 0)$
$(2,\ 0,\ 0,\ 0,\ 0)$	$(1,\ 0,\ -1,\ 0,\ 0)$

2.5 Linear combinations of roots

Definition 2.2. For two roots α, β in a root system Φ, the set of positive integral linear combinations of α, β that are also roots is denoted $(\alpha, \beta) = \{i\alpha + j\beta \in \Phi : i, j \in \mathbb{Z}_{\geq 1}\}$.

α	β	Pairs (i, j)
$(-1,\ -1)$	$(-1,\ 1)$	(1,1)
$(-1,\ -1)$	$(1,\ -1)$	(1,1)
$(-1,\ 0)$	$(0,\ -1)$	(1,1)
$(-1,\ 0)$	$(0,\ 1)$	(1,1)
$(-1,\ 1)$	$(1,\ 1)$	(1,1)
$(0,\ -1)$	$(1,\ 0)$	(1,1)
$(0,\ 1)$	$(1,\ 0)$	(1,1)
$(1,\ -1)$	$(1,\ 1)$	(1,1)
$(-2,\ 0)$	$(1,\ -1)$	(1,1), (1,2)
$(-2,\ 0)$	$(1,\ 1)$	(1,1), (1,2)
$(-1,\ -1)$	$(0,\ 2)$	(1,1), (2,1)
$(-1,\ -1)$	$(2,\ 0)$	(1,1), (2,1)
$(-1,\ 1)$	$(0,\ -2)$	(1,1), (2,1)
$(-1,\ 1)$	$(2,\ 0)$	(1,1), (2,1)
$(0,\ -2)$	$(1,\ 1)$	(1,1), (1,2)
$(0,\ 2)$	$(1,\ -1)$	(1,1), (1,2)
$(-1,\ -1)$	$(0,\ 1)$	(1,1), (1,2), (2,2)
$(-1,\ -1)$	$(1,\ 0)$	(1,1), (1,2), (2,2)
$(-1,\ 0)$	$(1,\ -1)$	(1,1), (2,1), (2,2)
$(-1,\ 0)$	$(1,\ 1)$	(1,1), (2,1), (2,2)
$(-1,\ 1)$	$(0,\ -1)$	(1,1), (1,2), (2,2)
$(-1,\ 1)$	$(1,\ 0)$	(1,1), (1,2), (2,2)
$(0,\ -1)$	$(1,\ 1)$	(1,1), (2,1), (2,2)
$(0,\ 1)$	$(1,\ -1)$	(1,1), (2,1), (2,2)

3 Root spaces

3.1 Dimensions

For a root $\alpha \in \Phi$, the dimension of the associated root space V_α (and also the dimension of the root subgroup U_α) is denoted d_α. The table below lists these dimensions.

α	d_α
$(-2,\ 0,\ 0,\ 0,\ 0)$	1
$(0,\ -2,\ 0,\ 0,\ 0)$	1
$(0,\ 2,\ 0,\ 0,\ 0)$	1
$(2,\ 0,\ 0,\ 0,\ 0)$	1
$(-1,\ -1,\ 0,\ 0,\ 0)$	2
$(-1,\ 0,\ 0,\ 0,\ 0)$	2
$(-1,\ 1,\ 0,\ 0,\ 0)$	2
$(0,\ -1,\ 0,\ 0,\ 0)$	2
$(0,\ 1,\ 0,\ 0,\ 0)$	2
$(1,\ -1,\ 0,\ 0,\ 0)$	2
$(1,\ 0,\ 0,\ 0,\ 0)$	2
$(1,\ 1,\ 0,\ 0,\ 0)$	2

3.2 Root space table

The table below lists each root along with a generic element of the associated root space (inside the Lie algebra). These are computed as follows: given a character α, consider the set

$$V_\alpha = \left\{X \in \mathfrak{g} : t \cdot X \cdot t^{-1} = \alpha(t) \cdot X, \forall t \in T\right\}$$

This is a vector subspace of the Lie algebra $\mathfrak{g}$. When V_α is more than just the zero vector, α is called a **root** and the root system Φ is the set of all such roots. For this document, roots are computed as follows: compile a list of potential character vectors α, which are all possible integer vectors of length n with entries from $\{0, \pm 1, \pm 2\}$, then reduce that list of potentials by quotienting by the trivial characters of the torus. Next, instantiate a generic torus element t and a generic Lie algebra element X. For each potential character α, solve the equation

$$t \cdot X \cdot t^{-1} = \alpha(t) \cdot X$$

for X. If there is a nonzero solution space, add α to the set of roots.

Root	**Generic element of root space**
$(-2,\ 0,\ 0,\ 0,\ 0)$	$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ -u_0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
$(-1,\ -1,\ 0,\ 0,\ 0)$	$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & -\sqrt{d}u_0 + u_1 & 0 & 0 & 0 \\ \sqrt{d}u_0 + u_1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
$(-1,\ 0,\ 0,\ 0,\ 0)$	$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & c_0\left(-\sqrt{d}u_0 + du_1\right) \\ 0 & 0 & 0 & 0 & 0 \\ \sqrt{d}u_1 + u_0 & 0 & 0 & 0 & 0 \end{bmatrix}$

Root	Generic element of root space
$(-1,\ 1,\ 0,\ 0,\ 0)$	$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ \sqrt{d}u_1+u_0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \sqrt{d}u_1-u_0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
$(0,\ -2,\ 0,\ 0,\ 0)$	$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & -u_0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
$(0,\ -1,\ 0,\ 0,\ 0)$	$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & c_0\left(-\sqrt{d}u_1+du_0\right) \\ 0 & \sqrt{d}u_0+u_1 & 0 & 0 & 0 \end{bmatrix}$
$(0,\ 1,\ 0,\ 0,\ 0)$	$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & c_0\left(\sqrt{d}u_1-du_0\right) \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \sqrt{d}u_0+u_1 & 0 \end{bmatrix}$
$(0,\ 2,\ 0,\ 0,\ 0)$	$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -u_0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
$(1,\ -1,\ 0,\ 0,\ 0)$	$\begin{bmatrix} 0 & \sqrt{d}u_0+u_1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \sqrt{d}u_0-u_1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
$(1,\ 0,\ 0,\ 0,\ 0)$	$\begin{bmatrix} 0 & 0 & 0 & 0 & c_0\left(\sqrt{d}u_1-du_0\right) \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \sqrt{d}u_0+u_1 & 0 & 0 \end{bmatrix}$
$(1,\ 1,\ 0,\ 0,\ 0)$	$\begin{bmatrix} 0 & 0 & 0 & -\sqrt{d}u_1+u_0 & 0 \\ 0 & 0 & \sqrt{d}u_1+u_0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
$(2,\ 0,\ 0,\ 0,\ 0)$	$\begin{bmatrix} 0 & 0 & -u_0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

4 Root subgroups

4.1 Root subgroup table

For the purposes of this document, root subgroups are found by simply exponentiating the associated root space. That is, for a root $\alpha \in \Phi$, the root space is

$$V_\alpha = \{X \in \mathfrak{g} : t \cdot X \cdot t^{-1} = \alpha(t) \cdot X, \forall t \in T\}$$

and the root subgroup is

$$U_\alpha = \left\{e^X = 1 + X + \frac{X^2}{2!} + \frac{X^3}{3!} + \cdots : X \in V_\alpha\right\}$$

For the purposes of the groups considered in this package, the X^3 and higher terms of the exponential series always vanish, and even the quadratic term frequently vanishes, so we can simplify the description of U_α to:

$$U_\alpha = \left\{1 + X + \frac{X^2}{2} : X \in V_\alpha\right\}$$

The table below lists each root along with a generic element of the associated root subgroup.

Root	**Generic element of root subgroup**
$(-2,\ 0,\ 0,\ 0,\ 0)$	$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ -u_0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$
$(-1,\ -1,\ 0,\ 0,\ 0)$	$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & -\sqrt{d}u_0 + u_1 & 1 & 0 & 0 \\ \sqrt{d}u_0 + u_1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$
$(-1,\ 0,\ 0,\ 0,\ 0)$	$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ \frac{c_0\sqrt{d}\left(du_1^2 - u_0^2\right)}{2} & 0 & 1 & 0 & c_0\left(-\sqrt{d}u_0 + du_1\right) \\ 0 & 0 & 0 & 1 & 0 \\ \sqrt{d}u_1 + u_0 & 0 & 0 & 0 & 1 \end{bmatrix}$
$(-1,\ 1,\ 0,\ 0,\ 0)$	$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ \sqrt{d}u_1 + u_0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & \sqrt{d}u_1 - u_0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$
$(0,\ -2,\ 0,\ 0,\ 0)$	$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & -u_0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$
$(0,\ -1,\ 0,\ 0,\ 0)$	$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & \frac{c_0\sqrt{d}\left(du_0^2 - u_1^2\right)}{2} & 0 & 1 & c_0\left(-\sqrt{d}u_1 + du_0\right) \\ 0 & \sqrt{d}u_0 + u_1 & 0 & 0 & 1 \end{bmatrix}$

Root	Generic element of root subgroup
(0, 1, 0, 0, 0)	$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & \frac{c_0\sqrt{d}\left(-du_0^2+u_1^2\right)}{2} & c_0\left(\sqrt{d}u_1 - du_0\right) \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & \sqrt{d}u_0 + u_1 & 1 \end{bmatrix}$
(0, 2, 0, 0, 0)	$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & -u_0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$
(1, −1, 0, 0, 0)	$\begin{bmatrix} 1 & \sqrt{d}u_0 + u_1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & \sqrt{d}u_0 - u_1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$
(1, 0, 0, 0, 0)	$\begin{bmatrix} 1 & 0 & \frac{c_0\sqrt{d}\left(-du_0^2+u_1^2\right)}{2} & 0 & c_0\left(\sqrt{d}u_1 - du_0\right) \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & \sqrt{d}u_0 + u_1 & 0 & 1 \end{bmatrix}$
(1, 1, 0, 0, 0)	$\begin{bmatrix} 1 & 0 & 0 & -\sqrt{d}u_1 + u_0 & 0 \\ 0 & 1 & \sqrt{d}u_1 + u_0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$
(2, 0, 0, 0, 0)	$\begin{bmatrix} 1 & 0 & -u_0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$

4.2 Homomorphism property

For each root $\alpha \in \Phi$, the associated root subgroup map $X_\alpha : V_\alpha \to G$ is a pseudo-homomorphism with respect to the additive structure on V_α and the group operation (i.e. matrix multiplication) in G. That is, for any $u, v \in V_\alpha$,

$$X_\alpha(u) \cdot X_\alpha(v) = X_\alpha(u+v) \cdot \prod_{\substack{i>1 \\ i\alpha\in\Phi}} X_{i\alpha}\left(q_\alpha^i(u,v)\right) \tag{1}$$

for some homogeneous polynomial maps $q_\alpha^i : V_\alpha \times V_\alpha \to V_{i\alpha}$. Thus X_α is a group homomorphism if and only the product on the right hand side reduces to 1. The quantities $q_\alpha^i(u,v)$ are called **homomorphism defect coefficients**. For any root α of a root system Φ of any group considered in the Pinning-Sympy package, the set

$$\{i\alpha \in \Phi : i \in \mathbb{Z}_{\geq 2}\}$$

is either empty or a singleton set consisting of $\{2\alpha\}$. The non-empty case only arises when Φ is a non-reduced root system (of Dynkin type BC), and when α is a multipliable root.

The table below lists all multipliable roots and the associated homomorphism defect.

α	$q^i_\alpha(u,v)$
$(-1,\ 0)$	$c_0du_0v_1 - c_0du_1v_0$
$(0,\ -1)$	$-c_0du_0v_1 + c_0du_1v_0$
$(0,\ 1)$	$c_0du_0v_1 - c_0du_1v_0$
$(1,\ 0)$	$c_0du_0v_1 - c_0du_1v_0$

Here are the pseudo-homomorphism equations.

$$\alpha = (-1,\ 0,\ 0,\ 0,\ 0)$$

$$X_\alpha\left(\begin{bmatrix}u_0\\u_1\end{bmatrix}\right)X_\alpha\left(\begin{bmatrix}v_0\\v_1\end{bmatrix}\right) = X_\alpha\left(\begin{bmatrix}u_0\\u_1\end{bmatrix}+\begin{bmatrix}v_0\\v_1\end{bmatrix}\right)\cdot X_{2\alpha}\left(c_0du_0v_1 - c_0du_1v_0\right)$$

$$\begin{bmatrix}1 & 0 & 0 & 0 & 0\\ 0 & 1 & 0 & 0 & 0\\ \frac{c_0\sqrt{d}\left(du_1^2-u_0^2\right)}{2} & 0 & 1 & 0 & c_0\left(-\sqrt{d}u_0+du_1\right)\\ 0 & 0 & 0 & 1 & 0\\ \sqrt{d}u_1+u_0 & 0 & 0 & 0 & 1\end{bmatrix}\begin{bmatrix}1 & 0 & 0 & 0 & 0\\ 0 & 1 & 0 & 0 & 0\\ \frac{c_0\sqrt{d}\left(dv_1^2-v_0^2\right)}{2} & 0 & 1 & 0 & c_0\left(-\sqrt{d}v_0+dv_1\right)\\ 0 & 0 & 0 & 1 & 0\\ \sqrt{d}v_1+v_0 & 0 & 0 & 0 & 1\end{bmatrix} = \begin{bmatrix}1 & 0 & 0 & 0 & 0\\ 0 & 1 & 0 & 0 & 0\\ -\frac{c_0\left(\sqrt{d}(u_0+v_0)-d(u_1+v_1)\right)\left(\sqrt{d}(u_1+v_1)+u_0+v_0\right)}{2} & 0 & 1 & 0 & -c_0\left(\sqrt{d}\left(u_0+v_0\right)-d\left(u_1+v_1\right)\right)\\ 0 & 0 & 0 & 1 & 0\\ \sqrt{d}\left(u_1+v_1\right)+u_0+v_0 & 0 & 0 & 0 & 1\end{bmatrix}\begin{bmatrix}1 & 0 & 0 & 0 & 0\\ 0 & 1 & 0 & 0 & 0\\ c_0d\left(-u_0v_1+u_1v_0\right) & 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix}$$

$$\alpha = (0,\ -1,\ 0,\ 0,\ 0)$$

$$X_\alpha\left(\begin{bmatrix}u_0\\u_1\end{bmatrix}\right)X_\alpha\left(\begin{bmatrix}v_0\\v_1\end{bmatrix}\right) = X_\alpha\left(\begin{bmatrix}u_0\\u_1\end{bmatrix}+\begin{bmatrix}v_0\\v_1\end{bmatrix}\right)\cdot X_{2\alpha}\left(-c_0du_0v_1 + c_0du_1v_0\right)$$

$$\begin{bmatrix}1 & 0 & 0 & 0 & 0\\ 0 & 1 & 0 & 0 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & \frac{c_0\sqrt{d}\left(du_0^2-u_1^2\right)}{2} & 0 & 1 & c_0\left(-\sqrt{d}u_1+du_0\right)\\ 0 & \sqrt{d}u_0+u_1 & 0 & 0 & 1\end{bmatrix}\begin{bmatrix}1 & 0 & 0 & 0 & 0\\ 0 & 1 & 0 & 0 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & \frac{c_0\sqrt{d}\left(dv_0^2-v_1^2\right)}{2} & 0 & 1 & c_0\left(-\sqrt{d}v_1+dv_0\right)\\ 0 & \sqrt{d}v_0+v_1 & 0 & 0 & 1\end{bmatrix} = \begin{bmatrix}1 & 0 & 0 & 0 & 0\\ 0 & 1 & 0 & 0 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & -\frac{c_0\left(\sqrt{d}(u_1+v_1)-d(u_0+v_0)\right)\left(\sqrt{d}(u_0+v_0)+u_1+v_1\right)}{2} & 0 & 1 & -c_0\left(\sqrt{d}\left(u_1+v_1\right)-d\left(u_0+v_0\right)\right)\\ 0 & \sqrt{d}\left(u_0+v_0\right)+u_1+v_1 & 0 & 0 & 1\end{bmatrix}\begin{bmatrix}1 & 0 & 0 & 0 & 0\\ 0 & 1 & 0 & 0 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & c_0d\left(u_0v_1-u_1v_0\right) & 0 & 1 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix}$$

$$\alpha = (0,\ 1,\ 0,\ 0,\ 0)$$

$$X_\alpha\left(\begin{bmatrix}u_0\\u_1\end{bmatrix}\right)X_\alpha\left(\begin{bmatrix}v_0\\v_1\end{bmatrix}\right) = X_\alpha\left(\begin{bmatrix}u_0\\u_1\end{bmatrix}+\begin{bmatrix}v_0\\v_1\end{bmatrix}\right)\cdot X_{2\alpha}\left(c_0du_0v_1 - c_0du_1v_0\right)$$

$$\begin{bmatrix}1 & 0 & 0 & 0 & 0\\ 0 & 1 & 0 & \frac{c_0\sqrt{d}\left(-du_0^2+u_1^2\right)}{2} & c_0\left(\sqrt{d}u_1-du_0\right)\\ 0 & 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 1 & 0\\ 0 & 0 & 0 & \sqrt{d}u_0+u_1 & 1\end{bmatrix}\begin{bmatrix}1 & 0 & 0 & 0 & 0\\ 0 & 1 & 0 & \frac{c_0\sqrt{d}\left(-dv_0^2+v_1^2\right)}{2} & c_0\left(\sqrt{d}v_1-dv_0\right)\\ 0 & 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 1 & 0\\ 0 & 0 & 0 & \sqrt{d}v_0+v_1 & 1\end{bmatrix} = \begin{bmatrix}1 & 0 & 0 & 0 & 0\\ 0 & 1 & 0 & \frac{c_0\left(\sqrt{d}(u_1+v_1)-d(u_0+v_0)\right)\left(\sqrt{d}(u_0+v_0)+u_1+v_1\right)}{2} & c_0\left(\sqrt{d}\left(u_1+v_1\right)-d\left(u_0+v_0\right)\right)\\ 0 & 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 1 & 0\\ 0 & 0 & 0 & \sqrt{d}\left(u_0+v_0\right)+u_1+v_1 & 1\end{bmatrix}\begin{bmatrix}1 & 0 & 0 & 0 & 0\\ 0 & 1 & 0 & c_0d\left(-u_0v_1+u_1v_0\right) & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix}$$

$$\alpha = (1,\ 0,\ 0,\ 0,\ 0)$$

$$X_\alpha\left(\begin{bmatrix}u_0\\u_1\end{bmatrix}\right)X_\alpha\left(\begin{bmatrix}v_0\\v_1\end{bmatrix}\right) = X_\alpha\left(\begin{bmatrix}u_0\\u_1\end{bmatrix}+\begin{bmatrix}v_0\\v_1\end{bmatrix}\right)\cdot X_{2\alpha}\left(c_0du_0v_1 - c_0du_1v_0\right)$$

$$\begin{bmatrix}1 & 0 & \frac{c_0\sqrt{d}\left(-du_0^2+u_1^2\right)}{2} & 0 & c_0\left(\sqrt{d}u_1-du_0\right)\\ 0 & 1 & 0 & 0 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 1 & 0\\ 0 & 0 & \sqrt{d}u_0+u_1 & 0 & 1\end{bmatrix}\begin{bmatrix}1 & 0 & \frac{c_0\sqrt{d}\left(-dv_0^2+v_1^2\right)}{2} & 0 & c_0\left(\sqrt{d}v_1-dv_0\right)\\ 0 & 1 & 0 & 0 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 1 & 0\\ 0 & 0 & \sqrt{d}v_0+v_1 & 0 & 1\end{bmatrix} = \begin{bmatrix}1 & 0 & \frac{c_0\left(\sqrt{d}(u_1+v_1)-d(u_0+v_0)\right)\left(\sqrt{d}(u_0+v_0)+u_1+v_1\right)}{2} & 0 & c_0\left(\sqrt{d}\left(u_1+v_1\right)-d\left(u_0+v_0\right)\right)\\ 0 & 1 & 0 & 0 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 1 & 0\\ 0 & 0 & \sqrt{d}\left(u_0+v_0\right)+u_1+v_1 & 0 & 1\end{bmatrix}\begin{bmatrix}1 & 0 & c_0d\left(-u_0v_1+u_1v_0\right) & 0 & 0\\ 0 & 1 & 0 & 0 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix}$$

5 Commutators

The commutator formula says that for two roots $\alpha, \beta \in \Phi$, the commutator of two root group elements $X_\alpha(u)$ and $X_\beta(v)$ is controlled by the structure of Φ, in particular by which positive integer linear combinations $i\alpha + j\beta$ are elements of Φ. Recall that

$$(\alpha, \beta) = \{i\alpha + j\beta \in \Phi : i, j \in \mathbb{Z}_{\geq 1}\}$$

Specifically, we have the commutator formula

$$\Big[X_\alpha(u), X_\beta(v)\Big] = \prod_{(\alpha,\beta)} X_{i\alpha+j\beta}\Big(N_{ij}^{\alpha\beta}(u,v)\Big)$$

for some homogeneous polynomial functions $N_{ij}^{\alpha\beta} : V_\alpha \times V_\beta \to V_{i\alpha+j\beta}$. We repeat here the table of linear combinations of roots. The quantity $N_{ij}^{\alpha\beta}(u,v)$ is called a **commutator coefficient**. The table below gives these coefficients.

α	β	i	j	$i\alpha + j\beta$	$N_{ij}^{\alpha\beta}(u,v)$
$(-1,\ -1)$	$(-1,\ 1)$	1	1	$(-2,\ 0)$	$2du_0v_1 - 2u_1v_0$
$(-1,\ -1)$	$(1,\ -1)$	1	1	$(0,\ -2)$	$-2du_0v_0 - 2u_1v_1$
$(-1,\ 0)$	$(0,\ -1)$	1	1	$(-1,\ -1)$	$\begin{bmatrix} c_0\left(-du_1v_0 + u_0v_1\right) & c_0d\left(-u_0v_0 + u_1v_1\right)\end{bmatrix}$
$(-1,\ 0)$	$(0,\ 1)$	1	1	$(-1,\ 1)$	$\begin{bmatrix} c_0d\left(u_0v_0 - u_1v_1\right) & c_0\left(du_1v_0 - u_0v_1\right)\end{bmatrix}$
$(-1,\ 1)$	$(-1,\ -1)$	1	1	$(-2,\ 0)$	$-2du_1v_0 + 2u_0v_1$
$(-1,\ 1)$	$(1,\ 1)$	1	1	$(0,\ 2)$	$2du_1v_1 - 2u_0v_0$
$(0,\ -1)$	$(-1,\ 0)$	1	1	$(-1,\ -1)$	$\begin{bmatrix} c_0\left(du_0v_1 - u_1v_0\right) & c_0d\left(u_0v_0 - u_1v_1\right)\end{bmatrix}$
$(0,\ -1)$	$(1,\ 0)$	1	1	$(1,\ -1)$	$\begin{bmatrix} c_0\left(du_0v_0 - u_1v_1\right) & c_0d\left(-u_0v_1 + u_1v_0\right)\end{bmatrix}$
$(0,\ 1)$	$(-1,\ 0)$	1	1	$(-1,\ 1)$	$\begin{bmatrix} c_0d\left(-u_0v_0 + u_1v_1\right) & c_0\left(-du_0v_1 + u_1v_0\right)\end{bmatrix}$
$(0,\ 1)$	$(1,\ 0)$	1	1	$(1,\ 1)$	$\begin{bmatrix} c_0d\left(-u_0v_1 + u_1v_0\right) & c_0\left(-du_0v_0 + u_1v_1\right)\end{bmatrix}$
$(1,\ -1)$	$(-1,\ -1)$	1	1	$(0,\ -2)$	$2du_0v_0 + 2u_1v_1$
$(1,\ -1)$	$(1,\ 1)$	1	1	$(2,\ 0)$	$-2du_0v_1 - 2u_1v_0$
$(1,\ 0)$	$(0,\ -1)$	1	1	$(1,\ -1)$	$\begin{bmatrix} c_0\left(-du_0v_0 + u_1v_1\right) & c_0d\left(-u_0v_1 + u_1v_0\right)\end{bmatrix}$
$(1,\ 0)$	$(0,\ 1)$	1	1	$(1,\ 1)$	$\begin{bmatrix} c_0d\left(-u_0v_1 + u_1v_0\right) & c_0\left(du_0v_0 - u_1v_1\right)\end{bmatrix}$
$(1,\ 1)$	$(-1,\ 1)$	1	1	$(0,\ 2)$	$-2du_1v_1 + 2u_0v_0$
$(1,\ 1)$	$(1,\ -1)$	1	1	$(2,\ 0)$	$2du_1v_0 + 2u_0v_1$
$(-2,\ 0)$	$(1,\ -1)$	1	1	$(-1,\ -1)$	$\begin{bmatrix} u_0v_0 & -u_0v_1\end{bmatrix}$
$(-2,\ 0)$	$(1,\ -1)$	1	2	$(0,\ -2)$	$u_0\left(dv_0^2 - v_1^2\right)$
$(-2,\ 0)$	$(1,\ 1)$	1	1	$(-1,\ 1)$	$\begin{bmatrix} u_0v_0 & u_0v_1\end{bmatrix}$
$(-2,\ 0)$	$(1,\ 1)$	1	2	$(0,\ 2)$	$u_0\left(-dv_1^2 + v_0^2\right)$
$(-1,\ -1)$	$(0,\ 2)$	1	1	$(-1,\ 1)$	$\begin{bmatrix} u_1v_0 & u_0v_0\end{bmatrix}$
$(-1,\ -1)$	$(0,\ 2)$	2	1	$(-2,\ 0)$	$v_0\left(du_0^2 - u_1^2\right)$
$(-1,\ -1)$	$(2,\ 0)$	1	1	$(1,\ -1)$	$\begin{bmatrix} -u_0v_0 & u_1v_0\end{bmatrix}$
$(-1,\ -1)$	$(2,\ 0)$	2	1	$(0,\ -2)$	$v_0\left(du_0^2 - u_1^2\right)$
$(-1,\ 1)$	$(0,\ -2)$	1	1	$(-1,\ -1)$	$\begin{bmatrix} u_1v_0 & u_0v_0\end{bmatrix}$
$(-1,\ 1)$	$(0,\ -2)$	2	1	$(-2,\ 0)$	$v_0\left(-du_1^2 + u_0^2\right)$

α	β	i	j	$i\alpha + j\beta$	$N_{ij}^{\alpha\beta}(u,v)$
$(-1,\ 1)$	$(2,\ 0)$	1	1	$(1,\ 1)$	$\begin{bmatrix} -u_0v_0 & -u_1v_0 \end{bmatrix}$
$(-1,\ 1)$	$(2,\ 0)$	2	1	$(0,\ 2)$	$v_0\left(-du_1^2 + u_0^2\right)$
$(0,\ -2)$	$(-1,\ 1)$	1	1	$(-1,\ -1)$	$\begin{bmatrix} -u_0v_1 & -u_0v_0 \end{bmatrix}$
$(0,\ -2)$	$(-1,\ 1)$	1	2	$(-2,\ 0)$	$u_0\left(dv_1^2 - v_0^2\right)$
$(0,\ -2)$	$(1,\ 1)$	1	1	$(1,\ -1)$	$\begin{bmatrix} -u_0v_1 & u_0v_0 \end{bmatrix}$
$(0,\ -2)$	$(1,\ 1)$	1	2	$(2,\ 0)$	$u_0\left(-dv_1^2 + v_0^2\right)$
$(0,\ 2)$	$(-1,\ -1)$	1	1	$(-1,\ 1)$	$\begin{bmatrix} -u_0v_1 & -u_0v_0 \end{bmatrix}$
$(0,\ 2)$	$(-1,\ -1)$	1	2	$(-2,\ 0)$	$u_0\left(-dv_0^2 + v_1^2\right)$
$(0,\ 2)$	$(1,\ -1)$	1	1	$(1,\ 1)$	$\begin{bmatrix} u_0v_1 & -u_0v_0 \end{bmatrix}$
$(0,\ 2)$	$(1,\ -1)$	1	2	$(2,\ 0)$	$u_0\left(dv_0^2 - v_1^2\right)$
$(1,\ -1)$	$(-2,\ 0)$	1	1	$(-1,\ -1)$	$\begin{bmatrix} -u_0v_0 & u_1v_0 \end{bmatrix}$
$(1,\ -1)$	$(-2,\ 0)$	2	1	$(0,\ -2)$	$v_0\left(-du_0^2 + u_1^2\right)$
$(1,\ -1)$	$(0,\ 2)$	1	1	$(1,\ 1)$	$\begin{bmatrix} -u_1v_0 & u_0v_0 \end{bmatrix}$
$(1,\ -1)$	$(0,\ 2)$	2	1	$(2,\ 0)$	$v_0\left(-du_0^2 + u_1^2\right)$
$(1,\ 1)$	$(-2,\ 0)$	1	1	$(-1,\ 1)$	$\begin{bmatrix} -u_0v_0 & -u_1v_0 \end{bmatrix}$
$(1,\ 1)$	$(-2,\ 0)$	2	1	$(0,\ 2)$	$v_0\left(du_1^2 - u_0^2\right)$
$(1,\ 1)$	$(0,\ -2)$	1	1	$(1,\ -1)$	$\begin{bmatrix} u_1v_0 & -u_0v_0 \end{bmatrix}$
$(1,\ 1)$	$(0,\ -2)$	2	1	$(2,\ 0)$	$v_0\left(du_1^2 - u_0^2\right)$
$(2,\ 0)$	$(-1,\ -1)$	1	1	$(1,\ -1)$	$\begin{bmatrix} u_0v_0 & -u_0v_1 \end{bmatrix}$
$(2,\ 0)$	$(-1,\ -1)$	1	2	$(0,\ -2)$	$u_0\left(-dv_0^2 + v_1^2\right)$
$(2,\ 0)$	$(-1,\ 1)$	1	1	$(1,\ 1)$	$\begin{bmatrix} u_0v_0 & u_0v_1 \end{bmatrix}$
$(2,\ 0)$	$(-1,\ 1)$	1	2	$(0,\ 2)$	$u_0\left(dv_1^2 - v_0^2\right)$
$(-1,\ -1)$	$(0,\ 1)$	1	1	$(-1,\ 0)$	$\begin{bmatrix} -du_0v_0 - u_1v_1 & -u_0v_1 - u_1v_0 \end{bmatrix}$
$(-1,\ -1)$	$(0,\ 1)$	1	2	$(-1,\ 1)$	$\begin{bmatrix} \frac{c_0du_0\left(dv_0^2-v_1^2\right)}{2} & \frac{c_0u_1\left(dv_0^2-v_1^2\right)}{2} \end{bmatrix}$
$(-1,\ -1)$	$(0,\ 1)$	2	2	$(-2,\ 0)$	0
$(-1,\ -1)$	$(1,\ 0)$	1	1	$(0,\ -1)$	$\begin{bmatrix} u_0v_1 - u_1v_0 & du_0v_0 - u_1v_1 \end{bmatrix}$
$(-1,\ -1)$	$(1,\ 0)$	1	2	$(1,\ -1)$	$\begin{bmatrix} \frac{c_0u_1\left(dv_0^2-v_1^2\right)}{2} & \frac{c_0du_0\left(-dv_0^2+v_1^2\right)}{2} \end{bmatrix}$
$(-1,\ -1)$	$(1,\ 0)$	2	2	$(0,\ -2)$	0
$(-1,\ 0)$	$(1,\ -1)$	1	1	$(0,\ -1)$	$\begin{bmatrix} u_0v_0 + u_1v_1 & du_1v_0 + u_0v_1 \end{bmatrix}$
$(-1,\ 0)$	$(1,\ -1)$	2	1	$(-1,\ -1)$	$\begin{bmatrix} \frac{c_0v_1\left(-du_1^2+u_0^2\right)}{2} & \frac{c_0dv_0\left(du_1^2-u_0^2\right)}{2} \end{bmatrix}$
$(-1,\ 0)$	$(1,\ -1)$	2	2	$(0,\ -2)$	0
$(-1,\ 0)$	$(1,\ 1)$	1	1	$(0,\ 1)$	$\begin{bmatrix} -u_0v_1 + u_1v_0 & -du_1v_1 + u_0v_0 \end{bmatrix}$
$(-1,\ 0)$	$(1,\ 1)$	2	1	$(-1,\ 1)$	$\begin{bmatrix} \frac{c_0dv_1\left(du_1^2-u_0^2\right)}{2} & \frac{c_0v_0\left(du_1^2-u_0^2\right)}{2} \end{bmatrix}$
$(-1,\ 0)$	$(1,\ 1)$	2	2	$(0,\ 2)$	0
$(-1,\ 1)$	$(0,\ -1)$	1	1	$(-1,\ 0)$	$\begin{bmatrix} -du_1v_0 - u_0v_1 & -u_0v_0 - u_1v_1 \end{bmatrix}$
$(-1,\ 1)$	$(0,\ -1)$	1	2	$(-1,\ -1)$	$\begin{bmatrix} \frac{c_0u_0\left(-dv_0^2+v_1^2\right)}{2} & \frac{c_0du_1\left(-dv_0^2+v_1^2\right)}{2} \end{bmatrix}$
$(-1,\ 1)$	$(0,\ -1)$	2	2	$(-2,\ 0)$	0

α	β	i	j	$i\alpha + j\beta$	$N_{ij}^{\alpha\beta}(u, v)$
$(-1,\ 1)$	$(1,\ 0)$	1	1	$(0,\ 1)$	$\begin{bmatrix} u_0v_0 - u_1v_1 & -du_1v_0 + u_0v_1 \end{bmatrix}$
$(-1,\ 1)$	$(1,\ 0)$	1	2	$(1,\ 1)$	$\begin{bmatrix} \frac{c_0du_1\left(dv_0^2 - v_1^2\right)}{2} & \frac{c_0u_0\left(dv_0^2 - v_1^2\right)}{2} \end{bmatrix}$
$(-1,\ 1)$	$(1,\ 0)$	2	2	$(0,\ 2)$	0
$(0,\ -1)$	$(-1,\ 1)$	1	1	$(-1,\ 0)$	$\begin{bmatrix} du_0v_1 + u_1v_0 & u_0v_0 + u_1v_1 \end{bmatrix}$
$(0,\ -1)$	$(-1,\ 1)$	2	1	$(-1,\ -1)$	$\begin{bmatrix} \frac{c_0v_0\left(du_0^2 - u_1^2\right)}{2} & \frac{c_0dv_1\left(du_0^2 - u_1^2\right)}{2} \end{bmatrix}$
$(0,\ -1)$	$(-1,\ 1)$	2	2	$(-2,\ 0)$	0
$(0,\ -1)$	$(1,\ 1)$	1	1	$(1,\ 0)$	$\begin{bmatrix} u_0v_0 + u_1v_1 & du_0v_1 + u_1v_0 \end{bmatrix}$
$(0,\ -1)$	$(1,\ 1)$	2	1	$(1,\ -1)$	$\begin{bmatrix} \frac{c_0v_0\left(du_0^2 - u_1^2\right)}{2} & \frac{c_0dv_1\left(-du_0^2 + u_1^2\right)}{2} \end{bmatrix}$
$(0,\ -1)$	$(1,\ 1)$	2	2	$(2,\ 0)$	0
$(0,\ 1)$	$(-1,\ -1)$	1	1	$(-1,\ 0)$	$\begin{bmatrix} du_0v_0 + u_1v_1 & u_0v_1 + u_1v_0 \end{bmatrix}$
$(0,\ 1)$	$(-1,\ -1)$	2	1	$(-1,\ 1)$	$\begin{bmatrix} \frac{c_0dv_0\left(-du_0^2 + u_1^2\right)}{2} & \frac{c_0v_1\left(-du_0^2 + u_1^2\right)}{2} \end{bmatrix}$
$(0,\ 1)$	$(-1,\ -1)$	2	2	$(-2,\ 0)$	0
$(0,\ 1)$	$(1,\ -1)$	1	1	$(1,\ 0)$	$\begin{bmatrix} -u_0v_1 + u_1v_0 & du_0v_0 - u_1v_1 \end{bmatrix}$
$(0,\ 1)$	$(1,\ -1)$	2	1	$(1,\ 1)$	$\begin{bmatrix} \frac{c_0dv_0\left(-du_0^2 + u_1^2\right)}{2} & \frac{c_0v_1\left(du_0^2 - u_1^2\right)}{2} \end{bmatrix}$
$(0,\ 1)$	$(1,\ -1)$	2	2	$(2,\ 0)$	0
$(1,\ -1)$	$(-1,\ 0)$	1	1	$(0,\ -1)$	$\begin{bmatrix} -u_0v_0 - u_1v_1 & -du_0v_1 - u_1v_0 \end{bmatrix}$
$(1,\ -1)$	$(-1,\ 0)$	1	2	$(-1,\ -1)$	$\begin{bmatrix} \frac{c_0u_1\left(dv_1^2 - v_0^2\right)}{2} & \frac{c_0du_0\left(-dv_1^2 + v_0^2\right)}{2} \end{bmatrix}$
$(1,\ -1)$	$(-1,\ 0)$	2	2	$(0,\ -2)$	0
$(1,\ -1)$	$(0,\ 1)$	1	1	$(1,\ 0)$	$\begin{bmatrix} -u_0v_1 + u_1v_0 & -du_0v_0 + u_1v_1 \end{bmatrix}$
$(1,\ -1)$	$(0,\ 1)$	1	2	$(1,\ 1)$	$\begin{bmatrix} \frac{c_0du_0\left(dv_0^2 - v_1^2\right)}{2} & \frac{c_0u_1\left(-dv_0^2 + v_1^2\right)}{2} \end{bmatrix}$
$(1,\ -1)$	$(0,\ 1)$	2	2	$(2,\ 0)$	0
$(1,\ 0)$	$(-1,\ -1)$	1	1	$(0,\ -1)$	$\begin{bmatrix} u_0v_1 - u_1v_0 & -du_0v_0 + u_1v_1 \end{bmatrix}$
$(1,\ 0)$	$(-1,\ -1)$	2	1	$(1,\ -1)$	$\begin{bmatrix} \frac{c_0v_1\left(-du_0^2 + u_1^2\right)}{2} & \frac{c_0dv_0\left(du_0^2 - u_1^2\right)}{2} \end{bmatrix}$
$(1,\ 0)$	$(-1,\ -1)$	2	2	$(0,\ -2)$	0
$(1,\ 0)$	$(-1,\ 1)$	1	1	$(0,\ 1)$	$\begin{bmatrix} -u_0v_0 + u_1v_1 & du_0v_1 - u_1v_0 \end{bmatrix}$
$(1,\ 0)$	$(-1,\ 1)$	2	1	$(1,\ 1)$	$\begin{bmatrix} \frac{c_0dv_1\left(-du_0^2 + u_1^2\right)}{2} & \frac{c_0v_0\left(-du_0^2 + u_1^2\right)}{2} \end{bmatrix}$
$(1,\ 0)$	$(-1,\ 1)$	2	2	$(0,\ 2)$	0
$(1,\ 1)$	$(-1,\ 0)$	1	1	$(0,\ 1)$	$\begin{bmatrix} -u_0v_1 + u_1v_0 & du_1v_1 - u_0v_0 \end{bmatrix}$
$(1,\ 1)$	$(-1,\ 0)$	1	2	$(-1,\ 1)$	$\begin{bmatrix} \frac{c_0du_1\left(-dv_1^2 + v_0^2\right)}{2} & \frac{c_0u_0\left(-dv_1^2 + v_0^2\right)}{2} \end{bmatrix}$
$(1,\ 1)$	$(-1,\ 0)$	2	2	$(0,\ 2)$	0
$(1,\ 1)$	$(0,\ -1)$	1	1	$(1,\ 0)$	$\begin{bmatrix} -u_0v_0 - u_1v_1 & -du_1v_0 - u_0v_1 \end{bmatrix}$
$(1,\ 1)$	$(0,\ -1)$	1	2	$(1,\ -1)$	$\begin{bmatrix} \frac{c_0u_0\left(-dv_0^2 + v_1^2\right)}{2} & \frac{c_0du_1\left(dv_0^2 - v_1^2\right)}{2} \end{bmatrix}$
$(1,\ 1)$	$(0,\ -1)$	2	2	$(2,\ 0)$	0

5.1 Commutator identities

$$\alpha = (-2,\ 0,\ 0,\ 0,\ 0)\,, \quad \beta = (1,\ -1,\ 0,\ 0,\ 0)$$

$$\left[X_\alpha\left(u_0\right), X_\beta\left(\begin{bmatrix} v_0 \\ v_1 \end{bmatrix}\right)\right] = X_{\alpha+\beta}\left(\begin{bmatrix} u_0v_0 \\ -u_0v_1 \end{bmatrix}\right) X_{\alpha+2\beta}\left(u_0\left(dv_0^2 - v_1^2\right)\right)$$

$$\begin{bmatrix}1&0&0&0&0\\0&1&0&0&0\\-u_0&0&1&0&0\\0&0&0&1&0\\0&0&0&0&1\end{bmatrix}\begin{bmatrix}1&\sqrt{d}v_0+v_1&0&0&0\\0&1&0&0&0\\0&0&1&0&0\\0&0&\sqrt{d}v_0-v_1&1&0\\0&0&0&0&1\end{bmatrix}\begin{bmatrix}1&0&0&0&0\\0&1&0&0&0\\u_0&0&1&0&0\\0&0&0&1&0\\0&0&0&0&1\end{bmatrix}\begin{bmatrix}1&-\sqrt{d}v_0-v_1&0&0&0\\0&1&0&0&0\\0&0&1&0&0\\0&0&-\sqrt{d}v_0+v_1&1&0\\0&0&0&0&1\end{bmatrix}=\begin{bmatrix}1&0&0&0&0\\0&1&0&0&0\\0&-u_0\left(\sqrt{d}v_0+v_1\right)&1&0&0\\u_0\left(\sqrt{d}v_0-v_1\right)&u_0\left(-dv_0^2+v_1^2\right)&0&1&0\\0&0&0&0&1\end{bmatrix}$$

$$\alpha=(-2,\ 0,\ 0,\ 0,\ 0)\,,\quad \beta=(1,\ 1,\ 0,\ 0,\ 0)$$

$$\left[X_\alpha\left(u_0\right),X_\beta\left(\begin{bmatrix}v_0\\v_1\end{bmatrix}\right)\right]=X_{\alpha+\beta}\left(\begin{bmatrix}u_0v_0\\u_0v_1\end{bmatrix}\right)X_{\alpha+2\beta}\left(u_0\left(-dv_1^2+v_0^2\right)\right)$$

$$\begin{bmatrix}1&0&0&0&0\\0&1&0&0&0\\-u_0&0&1&0&0\\0&0&0&1&0\\0&0&0&0&1\end{bmatrix}\begin{bmatrix}1&0&0&-\sqrt{d}v_1+v_0&0\\0&1&\sqrt{d}v_1+v_0&0&0\\0&0&1&0&0\\0&0&0&1&0\\0&0&0&0&1\end{bmatrix}\begin{bmatrix}1&0&0&0&0\\0&1&0&0&0\\u_0&0&1&0&0\\0&0&0&1&0\\0&0&0&0&1\end{bmatrix}\begin{bmatrix}1&0&0&\sqrt{d}v_1-v_0&0\\0&1&-\sqrt{d}v_1-v_0&0&0\\0&0&1&0&0\\0&0&0&1&0\\0&0&0&0&1\end{bmatrix}=\begin{bmatrix}1&0&0&0&0\\u_0\left(\sqrt{d}v_1+v_0\right)&1&0&u_0\left(dv_1^2-v_0^2\right)&0\\0&0&1&u_0\left(\sqrt{d}v_1-v_0\right)&0\\0&0&0&1&0\\0&0&0&0&1\end{bmatrix}$$

$$\alpha=(-1,\ -1,\ 0,\ 0,\ 0)\,,\quad \beta=(-1,\ 1,\ 0,\ 0,\ 0)$$

$$\left[X_\alpha\left(\begin{bmatrix}u_0\\u_1\end{bmatrix}\right),X_\beta\left(\begin{bmatrix}v_0\\v_1\end{bmatrix}\right)\right]=X_{\alpha+\beta}\left(2du_0v_1-2u_1v_0\right)$$

$$\begin{bmatrix}1&0&0&0&0\\0&1&0&0&0\\0&-\sqrt{d}u_0+u_1&1&0&0\\\sqrt{d}u_0+u_1&0&0&1&0\\0&0&0&0&1\end{bmatrix}\begin{bmatrix}1&0&0&0&0\\\sqrt{d}v_1+v_0&1&0&0&0\\0&0&1&\sqrt{d}v_1-v_0&0\\0&0&0&1&0\\0&0&0&0&1\end{bmatrix}\begin{bmatrix}1&0&0&0&0\\0&1&0&0&0\\0&\sqrt{d}u_0-u_1&1&0&0\\-\sqrt{d}u_0-u_1&0&0&1&0\\0&0&0&0&1\end{bmatrix}\begin{bmatrix}1&0&0&0&0\\-\sqrt{d}v_1-v_0&1&0&0&0\\0&0&1&-\sqrt{d}v_1+v_0&0\\0&0&0&1&0\\0&0&0&0&1\end{bmatrix}=\begin{bmatrix}1&0&0&0&0\\0&1&0&0&0\\-2du_0v_1+2u_1v_0&0&1&0&0\\0&0&0&1&0\\0&0&0&0&1\end{bmatrix}$$

$$\alpha=(-1,\ -1,\ 0,\ 0,\ 0)\,,\quad \beta=(0,\ 1,\ 0,\ 0,\ 0)$$

$$\left[X_\alpha\left(\begin{bmatrix}u_0\\u_1\end{bmatrix}\right),X_\beta\left(\begin{bmatrix}v_0\\v_1\end{bmatrix}\right)\right]=X_{\alpha+\beta}\left(\begin{bmatrix}-du_0v_0-u_1v_1\\-u_0v_1-u_1v_0\end{bmatrix}\right)X_{\alpha+2\beta}\left(\begin{bmatrix}\frac{c_0du_0\left(dv_0^2-v_1^2\right)}{2}\\\frac{c_0u_1\left(dv_0^2-v_1^2\right)}{2}\end{bmatrix}\right)X_{2\alpha+2\beta}\left(0\right)$$

$$\begin{bmatrix}1&0&0&0&0\\0&1&0&0&0\\0&-\sqrt{d}u_0+u_1&1&0&0\\\sqrt{d}u_0+u_1&0&0&1&0\\0&0&0&0&1\end{bmatrix}\begin{bmatrix}1&0&0&0&0\\0&1&0&\frac{c_0\sqrt{d}\left(-dv_0^2+v_1^2\right)}{2}&c_0\left(\sqrt{d}v_1-dv_0\right)\\0&0&1&0&0\\0&0&0&1&0\\0&0&0&\sqrt{d}v_0+v_1&1\end{bmatrix}\begin{bmatrix}1&0&0&0&0\\0&1&0&0&0\\0&\sqrt{d}u_0-u_1&1&0&0\\-\sqrt{d}u_0-u_1&0&0&1&0\\0&0&0&0&1\end{bmatrix}\begin{bmatrix}1&0&0&0&0\\0&1&0&-\frac{c_0d^{\frac{3}{2}}v_0^2}{2}+\frac{c_0\sqrt{d}v_1^2}{2}&-c_0\sqrt{d}v_1+c_0dv_0\\0&0&1&0&0\\0&0&0&1&0\\0&0&0&-\sqrt{d}v_0-v_1&1\end{bmatrix}=\begin{bmatrix}1&0&0&0&0\\\frac{c_0\left(\sqrt{d}u_1+du_0\right)\left(dv_0^2-v_1^2\right)}{2}&1&0&0&0\\\frac{c_0\sqrt{d}\left(-d^2u_0^2v_0^2+du_0^2v_1^2+du_1^2v_0^2-u_1^2v_1^2\right)}{2}&0&1&\frac{c_0\left(\sqrt{d}u_1-du_0\right)\left(dv_0^2-v_1^2\right)}{2}&c_0\left(\sqrt{d}\left(du_0v_0+u_1v_1\right)-d\left(u_0v_1+u_1v_0\right)\right)\\0&0&0&1&0\\-\sqrt{d}\left(u_0v_1+u_1v_0\right)-du_0v_0-u_1v_1&0&0&0&1\end{bmatrix}$$

$$\alpha=(-1,\ -1,\ 0,\ 0,\ 0)\,,\quad \beta=(0,\ 2,\ 0,\ 0,\ 0)$$

$$\left[X_\alpha\left(\begin{bmatrix}u_0\\u_1\end{bmatrix}\right),X_\beta\left(v_0\right)\right]=X_{\alpha+\beta}\left(\begin{bmatrix}u_1v_0\\u_0v_0\end{bmatrix}\right)X_{2\alpha+\beta}\left(v_0\left(du_0^2-u_1^2\right)\right)$$

$$\begin{bmatrix}1&0&0&0&0\\0&1&0&0&0\\0&-\sqrt{d}u_0+u_1&1&0&0\\\sqrt{d}u_0+u_1&0&0&1&0\\0&0&0&0&1\end{bmatrix}\begin{bmatrix}1&0&0&0&0\\0&1&0&-v_0&0\\0&0&1&0&0\\0&0&0&1&0\\0&0&0&0&1\end{bmatrix}\begin{bmatrix}1&0&0&0&0\\0&1&0&0&0\\0&\sqrt{d}u_0-u_1&1&0&0\\-\sqrt{d}u_0-u_1&0&0&1&0\\0&0&0&0&1\end{bmatrix}\begin{bmatrix}1&0&0&0&0\\0&1&0&v_0&0\\0&0&1&0&0\\0&0&0&1&0\\0&0&0&0&1\end{bmatrix}=\begin{bmatrix}1&0&0&0&0\\v_0\left(\sqrt{d}u_0+u_1\right)&1&0&0&0\\v_0\left(-du_0^2+u_1^2\right)&0&1&v_0\left(\sqrt{d}u_0-u_1\right)&0\\0&0&0&1&0\\0&0&0&0&1\end{bmatrix}$$

$$\alpha=(-1,\ -1,\ 0,\ 0,\ 0)\,,\quad \beta=(1,\ -1,\ 0,\ 0,\ 0)$$

$$\left[X_\alpha\left(\begin{bmatrix}u_0\\u_1\end{bmatrix}\right),X_\beta\left(\begin{bmatrix}v_0\\v_1\end{bmatrix}\right)\right]=X_{\alpha+\beta}\left(-2du_0v_0-2u_1v_1\right)$$

$$\begin{bmatrix}1&0&0&0&0\\0&1&0&0&0\\0&-\sqrt{d}u_0+u_1&1&0&0\\\sqrt{d}u_0+u_1&0&0&1&0\\0&0&0&0&1\end{bmatrix}\begin{bmatrix}1&\sqrt{d}v_0+v_1&0&0&0\\0&1&0&0&0\\0&0&1&0&0\\0&0&\sqrt{d}v_0-v_1&1&0\\0&0&0&0&1\end{bmatrix}\begin{bmatrix}1&0&0&0&0\\0&1&0&0&0\\0&\sqrt{d}u_0-u_1&1&0&0\\-\sqrt{d}u_0-u_1&0&0&1&0\\0&0&0&0&1\end{bmatrix}\begin{bmatrix}1&-\sqrt{d}v_0-v_1&0&0&0\\0&1&0&0&0\\0&0&1&0&0\\0&0&-\sqrt{d}v_0+v_1&1&0\\0&0&0&0&1\end{bmatrix}=\begin{bmatrix}1&0&0&0&0\\0&1&0&0&0\\0&0&1&0&0\\0&2du_0v_0+2u_1v_1&0&1&0\\0&0&0&0&1\end{bmatrix}$$

$$\alpha=(-1,\ -1,\ 0,\ 0,\ 0)\,,\quad \beta=(1,\ 0,\ 0,\ 0,\ 0)$$

$$\left[X_\alpha\left(\begin{bmatrix}u_0\\u_1\end{bmatrix}\right),X_\beta\left(\begin{bmatrix}v_0\\v_1\end{bmatrix}\right)\right]=X_{\alpha+\beta}\left(\begin{bmatrix}u_0v_1-u_1v_0\\du_0v_0-u_1v_1\end{bmatrix}\right)X_{\alpha+2\beta}\left(\begin{bmatrix}\frac{c_0u_1\left(dv_0^2-v_1^2\right)}{2}\\\frac{c_0du_0\left(-dv_0^2+v_1^2\right)}{2}\end{bmatrix}\right)X_{2\alpha+2\beta}\left(0\right)$$

$$\begin{bmatrix}1&0&0&0&0\\0&1&0&0&0\\0&-\sqrt{d}u_0+u_1&1&0&0\\\sqrt{d}u_0+u_1&0&0&1&0\\0&0&0&0&1\end{bmatrix}\begin{bmatrix}1&0&\frac{c_0\sqrt{d}\left(-dv_0^2+v_1^2\right)}{2}&0&c_0\left(\sqrt{d}v_1-dv_0\right)\\0&1&0&0&0\\0&0&1&0&0\\0&0&0&1&0\\0&0&\sqrt{d}v_0+v_1&0&1\end{bmatrix}\begin{bmatrix}1&0&0&0&0\\0&1&0&0&0\\0&\sqrt{d}u_0-u_1&1&0&0\\-\sqrt{d}u_0-u_1&0&0&1&0\\0&0&0&0&1\end{bmatrix}\begin{bmatrix}1&0&-\frac{c_0d^{\frac{3}{2}}v_0^2}{2}+\frac{c_0\sqrt{d}v_1^2}{2}&0&-c_0\sqrt{d}v_1+c_0dv_0\\0&1&0&0&0\\0&0&1&0&0\\0&0&0&1&0\\0&0&-\sqrt{d}v_0-v_1&0&1\end{bmatrix}=\begin{bmatrix}1&\frac{c_0\left(\sqrt{d}u_1-du_0\right)\left(dv_0^2-v_1^2\right)}{2}&0&0&0\\0&1&0&0&0\\0&0&1&0&0\\0&\frac{c_0\sqrt{d}\left(-d^2u_0^2v_0^2+du_0^2v_1^2+du_1^2v_0^2-u_1^2v_1^2\right)}{2}&\frac{c_0\left(\sqrt{d}u_1+du_0\right)\left(dv_0^2-v_1^2\right)}{2}&1&-c_0\left(\sqrt{d}\left(du_0v_0-u_1v_1\right)-d\left(u_0v_1-u_1v_0\right)\right)\\0&\sqrt{d}\left(u_0v_1-u_1v_0\right)+du_0v_0-u_1v_1&0&0&1\end{bmatrix}$$

$$\alpha=(-1,\ -1,\ 0,\ 0,\ 0)\,,\quad \beta=(2,\ 0,\ 0,\ 0,\ 0)$$

$$\left[X_{\alpha}\left(\begin{bmatrix}u_0\\u_1\end{bmatrix}\right), X_{\beta}\left(v_0\right)\right] = X_{\alpha+\beta}\left(\begin{bmatrix}-u_0v_0\\u_1v_0\end{bmatrix}\right) X_{2\alpha+\beta}\left(v_0\left(du_0^2-u_1^2\right)\right)$$

$$\begin{bmatrix}1 & 0 & 0 & 0 & 0\\ 0 & 1 & 0 & 0 & 0\\ 0 & -\sqrt{d}u_0+u_1 & 1 & 0 & 0\\ \sqrt{d}u_0+u_1 & 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix}\begin{bmatrix}1 & 0 & -v_0 & 0 & 0\\ 0 & 1 & 0 & 0 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix}\begin{bmatrix}1 & 0 & 0 & 0 & 0\\ 0 & 1 & 0 & 0 & 0\\ 0 & \sqrt{d}u_0-u_1 & 1 & 0 & 0\\ -\sqrt{d}u_0-u_1 & 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix}\begin{bmatrix}1 & 0 & v_0 & 0 & 0\\ 0 & 1 & 0 & 0 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix} = \begin{bmatrix}1 & v_0\left(-\sqrt{d}u_0+u_1\right) & 0 & 0 & 0\\ 0 & 1 & 0 & 0 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & v_0\left(-du_0^2+u_1^2\right) & -v_0\left(\sqrt{d}u_0+u_1\right) & 1 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix}$$

$$\alpha = (-1,\ 0,\ 0,\ 0,\ 0), \quad \beta = (0,\ -1,\ 0,\ 0,\ 0)$$

$$\left[X_{\alpha}\left(\begin{bmatrix}u_0\\u_1\end{bmatrix}\right), X_{\beta}\left(\begin{bmatrix}v_0\\v_1\end{bmatrix}\right)\right] = X_{\alpha+\beta}\left(\begin{bmatrix}c_0\left(-du_1v_0+u_0v_1\right)\\c_0d\left(-u_0v_0+u_1v_1\right)\end{bmatrix}\right)$$

$$\begin{bmatrix}1 & 0 & 0 & 0 & 0\\ 0 & 1 & 0 & 0 & 0\\ \frac{c_0\sqrt{d}\left(du_1^2-u_0^2\right)}{2} & 0 & 1 & 0 & c_0\left(-\sqrt{d}u_0+du_1\right)\\ 0 & 0 & 0 & 1 & 0\\ \sqrt{d}u_1+u_0 & 0 & 0 & 0 & 1\end{bmatrix}\begin{bmatrix}1 & 0 & 0 & 0 & 0\\ 0 & 1 & 0 & 0 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & \frac{c_0\sqrt{d}\left(dv_0^2-v_1^2\right)}{2} & 0 & 1 & c_0\left(-\sqrt{d}v_1+dv_0\right)\\ 0 & \sqrt{d}v_0+v_1 & 0 & 0 & 1\end{bmatrix}\begin{bmatrix}1 & 0 & 0 & 0 & 0\\ 0 & 1 & 0 & 0 & 0\\ \frac{c_0d^{\frac{3}{2}}u_1^2}{2}-\frac{c_0\sqrt{d}u_0^2}{2} & 0 & 1 & 0 & c_0\sqrt{d}u_0-c_0du_1\\ 0 & 0 & 0 & 1 & 0\\ -\sqrt{d}u_1-u_0 & 0 & 0 & 0 & 1\end{bmatrix}\begin{bmatrix}1 & 0 & 0 & 0 & 0\\ 0 & 1 & 0 & 0 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & \frac{c_0d^{\frac{3}{2}}v_0^2}{2}-\frac{c_0\sqrt{d}v_1^2}{2} & 0 & 1 & c_0\sqrt{d}v_1-c_0dv_0\\ 0 & -\sqrt{d}v_0-v_1 & 0 & 0 & 1\end{bmatrix} = \begin{bmatrix}1 & 0 & 0 & 0 & 0\\ 0 & 1 & 0 & 0 & 0\\ 0 & c_0\left(\sqrt{d}\left(du_1v_0-u_0v_1\right)-d\left(u_0v_0-u_1v_1\right)\right) & 1 & 0 & 0\\ -c_0\left(\sqrt{d}\left(du_1v_0-u_0v_1\right)+d\left(u_0v_0-u_1v_1\right)\right) & 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix}$$

$$\alpha = (-1,\ 0,\ 0,\ 0,\ 0), \quad \beta = (0,\ 1,\ 0,\ 0,\ 0)$$

$$\left[X_{\alpha}\left(\begin{bmatrix}u_0\\u_1\end{bmatrix}\right), X_{\beta}\left(\begin{bmatrix}v_0\\v_1\end{bmatrix}\right)\right] = X_{\alpha+\beta}\left(\begin{bmatrix}c_0d\left(u_0v_0-u_1v_1\right)\\c_0\left(du_1v_0-u_0v_1\right)\end{bmatrix}\right)$$

$$\begin{bmatrix}1 & 0 & 0 & 0 & 0\\ 0 & 1 & 0 & 0 & 0\\ \frac{c_0\sqrt{d}\left(du_1^2-u_0^2\right)}{2} & 0 & 1 & 0 & c_0\left(-\sqrt{d}u_0+du_1\right)\\ 0 & 0 & 0 & 1 & 0\\ \sqrt{d}u_1+u_0 & 0 & 0 & 0 & 1\end{bmatrix}\begin{bmatrix}1 & 0 & 0 & 0 & 0\\ 0 & 1 & 0 & \frac{c_0\sqrt{d}\left(-dv_0^2+v_1^2\right)}{2} & c_0\left(\sqrt{d}v_1-dv_0\right)\\ 0 & 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 1 & 0\\ 0 & 0 & 0 & \sqrt{d}v_0+v_1 & 1\end{bmatrix}\begin{bmatrix}1 & 0 & 0 & 0 & 0\\ 0 & 1 & 0 & 0 & 0\\ \frac{c_0d^{\frac{3}{2}}u_1^2}{2}-\frac{c_0\sqrt{d}u_0^2}{2} & 0 & 1 & 0 & c_0\sqrt{d}u_0-c_0du_1\\ 0 & 0 & 0 & 1 & 0\\ -\sqrt{d}u_1-u_0 & 0 & 0 & 0 & 1\end{bmatrix}\begin{bmatrix}1 & 0 & 0 & 0 & 0\\ 0 & 1 & 0 & -\frac{c_0d^{\frac{3}{2}}v_0^2}{2}+\frac{c_0\sqrt{d}v_1^2}{2} & -c_0\sqrt{d}v_1+c_0dv_0\\ 0 & 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 1 & 0\\ 0 & 0 & 0 & -\sqrt{d}v_0-v_1 & 1\end{bmatrix} = \begin{bmatrix}1 & 0 & 0 & 0 & 0\\ c_0\left(\sqrt{d}\left(du_1v_0-u_0v_1\right)+d\left(u_0v_0-u_1v_1\right)\right) & 1 & 0 & 0 & 0\\ 0 & 0 & 1 & c_0\left(\sqrt{d}\left(du_1v_0-u_0v_1\right)-d\left(u_0v_0-u_1v_1\right)\right) & 0\\ 0 & 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix}$$

$$\alpha = (-1,\ 0,\ 0,\ 0,\ 0), \quad \beta = (1,\ -1,\ 0,\ 0,\ 0)$$

$$\left[X_{\alpha}\left(\begin{bmatrix}u_0\\u_1\end{bmatrix}\right), X_{\beta}\left(\begin{bmatrix}v_0\\v_1\end{bmatrix}\right)\right] = X_{\alpha+\beta}\left(\begin{bmatrix}u_0v_0+u_1v_1\\du_1v_0+u_0v_1\end{bmatrix}\right) X_{2\alpha+\beta}\left(\begin{bmatrix}\frac{c_0v_1\left(-du_1^2+u_0^2\right)}{2}\\\frac{c_0dv_0\left(du_1^2-u_0^2\right)}{2}\end{bmatrix}\right) X_{2\alpha+2\beta}\left(0\right)$$

$$\begin{bmatrix}1 & 0 & 0 & 0 & 0\\ 0 & 1 & 0 & 0 & 0\\ \frac{c_0\sqrt{d}\left(du_1^2-u_0^2\right)}{2} & 0 & 1 & 0 & c_0\left(-\sqrt{d}u_0+du_1\right)\\ 0 & 0 & 0 & 1 & 0\\ \sqrt{d}u_1+u_0 & 0 & 0 & 0 & 1\end{bmatrix}\begin{bmatrix}1 & \sqrt{d}v_0+v_1 & 0 & 0 & 0\\ 0 & 1 & 0 & 0 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & 0 & \sqrt{d}v_0-v_1 & 1 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix}\begin{bmatrix}1 & 0 & 0 & 0 & 0\\ 0 & 1 & 0 & 0 & 0\\ \frac{c_0d^{\frac{3}{2}}u_1^2}{2}-\frac{c_0\sqrt{d}u_0^2}{2} & 0 & 1 & 0 & c_0\sqrt{d}u_0-c_0du_1\\ 0 & 0 & 0 & 1 & 0\\ -\sqrt{d}u_1-u_0 & 0 & 0 & 0 & 1\end{bmatrix}\begin{bmatrix}1 & -\sqrt{d}v_0-v_1 & 0 & 0 & 0\\ 0 & 1 & 0 & 0 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & 0 & -\sqrt{d}v_0+v_1 & 1 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix} = \begin{bmatrix}1 & 0 & 0 & 0 & 0\\ 0 & 1 & 0 & 0 & 0\\ 0 & \frac{c_0\left(\sqrt{d}v_1+dv_0\right)\left(du_1^2-u_0^2\right)}{2} & 1 & 0 & 0\\ -\frac{c_0\left(\sqrt{d}v_1-dv_0\right)\left(du_1^2-u_0^2\right)}{2} & \frac{c_0\sqrt{d}\left(-d^2u_1^2v_0^2+du_0^2v_0^2+du_1^2v_1^2-u_0^2v_1^2\right)}{2} & 0 & 1 & -c_0\left(\sqrt{d}\left(du_1v_0+u_0v_1\right)-d\left(u_0v_0+u_1v_1\right)\right)\\ 0 & \sqrt{d}\left(u_0v_0+u_1v_1\right)+du_1v_0+u_0v_1 & 0 & 0 & 1\end{bmatrix}$$

$$\alpha = (-1,\ 0,\ 0,\ 0,\ 0), \quad \beta = (1,\ 1,\ 0,\ 0,\ 0)$$

$$\left[X_{\alpha}\left(\begin{bmatrix}u_0\\u_1\end{bmatrix}\right), X_{\beta}\left(\begin{bmatrix}v_0\\v_1\end{bmatrix}\right)\right] = X_{\alpha+\beta}\left(\begin{bmatrix}-u_0v_1+u_1v_0\\-du_1v_1+u_0v_0\end{bmatrix}\right) X_{2\alpha+\beta}\left(\begin{bmatrix}\frac{c_0dv_1\left(du_1^2-u_0^2\right)}{2}\\\frac{c_0v_0\left(du_1^2-u_0^2\right)}{2}\end{bmatrix}\right) X_{2\alpha+2\beta}\left(0\right)$$

$$\begin{bmatrix}1 & 0 & 0 & 0 & 0\\ 0 & 1 & 0 & 0 & 0\\ \frac{c_0\sqrt{d}\left(du_1^2-u_0^2\right)}{2} & 0 & 1 & 0 & c_0\left(-\sqrt{d}u_0+du_1\right)\\ 0 & 0 & 0 & 1 & 0\\ \sqrt{d}u_1+u_0 & 0 & 0 & 0 & 1\end{bmatrix}\begin{bmatrix}1 & 0 & 0 & -\sqrt{d}v_1+v_0 & 0\\ 0 & 1 & \sqrt{d}v_1+v_0 & 0 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix}\begin{bmatrix}1 & 0 & 0 & 0 & 0\\ 0 & 1 & 0 & 0 & 0\\ \frac{c_0d^{\frac{3}{2}}u_1^2}{2}-\frac{c_0\sqrt{d}u_0^2}{2} & 0 & 1 & 0 & c_0\sqrt{d}u_0-c_0du_1\\ 0 & 0 & 0 & 1 & 0\\ -\sqrt{d}u_1-u_0 & 0 & 0 & 0 & 1\end{bmatrix}\begin{bmatrix}1 & 0 & 0 & \sqrt{d}v_1-v_0 & 0\\ 0 & 1 & -\sqrt{d}v_1-v_0 & 0 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix} = \begin{bmatrix}1 & 0 & 0 & 0 & 0\\ \frac{c_0\left(\sqrt{d}v_0+dv_1\right)\left(du_1^2-u_0^2\right)}{2} & 1 & 0 & \frac{c_0\sqrt{d}\left(d^2u_1^2v_1^2-du_0^2v_1^2-du_1^2v_0^2+u_0^2v_0^2\right)}{2} & -c_0\left(\sqrt{d}\left(du_1v_1-u_0v_0\right)-d\left(u_0v_1-u_1v_0\right)\right)\\ 0 & 0 & 1 & \frac{c_0\left(\sqrt{d}v_0-dv_1\right)\left(du_1^2-u_0^2\right)}{2} & 0\\ 0 & 0 & 0 & 1 & 0\\ 0 & 0 & 0 & \sqrt{d}\left(-u_0v_1+u_1v_0\right)-du_1v_1+u_0v_0 & 1\end{bmatrix}$$

$$\alpha = (-1,\ 1,\ 0,\ 0,\ 0), \quad \beta = (-1,\ -1,\ 0,\ 0,\ 0)$$

$$\left[X_{\alpha}\left(\begin{bmatrix}u_0\\u_1\end{bmatrix}\right), X_{\beta}\left(\begin{bmatrix}v_0\\v_1\end{bmatrix}\right)\right] = X_{\alpha+\beta}\left(-2du_1v_0+2u_0v_1\right)$$

$$\begin{bmatrix}1 & 0 & 0 & 0 & 0\\ \sqrt{d}u_1+u_0 & 1 & 0 & 0 & 0\\ 0 & 0 & 1 & \sqrt{d}u_1-u_0 & 0\\ 0 & 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix}\begin{bmatrix}1 & 0 & 0 & 0 & 0\\ 0 & 1 & 0 & 0 & 0\\ 0 & -\sqrt{d}v_0+v_1 & 1 & 0 & 0\\ \sqrt{d}v_0+v_1 & 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix}\begin{bmatrix}1 & 0 & 0 & 0 & 0\\ -\sqrt{d}u_1-u_0 & 1 & 0 & 0 & 0\\ 0 & 0 & 1 & -\sqrt{d}u_1+u_0 & 0\\ 0 & 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix}\begin{bmatrix}1 & 0 & 0 & 0 & 0\\ 0 & 1 & 0 & 0 & 0\\ 0 & \sqrt{d}v_0-v_1 & 1 & 0 & 0\\ -\sqrt{d}v_0-v_1 & 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix} = \begin{bmatrix}1 & 0 & 0 & 0 & 0\\ 0 & 1 & 0 & 0 & 0\\ 2du_1v_0-2u_0v_1 & 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix}$$

$$\alpha = (-1,\ 1,\ 0,\ 0,\ 0), \quad \beta = (0,\ -2,\ 0,\ 0,\ 0)$$

$$\left[X_{\alpha}\left(\begin{bmatrix}u_0\\u_1\end{bmatrix}\right), X_{\beta}\left(v_0\right)\right] = X_{\alpha+\beta}\left(\begin{bmatrix}u_1v_0\\u_0v_0\end{bmatrix}\right) X_{2\alpha+\beta}\left(v_0\left(-du_1^2+u_0^2\right)\right)$$

$$\begin{bmatrix}1 & 0 & 0 & 0 & 0\\ \sqrt{d}u_1+u_0 & 1 & 0 & 0 & 0\\ 0 & 0 & 1 & \sqrt{d}u_1-u_0 & 0\\ 0 & 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix}\begin{bmatrix}1 & 0 & 0 & 0 & 0\\ 0 & 1 & 0 & 0 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & -v_0 & 0 & 1 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix}\begin{bmatrix}1 & 0 & 0 & 0 & 0\\ -\sqrt{d}u_1-u_0 & 1 & 0 & 0 & 0\\ 0 & 0 & 1 & -\sqrt{d}u_1+u_0 & 0\\ 0 & 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix}\begin{bmatrix}1 & 0 & 0 & 0 & 0\\ 0 & 1 & 0 & 0 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & v_0 & 0 & 1 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix} = \begin{bmatrix}1 & 0 & 0 & 0 & 0\\ 0 & 1 & 0 & 0 & 0\\ v_0\left(du_1^2-u_0^2\right) & v_0\left(-\sqrt{d}u_1+u_0\right) & 1 & 0 & 0\\ v_0\left(\sqrt{d}u_1+u_0\right) & 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix}$$

$$\alpha = (-1,\ 1,\ 0,\ 0,\ 0), \quad \beta = (0,\ -1,\ 0,\ 0,\ 0)$$

$$\left[X_\alpha\left(\begin{bmatrix}u_0\\u_1\end{bmatrix}\right),X_\beta\left(\begin{bmatrix}v_0\\v_1\end{bmatrix}\right)\right]=X_{\alpha+\beta}\left(\begin{bmatrix}-du_1v_0-u_0v_1\\-u_0v_0-u_1v_1\end{bmatrix}\right)X_{\alpha+2\beta}\left(\begin{bmatrix}\frac{c_0u_0\left(-dv_0^2+v_1^2\right)}{2}\\\frac{c_0du_1\left(-dv_0^2+v_1^2\right)}{2}\end{bmatrix}\right)X_{2\alpha+2\beta}\left(0\right)$$

$$\begin{bmatrix}1&0&0&0&0\\\sqrt{d}u_1+u_0&1&0&0&0\\0&0&1&\sqrt{d}u_1-u_0&0\\0&0&0&1&0\\0&0&0&0&1\end{bmatrix}\begin{bmatrix}1&0&0&0&0\\0&1&0&0&0\\0&0&1&0&0\\0&\frac{c_0\sqrt{d}\left(dv_0^2-v_1^2\right)}{2}&0&1&c_0\left(-\sqrt{d}v_1+dv_0\right)\\0&\sqrt{d}v_0+v_1&0&0&1\end{bmatrix}\begin{bmatrix}1&0&0&0&0\\-\sqrt{d}u_1-u_0&1&0&0&0\\0&0&1&-\sqrt{d}u_1+u_0&0\\0&0&0&1&0\\0&0&0&0&1\end{bmatrix}\begin{bmatrix}1&0&0&0&0\\0&1&0&0&0\\0&0&1&0&0\\0&\frac{c_0d^{\frac{3}{2}}v_0^2}{2}-\frac{c_0\sqrt{d}v_1^2}{2}&0&1&c_0\sqrt{d}v_1-c_0dv_0\\0&-\sqrt{d}v_0-v_1&0&0&1\end{bmatrix}=\begin{bmatrix}1&0&0&0&0\\0&1&0&0&0\\\frac{c_0\sqrt{d}\left(-d^2u_1^2v_0^2+du_0^2v_0^2+du_1^2v_1^2-u_0^2v_1^2\right)}{2}&\frac{c_0\left(\sqrt{d}u_0-du_1\right)\left(dv_0^2-v_1^2\right)}{2}&1&0&c_0\left(\sqrt{d}\left(du_1v_0+u_0v_1\right)-d\left(u_0v_0+u_1v_1\right)\right)\\-\frac{c_0\left(\sqrt{d}u_0+du_1\right)\left(dv_0^2-v_1^2\right)}{2}&0&0&1&0\\-\sqrt{d}\left(u_0v_0+u_1v_1\right)-du_1v_0-u_0v_1&0&0&0&1\end{bmatrix}$$

$$\alpha=\left(-1,\ 1,\ 0,\ 0,\ 0\right),\quad\beta=\left(1,\ 0,\ 0,\ 0,\ 0\right)$$

$$\left[X_\alpha\left(\begin{bmatrix}u_0\\u_1\end{bmatrix}\right),X_\beta\left(\begin{bmatrix}v_0\\v_1\end{bmatrix}\right)\right]=X_{\alpha+\beta}\left(\begin{bmatrix}u_0v_0-u_1v_1\\-du_1v_0+u_0v_1\end{bmatrix}\right)X_{\alpha+2\beta}\left(\begin{bmatrix}\frac{c_0du_1\left(dv_0^2-v_1^2\right)}{2}\\\frac{c_0u_0\left(dv_0^2-v_1^2\right)}{2}\end{bmatrix}\right)X_{2\alpha+2\beta}\left(0\right)$$

$$\begin{bmatrix}1&0&0&0&0\\\sqrt{d}u_1+u_0&1&0&0&0\\0&0&1&\sqrt{d}u_1-u_0&0\\0&0&0&1&0\\0&0&0&0&1\end{bmatrix}\begin{bmatrix}1&0&\frac{c_0\sqrt{d}\left(-dv_0^2+v_1^2\right)}{2}&0&c_0\left(\sqrt{d}v_1-dv_0\right)\\0&1&0&0&0\\0&0&1&0&0\\0&0&0&1&0\\0&0&\sqrt{d}v_0+v_1&0&1\end{bmatrix}\begin{bmatrix}1&0&0&0&0\\-\sqrt{d}u_1-u_0&1&0&0&0\\0&0&1&-\sqrt{d}u_1+u_0&0\\0&0&0&1&0\\0&0&0&0&1\end{bmatrix}\begin{bmatrix}1&0&-\frac{c_0d^{\frac{3}{2}}v_0^2}{2}+\frac{c_0\sqrt{d}v_1^2}{2}&0&-c_0\sqrt{d}v_1+c_0dv_0\\0&1&0&0&0\\0&0&1&0&0\\0&0&0&1&0\\0&0&-\sqrt{d}v_0-v_1&0&1\end{bmatrix}=\begin{bmatrix}1&0&0&-\frac{c_0\left(\sqrt{d}u_0-du_1\right)\left(dv_0^2-v_1^2\right)}{2}&0\\0&1&\frac{c_0\left(\sqrt{d}u_0+du_1\right)\left(dv_0^2-v_1^2\right)}{2}&\frac{c_0\sqrt{d}\left(d^2u_1^2v_0^2-du_0^2v_0^2-du_1^2v_1^2+u_0^2v_1^2\right)}{2}&-c_0\left(\sqrt{d}\left(du_1v_0-u_0v_1\right)+d\left(u_0v_0-u_1v_1\right)\right)\\0&0&1&0&0\\0&0&0&1&0\\0&0&0&\sqrt{d}\left(u_0v_0-u_1v_1\right)-du_1v_0+u_0v_1&1\end{bmatrix}$$

$$\alpha=\left(-1,\ 1,\ 0,\ 0,\ 0\right),\quad\beta=\left(1,\ 1,\ 0,\ 0,\ 0\right)$$

$$\left[X_\alpha\left(\begin{bmatrix}u_0\\u_1\end{bmatrix}\right),X_\beta\left(\begin{bmatrix}v_0\\v_1\end{bmatrix}\right)\right]=X_{\alpha+\beta}\left(2du_1v_1-2u_0v_0\right)$$

$$\begin{bmatrix}1&0&0&0&0\\\sqrt{d}u_1+u_0&1&0&0&0\\0&0&1&\sqrt{d}u_1-u_0&0\\0&0&0&1&0\\0&0&0&0&1\end{bmatrix}\begin{bmatrix}1&0&0&-\sqrt{d}v_1+v_0&0\\0&1&\sqrt{d}v_1+v_0&0&0\\0&0&1&0&0\\0&0&0&1&0\\0&0&0&0&1\end{bmatrix}\begin{bmatrix}1&0&0&0&0\\-\sqrt{d}u_1-u_0&1&0&0&0\\0&0&1&-\sqrt{d}u_1+u_0&0\\0&0&0&1&0\\0&0&0&0&1\end{bmatrix}\begin{bmatrix}1&0&0&\sqrt{d}v_1-v_0&0\\0&1&-\sqrt{d}v_1-v_0&0&0\\0&0&1&0&0\\0&0&0&1&0\\0&0&0&0&1\end{bmatrix}=\begin{bmatrix}1&0&0&0&0\\0&1&0&-2du_1v_1+2u_0v_0&0\\0&0&1&0&0\\0&0&0&1&0\\0&0&0&0&1\end{bmatrix}$$

$$\alpha=\left(-1,\ 1,\ 0,\ 0,\ 0\right),\quad\beta=\left(2,\ 0,\ 0,\ 0,\ 0\right)$$

$$\left[X_\alpha\left(\begin{bmatrix}u_0\\u_1\end{bmatrix}\right),X_\beta\left(v_0\right)\right]=X_{\alpha+\beta}\left(\begin{bmatrix}-u_0v_0\\-u_1v_0\end{bmatrix}\right)X_{2\alpha+\beta}\left(v_0\left(-du_1^2+u_0^2\right)\right)$$

$$\begin{bmatrix}1&0&0&0&0\\\sqrt{d}u_1+u_0&1&0&0&0\\0&0&1&\sqrt{d}u_1-u_0&0\\0&0&0&1&0\\0&0&0&0&1\end{bmatrix}\begin{bmatrix}1&0&-v_0&0&0\\0&1&0&0&0\\0&0&1&0&0\\0&0&0&1&0\\0&0&0&0&1\end{bmatrix}\begin{bmatrix}1&0&0&0&0\\-\sqrt{d}u_1-u_0&1&0&0&0\\0&0&1&-\sqrt{d}u_1+u_0&0\\0&0&0&1&0\\0&0&0&0&1\end{bmatrix}\begin{bmatrix}1&0&v_0&0&0\\0&1&0&0&0\\0&0&1&0&0\\0&0&0&1&0\\0&0&0&0&1\end{bmatrix}=\begin{bmatrix}1&0&0&v_0\left(\sqrt{d}u_1-u_0\right)&0\\0&1&-v_0\left(\sqrt{d}u_1+u_0\right)&v_0\left(du_1^2-u_0^2\right)&0\\0&0&1&0&0\\0&0&0&1&0\\0&0&0&0&1\end{bmatrix}$$

$$\alpha=\left(0,\ -2,\ 0,\ 0,\ 0\right),\quad\beta=\left(-1,\ 1,\ 0,\ 0,\ 0\right)$$

$$\left[X_\alpha\left(u_0\right),X_\beta\left(\begin{bmatrix}v_0\\v_1\end{bmatrix}\right)\right]=X_{\alpha+\beta}\left(\begin{bmatrix}-u_0v_1\\-u_0v_0\end{bmatrix}\right)X_{\alpha+2\beta}\left(u_0\left(dv_1^2-v_0^2\right)\right)$$

$$\begin{bmatrix}1&0&0&0&0\\0&1&0&0&0\\0&0&1&0&0\\0&-u_0&0&1&0\\0&0&0&0&1\end{bmatrix}\begin{bmatrix}1&0&0&0&0\\\sqrt{d}v_1+v_0&1&0&0&0\\0&0&1&\sqrt{d}v_1-v_0&0\\0&0&0&1&0\\0&0&0&0&1\end{bmatrix}\begin{bmatrix}1&0&0&0&0\\0&1&0&0&0\\0&0&1&0&0\\0&u_0&0&1&0\\0&0&0&0&1\end{bmatrix}\begin{bmatrix}1&0&0&0&0\\-\sqrt{d}v_1-v_0&1&0&0&0\\0&0&1&-\sqrt{d}v_1+v_0&0\\0&0&0&1&0\\0&0&0&0&1\end{bmatrix}=\begin{bmatrix}1&0&0&0&0\\0&1&0&0&0\\u_0\left(-dv_1^2+v_0^2\right)&u_0\left(\sqrt{d}v_1-v_0\right)&1&0&0\\-u_0\left(\sqrt{d}v_1+v_0\right)&0&0&1&0\\0&0&0&0&1\end{bmatrix}$$

$$\alpha=\left(0,\ -2,\ 0,\ 0,\ 0\right),\quad\beta=\left(1,\ 1,\ 0,\ 0,\ 0\right)$$

$$\left[X_\alpha\left(u_0\right),X_\beta\left(\begin{bmatrix}v_0\\v_1\end{bmatrix}\right)\right]=X_{\alpha+\beta}\left(\begin{bmatrix}-u_0v_1\\u_0v_0\end{bmatrix}\right)X_{\alpha+2\beta}\left(u_0\left(-dv_1^2+v_0^2\right)\right)$$

$$\begin{bmatrix}1&0&0&0&0\\0&1&0&0&0\\0&0&1&0&0\\0&-u_0&0&1&0\\0&0&0&0&1\end{bmatrix}\begin{bmatrix}1&0&0&-\sqrt{d}v_1+v_0&0\\0&1&\sqrt{d}v_1+v_0&0&0\\0&0&1&0&0\\0&0&0&1&0\\0&0&0&0&1\end{bmatrix}\begin{bmatrix}1&0&0&0&0\\0&1&0&0&0\\0&0&1&0&0\\0&u_0&0&1&0\\0&0&0&0&1\end{bmatrix}\begin{bmatrix}1&0&0&\sqrt{d}v_1-v_0&0\\0&1&-\sqrt{d}v_1-v_0&0&0\\0&0&1&0&0\\0&0&0&1&0\\0&0&0&0&1\end{bmatrix}=\begin{bmatrix}1&u_0\left(-\sqrt{d}v_1+v_0\right)&u_0\left(dv_1^2-v_0^2\right)&0&0\\0&1&0&0&0\\0&0&1&0&0\\0&0&-u_0\left(\sqrt{d}v_1+v_0\right)&1&0\\0&0&0&0&1\end{bmatrix}$$

$$\alpha=\left(0,\ -1,\ 0,\ 0,\ 0\right),\quad\beta=\left(-1,\ 0,\ 0,\ 0,\ 0\right)$$

$$\left[X_\alpha\left(\begin{bmatrix}u_0\\u_1\end{bmatrix}\right),X_\beta\left(\begin{bmatrix}v_0\\v_1\end{bmatrix}\right)\right]=X_{\alpha+\beta}\left(\begin{bmatrix}c_0\left(du_0v_1-u_1v_0\right)\\c_0d\left(u_0v_0-u_1v_1\right)\end{bmatrix}\right)$$

$$\begin{bmatrix}1&0&0&0&0\\0&1&0&0&0\\0&0&1&0&0\\0&\frac{c_0\sqrt{d}\left(du_0^2-u_1^2\right)}{2}&0&1&c_0\left(-\sqrt{d}u_1+du_0\right)\\0&\sqrt{d}u_0+u_1&0&0&1\end{bmatrix}\begin{bmatrix}1&0&0&0&0\\0&1&0&0&0\\\frac{c_0\sqrt{d}\left(dv_1^2-v_0^2\right)}{2}&0&1&0&c_0\left(-\sqrt{d}v_0+dv_1\right)\\0&0&0&1&0\\\sqrt{d}v_1+v_0&0&0&0&1\end{bmatrix}\begin{bmatrix}1&0&0&0&0\\0&1&0&0&0\\0&0&1&0&0\\0&\frac{c_0d^{\frac{3}{2}}u_0^2}{2}-\frac{c_0\sqrt{d}u_1^2}{2}&0&1&c_0\sqrt{d}u_1-c_0du_0\\0&-\sqrt{d}u_0-u_1&0&0&1\end{bmatrix}\begin{bmatrix}1&0&0&0&0\\0&1&0&0&0\\\frac{c_0d^{\frac{3}{2}}v_1^2}{2}-\frac{c_0\sqrt{d}v_0^2}{2}&0&1&0&c_0\sqrt{d}v_0-c_0dv_1\\0&0&0&1&0\\-\sqrt{d}v_1-v_0&0&0&0&1\end{bmatrix}=\begin{bmatrix}1&0&0&0&0\\0&1&0&0&0\\0&-c_0\left(\sqrt{d}\left(du_0v_1-u_1v_0\right)-d\left(u_0v_0-u_1v_1\right)\right)&1&0&0\\c_0\left(\sqrt{d}\left(du_0v_1-u_1v_0\right)+d\left(u_0v_0-u_1v_1\right)\right)&0&0&1&0\\0&0&0&0&1\end{bmatrix}$$

$$\alpha = (0,\ -1,\ 0,\ 0,\ 0)\,, \quad \beta = (-1,\ 1,\ 0,\ 0,\ 0)$$

$$\left[X_\alpha\left(\begin{bmatrix}u_0\\u_1\end{bmatrix}\right), X_\beta\left(\begin{bmatrix}v_0\\v_1\end{bmatrix}\right)\right] = X_{\alpha+\beta}\left(\begin{bmatrix}du_0v_1+u_1v_0\\u_0v_0+u_1v_1\end{bmatrix}\right) X_{2\alpha+\beta}\left(\begin{bmatrix}\frac{c_0v_0\left(du_0^2-u_1^2\right)}{2}\\ \frac{c_0dv_1\left(du_0^2-u_1^2\right)}{2}\end{bmatrix}\right) X_{2\alpha+2\beta}\left(0\right)$$

$$\begin{bmatrix}1 & 0 & 0 & 0 & 0\\ 0 & 1 & 0 & 0 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & \frac{c_0\sqrt{d}\left(du_0^2-u_1^2\right)}{2} & 0 & 1 & c_0\left(-\sqrt{d}u_1+du_0\right)\\ 0 & \sqrt{d}u_0+u_1 & 0 & 0 & 1\end{bmatrix}\begin{bmatrix}1 & 0 & 0 & 0 & 0\\ \sqrt{d}v_1+v_0 & 1 & 0 & 0 & 0\\ 0 & 0 & 1 & \sqrt{d}v_1-v_0 & 0\\ 0 & 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix}\begin{bmatrix}1 & 0 & 0 & 0 & 0\\ 0 & 1 & 0 & 0 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & \frac{c_0d^{\frac{3}{2}}u_0^2}{2}-\frac{c_0\sqrt{d}u_1^2}{2} & 0 & 1 & c_0\sqrt{d}u_1-c_0du_0\\ 0 & -\sqrt{d}u_0-u_1 & 0 & 0 & 1\end{bmatrix}\begin{bmatrix}1 & 0 & 0 & 0 & 0\\ -\sqrt{d}v_1-v_0 & 1 & 0 & 0 & 0\\ 0 & 0 & 1 & -\sqrt{d}v_1+v_0 & 0\\ 0 & 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix} = \begin{bmatrix}1 & 0 & 0 & 0 & 0\\ 0 & 1 & 0 & 0 & 0\\ \frac{c_0\sqrt{d}\left(-d^2u_0^2v_1^2+du_0^2v_0^2+du_1^2v_1^2-u_1^2v_0^2\right)}{2} & -\frac{c_0\left(\sqrt{d}v_0-dv_1\right)\left(du_0^2-u_1^2\right)}{2} & 1 & 0 & -c_0\left(\sqrt{d}\left(du_0v_1+u_1v_0\right)-d\left(u_0v_0+u_1v_1\right)\right)\\ \frac{c_0\left(\sqrt{d}v_0+dv_1\right)\left(du_0^2-u_1^2\right)}{2} & 0 & 0 & 1 & 0\\ \sqrt{d}\left(u_0v_0+u_1v_1\right)+du_0v_1+u_1v_0 & 0 & 0 & 0 & 1\end{bmatrix}$$

$$\alpha = (0,\ -1,\ 0,\ 0,\ 0)\,, \quad \beta = (1,\ 0,\ 0,\ 0,\ 0)$$

$$\left[X_\alpha\left(\begin{bmatrix}u_0\\u_1\end{bmatrix}\right), X_\beta\left(\begin{bmatrix}v_0\\v_1\end{bmatrix}\right)\right] = X_{\alpha+\beta}\left(\begin{bmatrix}c_0\left(du_0v_0-u_1v_1\right)\\c_0d\left(-u_0v_1+u_1v_0\right)\end{bmatrix}\right)$$

$$\begin{bmatrix}1 & 0 & 0 & 0 & 0\\ 0 & 1 & 0 & 0 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & \frac{c_0\sqrt{d}\left(du_0^2-u_1^2\right)}{2} & 0 & 1 & c_0\left(-\sqrt{d}u_1+du_0\right)\\ 0 & \sqrt{d}u_0+u_1 & 0 & 0 & 1\end{bmatrix}\begin{bmatrix}1 & 0 & \frac{c_0\sqrt{d}\left(-dv_0^2+v_1^2\right)}{2} & 0 & c_0\left(\sqrt{d}v_1-dv_0\right)\\ 0 & 1 & 0 & 0 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 1 & 0\\ 0 & 0 & \sqrt{d}v_0+v_1 & 0 & 1\end{bmatrix}\begin{bmatrix}1 & 0 & 0 & 0 & 0\\ 0 & 1 & 0 & 0 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & \frac{c_0d^{\frac{3}{2}}u_0^2}{2}-\frac{c_0\sqrt{d}u_1^2}{2} & 0 & 1 & c_0\sqrt{d}u_1-c_0du_0\\ 0 & -\sqrt{d}u_0-u_1 & 0 & 0 & 1\end{bmatrix}\begin{bmatrix}1 & 0 & -\frac{c_0d^{\frac{3}{2}}v_0^2}{2}+\frac{c_0\sqrt{d}v_1^2}{2} & 0 & -c_0\sqrt{d}v_1+c_0dv_0\\ 0 & 1 & 0 & 0 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 1 & 0\\ 0 & 0 & -\sqrt{d}v_0-v_1 & 0 & 1\end{bmatrix} = \begin{bmatrix}1 & c_0\left(\sqrt{d}\left(du_0v_0-u_1v_1\right)-d\left(u_0v_1-u_1v_0\right)\right) & 0 & 0 & 0\\ 0 & 1 & 0 & 0 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & 0 & c_0\left(\sqrt{d}\left(du_0v_0-u_1v_1\right)+d\left(u_0v_1-u_1v_0\right)\right) & 1 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix}$$

$$\alpha = (0,\ -1,\ 0,\ 0,\ 0)\,, \quad \beta = (1,\ 1,\ 0,\ 0,\ 0)$$

$$\left[X_\alpha\left(\begin{bmatrix}u_0\\u_1\end{bmatrix}\right), X_\beta\left(\begin{bmatrix}v_0\\v_1\end{bmatrix}\right)\right] = X_{\alpha+\beta}\left(\begin{bmatrix}u_0v_0+u_1v_1\\du_0v_1+u_1v_0\end{bmatrix}\right) X_{2\alpha+\beta}\left(\begin{bmatrix}\frac{c_0v_0\left(du_0^2-u_1^2\right)}{2}\\ \frac{c_0dv_1\left(-du_0^2+u_1^2\right)}{2}\end{bmatrix}\right) X_{2\alpha+2\beta}\left(0\right)$$

$$\begin{bmatrix}1 & 0 & 0 & 0 & 0\\ 0 & 1 & 0 & 0 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & \frac{c_0\sqrt{d}\left(du_0^2-u_1^2\right)}{2} & 0 & 1 & c_0\left(-\sqrt{d}u_1+du_0\right)\\ 0 & \sqrt{d}u_0+u_1 & 0 & 0 & 1\end{bmatrix}\begin{bmatrix}1 & 0 & 0 & -\sqrt{d}v_1+v_0 & 0\\ 0 & 1 & \sqrt{d}v_1+v_0 & 0 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix}\begin{bmatrix}1 & 0 & 0 & 0 & 0\\ 0 & 1 & 0 & 0 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & \frac{c_0d^{\frac{3}{2}}u_0^2}{2}-\frac{c_0\sqrt{d}u_1^2}{2} & 0 & 1 & c_0\sqrt{d}u_1-c_0du_0\\ 0 & -\sqrt{d}u_0-u_1 & 0 & 0 & 1\end{bmatrix}\begin{bmatrix}1 & 0 & 0 & \sqrt{d}v_1-v_0 & 0\\ 0 & 1 & -\sqrt{d}v_1-v_0 & 0 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix} = \begin{bmatrix}1 & \frac{c_0\left(\sqrt{d}v_0-dv_1\right)\left(du_0^2-u_1^2\right)}{2} & \frac{c_0\sqrt{d}\left(d^2u_0^2v_1^2-du_0^2v_0^2-du_1^2v_1^2+u_1^2v_0^2\right)}{2} & 0 & c_0\left(\sqrt{d}\left(du_0v_1+u_1v_0\right)-d\left(u_0v_0+u_1v_1\right)\right)\\ 0 & 1 & 0 & 0 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & 0 & \frac{c_0\left(\sqrt{d}v_0+dv_1\right)\left(du_0^2-u_1^2\right)}{2} & 1 & 0\\ 0 & 0 & \sqrt{d}\left(u_0v_0+u_1v_1\right)+du_0v_1+u_1v_0 & 0 & 1\end{bmatrix}$$

$$\alpha = (0,\ 1,\ 0,\ 0,\ 0)\,, \quad \beta = (-1,\ -1,\ 0,\ 0,\ 0)$$

$$\left[X_\alpha\left(\begin{bmatrix}u_0\\u_1\end{bmatrix}\right), X_\beta\left(\begin{bmatrix}v_0\\v_1\end{bmatrix}\right)\right] = X_{\alpha+\beta}\left(\begin{bmatrix}du_0v_0+u_1v_1\\u_0v_1+u_1v_0\end{bmatrix}\right) X_{2\alpha+\beta}\left(\begin{bmatrix}\frac{c_0dv_0\left(-du_0^2+u_1^2\right)}{2}\\ \frac{c_0v_1\left(-du_0^2+u_1^2\right)}{2}\end{bmatrix}\right) X_{2\alpha+2\beta}\left(0\right)$$

$$\begin{bmatrix}1 & 0 & 0 & 0 & 0\\ 0 & 1 & 0 & \frac{c_0\sqrt{d}\left(-du_0^2+u_1^2\right)}{2} & c_0\left(\sqrt{d}u_1-du_0\right)\\ 0 & 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 1 & 0\\ 0 & 0 & 0 & \sqrt{d}u_0+u_1 & 1\end{bmatrix}\begin{bmatrix}1 & 0 & 0 & 0 & 0\\ 0 & 1 & 0 & 0 & 0\\ 0 & -\sqrt{d}v_0+v_1 & 1 & 0 & 0\\ \sqrt{d}v_0+v_1 & 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix}\begin{bmatrix}1 & 0 & 0 & 0 & 0\\ 0 & 1 & 0 & -\frac{c_0d^{\frac{3}{2}}u_0^2}{2}+\frac{c_0\sqrt{d}u_1^2}{2} & -c_0\sqrt{d}u_1+c_0du_0\\ 0 & 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 1 & 0\\ 0 & 0 & 0 & -\sqrt{d}u_0-u_1 & 1\end{bmatrix}\begin{bmatrix}1 & 0 & 0 & 0 & 0\\ 0 & 1 & 0 & 0 & 0\\ 0 & \sqrt{d}v_0-v_1 & 1 & 0 & 0\\ -\sqrt{d}v_0-v_1 & 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix} = \begin{bmatrix}1 & 0 & 0 & 0 & 0\\ -\frac{c_0\left(\sqrt{d}v_1+dv_0\right)\left(du_0^2-u_1^2\right)}{2} & 1 & 0 & 0 & 0\\ \frac{c_0\sqrt{d}\left(-d^2u_0^2v_0^2+du_0^2v_1^2+du_1^2v_0^2-u_1^2v_1^2\right)}{2} & 0 & 1 & -\frac{c_0\left(\sqrt{d}v_1-dv_0\right)\left(du_0^2-u_1^2\right)}{2} & -c_0\left(\sqrt{d}\left(du_0v_0+u_1v_1\right)-d\left(u_0v_1+u_1v_0\right)\right)\\ 0 & 0 & 0 & 1 & 0\\ \sqrt{d}\left(u_0v_1+u_1v_0\right)+du_0v_0+u_1v_1 & 0 & 0 & 0 & 1\end{bmatrix}$$

$$\alpha = (0,\ 1,\ 0,\ 0,\ 0)\,, \quad \beta = (-1,\ 0,\ 0,\ 0,\ 0)$$

$$\left[X_\alpha\left(\begin{bmatrix}u_0\\u_1\end{bmatrix}\right), X_\beta\left(\begin{bmatrix}v_0\\v_1\end{bmatrix}\right)\right] = X_{\alpha+\beta}\left(\begin{bmatrix}c_0d\left(-u_0v_0+u_1v_1\right)\\c_0\left(-du_0v_1+u_1v_0\right)\end{bmatrix}\right)$$

$$\begin{bmatrix}1 & 0 & 0 & 0 & 0\\ 0 & 1 & 0 & \frac{c_0\sqrt{d}\left(-du_0^2+u_1^2\right)}{2} & c_0\left(\sqrt{d}u_1-du_0\right)\\ 0 & 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 1 & 0\\ 0 & 0 & 0 & \sqrt{d}u_0+u_1 & 1\end{bmatrix}\begin{bmatrix}1 & 0 & 0 & 0 & 0\\ 0 & 1 & 0 & 0 & 0\\ \frac{c_0\sqrt{d}\left(dv_1^2-v_0^2\right)}{2} & 0 & 1 & 0 & c_0\left(-\sqrt{d}v_0+dv_1\right)\\ 0 & 0 & 0 & 1 & 0\\ \sqrt{d}v_1+v_0 & 0 & 0 & 0 & 1\end{bmatrix}\begin{bmatrix}1 & 0 & 0 & 0 & 0\\ 0 & 1 & 0 & -\frac{c_0d^{\frac{3}{2}}u_0^2}{2}+\frac{c_0\sqrt{d}u_1^2}{2} & -c_0\sqrt{d}u_1+c_0du_0\\ 0 & 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 1 & 0\\ 0 & 0 & 0 & -\sqrt{d}u_0-u_1 & 1\end{bmatrix}\begin{bmatrix}1 & 0 & 0 & 0 & 0\\ 0 & 1 & 0 & 0 & 0\\ \frac{c_0d^{\frac{3}{2}}v_1^2}{2}-\frac{c_0\sqrt{d}v_0^2}{2} & 0 & 1 & 0 & c_0\sqrt{d}v_0-c_0dv_1\\ 0 & 0 & 0 & 1 & 0\\ -\sqrt{d}v_1-v_0 & 0 & 0 & 0 & 1\end{bmatrix} = \begin{bmatrix}1 & 0 & 0 & 0 & 0\\ -c_0\left(\sqrt{d}\left(du_0v_1-u_1v_0\right)+d\left(u_0v_0-u_1v_1\right)\right) & 1 & 0 & 0 & 0\\ 0 & 0 & 1 & -c_0\left(\sqrt{d}\left(du_0v_1-u_1v_0\right)-d\left(u_0v_0-u_1v_1\right)\right) & 0\\ 0 & 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix}$$

$$\alpha = (0,\ 1,\ 0,\ 0,\ 0)\,, \quad \beta = (1,\ -1,\ 0,\ 0,\ 0)$$

$$\left[X_\alpha\left(\begin{bmatrix}u_0\\u_1\end{bmatrix}\right), X_\beta\left(\begin{bmatrix}v_0\\v_1\end{bmatrix}\right)\right] = X_{\alpha+\beta}\left(\begin{bmatrix}-u_0v_1+u_1v_0\\du_0v_0-u_1v_1\end{bmatrix}\right) X_{2\alpha+\beta}\left(\begin{bmatrix}\frac{c_0dv_0\left(-du_0^2+u_1^2\right)}{2}\\ \frac{c_0v_1\left(du_0^2-u_1^2\right)}{2}\end{bmatrix}\right) X_{2\alpha+2\beta}\left(0\right)$$

$$\begin{bmatrix}1 & 0 & 0 & 0 & 0\\ 0 & 1 & 0 & \frac{c_0\sqrt{d}\left(-du_0^2+u_1^2\right)}{2} & c_0\left(\sqrt{d}u_1-du_0\right)\\ 0 & 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 1 & 0\\ 0 & 0 & 0 & \sqrt{d}u_0+u_1 & 1\end{bmatrix}\begin{bmatrix}1 & \sqrt{d}v_0+v_1 & 0 & 0 & 0\\ 0 & 1 & 0 & 0 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & 0 & \sqrt{d}v_0-v_1 & 1 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix}\begin{bmatrix}1 & 0 & 0 & 0 & 0\\ 0 & 1 & 0 & -\frac{c_0d^{\frac{3}{2}}u_0^2}{2}+\frac{c_0\sqrt{d}u_1^2}{2} & -c_0\sqrt{d}u_1+c_0du_0\\ 0 & 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 1 & 0\\ 0 & 0 & 0 & -\sqrt{d}u_0-u_1 & 1\end{bmatrix}\begin{bmatrix}1 & -\sqrt{d}v_0-v_1 & 0 & 0 & 0\\ 0 & 1 & 0 & 0 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & 0 & -\sqrt{d}v_0+v_1 & 1 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix} = \begin{bmatrix}1 & 0 & \frac{c_0\sqrt{d}\left(d^2u_0^2v_0^2-du_0^2v_1^2-du_1^2v_0^2+u_1^2v_1^2\right)}{2} & -\frac{c_0\left(\sqrt{d}v_1+dv_0\right)\left(du_0^2-u_1^2\right)}{2} & c_0\left(\sqrt{d}\left(du_0v_0-u_1v_1\right)+d\left(u_0v_1-u_1v_0\right)\right)\\ 0 & 1 & \frac{c_0\left(\sqrt{d}v_1-dv_0\right)\left(du_0^2-u_1^2\right)}{2} & 0 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 1 & 0\\ 0 & 0 & \sqrt{d}\left(-u_0v_1+u_1v_0\right)+du_0v_0-u_1v_1 & 0 & 1\end{bmatrix}$$

$$\alpha = (0,\ 1,\ 0,\ 0,\ 0)\,, \quad \beta = (1,\ 0,\ 0,\ 0,\ 0)$$

$$\left[X_\alpha\left(\begin{bmatrix}u_0\\u_1\end{bmatrix}\right), X_\beta\left(\begin{bmatrix}v_0\\v_1\end{bmatrix}\right)\right] = X_{\alpha+\beta}\left(\begin{bmatrix}c_0d\left(-u_0v_1+u_1v_0\right)\\c_0\left(-du_0v_0+u_1v_1\right)\end{bmatrix}\right)$$

$$\begin{bmatrix}1 & 0 & 0 & 0 & 0\\ 0 & 1 & 0 & \frac{c_0\sqrt{d}\left(-du_0^2+u_1^2\right)}{2} & c_0\left(\sqrt{d}u_1-du_0\right)\\ 0 & 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 1 & 0\\ 0 & 0 & 0 & \sqrt{d}u_0+u_1 & 1\end{bmatrix}\begin{bmatrix}1 & 0 & \frac{c_0\sqrt{d}\left(-dv_0^2+v_1^2\right)}{2} & 0 & c_0\left(\sqrt{d}v_1-dv_0\right)\\ 0 & 1 & 0 & 0 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 1 & 0\\ 0 & 0 & \sqrt{d}v_0+v_1 & 0 & 1\end{bmatrix}\begin{bmatrix}1 & 0 & 0 & 0 & 0\\ 0 & 1 & 0 & -\frac{c_0d^{\frac{3}{2}}u_0^2}{2}+\frac{c_0\sqrt{d}u_1^2}{2} & -c_0\sqrt{d}u_1+c_0du_0\\ 0 & 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 1 & 0\\ 0 & 0 & 0 & -\sqrt{d}u_0-u_1 & 1\end{bmatrix}\begin{bmatrix}1 & 0 & -\frac{c_0d^{\frac{3}{2}}v_0^2}{2}+\frac{c_0\sqrt{d}v_1^2}{2} & 0 & -c_0\sqrt{d}v_1+c_0dv_0\\ 0 & 1 & 0 & 0 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 1 & 0\\ 0 & 0 & -\sqrt{d}v_0-v_1 & 0 & 1\end{bmatrix} = \begin{bmatrix}1 & 0 & 0 & c_0\left(\sqrt{d}\left(du_0v_0-u_1v_1\right)-d\left(u_0v_1-u_1v_0\right)\right) & 0\\ 0 & 1 & -c_0\left(\sqrt{d}\left(du_0v_0-u_1v_1\right)+d\left(u_0v_1-u_1v_0\right)\right) & 0 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix}$$

$$\alpha = (0,\ 2,\ 0,\ 0,\ 0)\,, \quad \beta = (-1,\ -1,\ 0,\ 0,\ 0)$$

$$\left[X_{\alpha}\left(u_0\right), X_{\beta}\left(\begin{bmatrix}v_0\\ v_1\end{bmatrix}\right)\right] = X_{\alpha+\beta}\left(\begin{bmatrix}-u_0v_1\\ -u_0v_0\end{bmatrix}\right) X_{\alpha+2\beta}\left(u_0\left(-dv_0^2+v_1^2\right)\right)$$

$$\begin{bmatrix}1&0&0&0&0\\0&1&0&-u_0&0\\0&0&1&0&0\\0&0&0&1&0\\0&0&0&0&1\end{bmatrix}\begin{bmatrix}1&0&0&0&0\\0&1&0&0&0\\0&-\sqrt{d}v_0+v_1&1&0&0\\\sqrt{d}v_0+v_1&0&0&1&0\\0&0&0&0&1\end{bmatrix}\begin{bmatrix}1&0&0&0&0\\0&1&0&u_0&0\\0&0&1&0&0\\0&0&0&1&0\\0&0&0&0&1\end{bmatrix}\begin{bmatrix}1&0&0&0&0\\0&1&0&0&0\\0&\sqrt{d}v_0-v_1&1&0&0\\-\sqrt{d}v_0-v_1&0&0&1&0\\0&0&0&0&1\end{bmatrix}=\begin{bmatrix}1&0&0&0&0\\-u_0\left(\sqrt{d}v_0+v_1\right)&1&0&0&0\\u_0\left(dv_0^2-v_1^2\right)&0&1&u_0\left(-\sqrt{d}v_0+v_1\right)&0\\0&0&0&1&0\\0&0&0&0&1\end{bmatrix}$$

$$\alpha=(0,\ 2,\ 0,\ 0,\ 0),\quad \beta=(1,\ -1,\ 0,\ 0,\ 0)$$

$$\left[X_{\alpha}\left(u_0\right), X_{\beta}\left(\begin{bmatrix}v_0\\ v_1\end{bmatrix}\right)\right] = X_{\alpha+\beta}\left(\begin{bmatrix}u_0v_1\\ -u_0v_0\end{bmatrix}\right) X_{\alpha+2\beta}\left(u_0\left(dv_0^2-v_1^2\right)\right)$$

$$\begin{bmatrix}1&0&0&0&0\\0&1&0&-u_0&0\\0&0&1&0&0\\0&0&0&1&0\\0&0&0&0&1\end{bmatrix}\begin{bmatrix}1&\sqrt{d}v_0+v_1&0&0&0\\0&1&0&0&0\\0&0&1&0&0\\0&0&\sqrt{d}v_0-v_1&1&0\\0&0&0&0&1\end{bmatrix}\begin{bmatrix}1&0&0&0&0\\0&1&0&u_0&0\\0&0&1&0&0\\0&0&0&1&0\\0&0&0&0&1\end{bmatrix}\begin{bmatrix}1&-\sqrt{d}v_0-v_1&0&0&0\\0&1&0&0&0\\0&0&1&0&0\\0&0&-\sqrt{d}v_0+v_1&1&0\\0&0&0&0&1\end{bmatrix}=\begin{bmatrix}1&0&u_0\left(-dv_0^2+v_1^2\right)&u_0\left(\sqrt{d}v_0+v_1\right)&0\\0&1&u_0\left(-\sqrt{d}v_0+v_1\right)&0&0\\0&0&1&0&0\\0&0&0&1&0\\0&0&0&0&1\end{bmatrix}$$

$$\alpha=(1,\ -1,\ 0,\ 0,\ 0),\quad \beta=(-2,\ 0,\ 0,\ 0,\ 0)$$

$$\left[X_{\alpha}\left(\begin{bmatrix}u_0\\ u_1\end{bmatrix}\right), X_{\beta}\left(v_0\right)\right] = X_{\alpha+\beta}\left(\begin{bmatrix}-u_0v_0\\ u_1v_0\end{bmatrix}\right) X_{2\alpha+\beta}\left(v_0\left(-du_0^2+u_1^2\right)\right)$$

$$\begin{bmatrix}1&\sqrt{d}u_0+u_1&0&0&0\\0&1&0&0&0\\0&0&1&0&0\\0&0&\sqrt{d}u_0-u_1&1&0\\0&0&0&0&1\end{bmatrix}\begin{bmatrix}1&0&0&0&0\\0&1&0&0&0\\-v_0&0&1&0&0\\0&0&0&1&0\\0&0&0&0&1\end{bmatrix}\begin{bmatrix}1&-\sqrt{d}u_0-u_1&0&0&0\\0&1&0&0&0\\0&0&1&0&0\\0&0&-\sqrt{d}u_0+u_1&1&0\\0&0&0&0&1\end{bmatrix}\begin{bmatrix}1&0&0&0&0\\0&1&0&0&0\\v_0&0&1&0&0\\0&0&0&1&0\\0&0&0&0&1\end{bmatrix}=\begin{bmatrix}1&0&0&0&0\\0&1&0&0&0\\0&v_0\left(\sqrt{d}u_0+u_1\right)&1&0&0\\v_0\left(-\sqrt{d}u_0+u_1\right)&v_0\left(du_0^2-u_1^2\right)&0&1&0\\0&0&0&0&1\end{bmatrix}$$

$$\alpha=(1,\ -1,\ 0,\ 0,\ 0),\quad \beta=(-1,\ -1,\ 0,\ 0,\ 0)$$

$$\left[X_{\alpha}\left(\begin{bmatrix}u_0\\ u_1\end{bmatrix}\right), X_{\beta}\left(\begin{bmatrix}v_0\\ v_1\end{bmatrix}\right)\right] = X_{\alpha+\beta}\left(2du_0v_0+2u_1v_1\right)$$

$$\begin{bmatrix}1&\sqrt{d}u_0+u_1&0&0&0\\0&1&0&0&0\\0&0&1&0&0\\0&0&\sqrt{d}u_0-u_1&1&0\\0&0&0&0&1\end{bmatrix}\begin{bmatrix}1&0&0&0&0\\0&1&0&0&0\\0&-\sqrt{d}v_0+v_1&1&0&0\\\sqrt{d}v_0+v_1&0&0&1&0\\0&0&0&0&1\end{bmatrix}\begin{bmatrix}1&-\sqrt{d}u_0-u_1&0&0&0\\0&1&0&0&0\\0&0&1&0&0\\0&0&-\sqrt{d}u_0+u_1&1&0\\0&0&0&0&1\end{bmatrix}\begin{bmatrix}1&0&0&0&0\\0&1&0&0&0\\0&\sqrt{d}v_0-v_1&1&0&0\\-\sqrt{d}v_0-v_1&0&0&1&0\\0&0&0&0&1\end{bmatrix}=\begin{bmatrix}1&0&0&0&0\\0&1&0&0&0\\0&0&1&0&0\\0&-2du_0v_0-2u_1v_1&0&1&0\\0&0&0&0&1\end{bmatrix}$$

$$\alpha=(1,\ -1,\ 0,\ 0,\ 0),\quad \beta=(-1,\ 0,\ 0,\ 0,\ 0)$$

$$\left[X_{\alpha}\left(\begin{bmatrix}u_0\\ u_1\end{bmatrix}\right), X_{\beta}\left(\begin{bmatrix}v_0\\ v_1\end{bmatrix}\right)\right] = X_{\alpha+\beta}\left(\begin{bmatrix}-u_0v_0-u_1v_1\\ -du_0v_1-u_1v_0\end{bmatrix}\right) X_{\alpha+2\beta}\left(\begin{bmatrix}\frac{c_0u_1\left(dv_1^2-v_0^2\right)}{2}\\ \frac{c_0du_0\left(-dv_1^2+v_0^2\right)}{2}\end{bmatrix}\right) X_{2\alpha+2\beta}\left(0\right)$$

$$\begin{bmatrix}1&\sqrt{d}u_0+u_1&0&0&0\\0&1&0&0&0\\0&0&1&0&0\\0&0&\sqrt{d}u_0-u_1&1&0\\0&0&0&0&1\end{bmatrix}\begin{bmatrix}1&0&0&0&0\\0&1&0&0&0\\\frac{c_0\sqrt{d}\left(dv_1^2-v_0^2\right)}{2}&0&1&0&c_0\left(-\sqrt{d}v_0+dv_1\right)\\0&0&0&1&0\\\sqrt{d}v_1+v_0&0&0&0&1\end{bmatrix}\begin{bmatrix}1&-\sqrt{d}u_0-u_1&0&0&0\\0&1&0&0&0\\0&0&1&0&0\\0&0&-\sqrt{d}u_0+u_1&1&0\\0&0&0&0&1\end{bmatrix}\begin{bmatrix}1&0&0&0&0\\0&1&0&0&0\\\frac{c_0d^{\frac{3}{2}}v_1^2}{2}-\frac{c_0\sqrt{d}v_0^2}{2}&0&1&0&c_0\sqrt{d}v_0-c_0dv_1\\0&0&0&1&0\\-\sqrt{d}v_1-v_0&0&0&0&1\end{bmatrix}=\begin{bmatrix}1&0&0&0&0\\0&1&0&0&0\\0&-\frac{c_0\left(\sqrt{d}u_1+du_0\right)\left(dv_1^2-v_0^2\right)}{2}&1&0&0\\\frac{c_0\left(\sqrt{d}u_1-du_0\right)\left(dv_1^2-v_0^2\right)}{2}&\frac{c_0\sqrt{d}\left(-d^2u_0^2v_1^2+du_0^2v_0^2+du_1^2v_1^2-u_1^2v_0^2\right)}{2}&0&1&c_0\left(\sqrt{d}\left(du_0v_1+u_1v_0\right)-d\left(u_0v_0+u_1v_1\right)\right)\\0&-\sqrt{d}\left(u_0v_0+u_1v_1\right)-du_0v_1-u_1v_0&0&0&1\end{bmatrix}$$

$$\alpha=(1,\ -1,\ 0,\ 0,\ 0),\quad \beta=(0,\ 1,\ 0,\ 0,\ 0)$$

$$\left[X_{\alpha}\left(\begin{bmatrix}u_0\\ u_1\end{bmatrix}\right), X_{\beta}\left(\begin{bmatrix}v_0\\ v_1\end{bmatrix}\right)\right] = X_{\alpha+\beta}\left(\begin{bmatrix}-u_0v_1+u_1v_0\\ -du_0v_0+u_1v_1\end{bmatrix}\right) X_{\alpha+2\beta}\left(\begin{bmatrix}\frac{c_0du_0\left(dv_0^2-v_1^2\right)}{2}\\ \frac{c_0u_1\left(-dv_0^2+v_1^2\right)}{2}\end{bmatrix}\right) X_{2\alpha+2\beta}\left(0\right)$$

$$\begin{bmatrix}1&\sqrt{d}u_0+u_1&0&0&0\\0&1&0&0&0\\0&0&1&0&0\\0&0&\sqrt{d}u_0-u_1&1&0\\0&0&0&0&1\end{bmatrix}\begin{bmatrix}1&0&0&0&0\\0&1&0&\frac{c_0\sqrt{d}\left(-dv_0^2+v_1^2\right)}{2}&c_0\left(\sqrt{d}v_1-dv_0\right)\\0&0&1&0&0\\0&0&0&1&0\\0&0&0&\sqrt{d}v_0+v_1&1\end{bmatrix}\begin{bmatrix}1&-\sqrt{d}u_0-u_1&0&0&0\\0&1&0&0&0\\0&0&1&0&0\\0&0&-\sqrt{d}u_0+u_1&1&0\\0&0&0&0&1\end{bmatrix}\begin{bmatrix}1&0&0&0&0\\0&1&0&-\frac{c_0d^{\frac{3}{2}}v_0^2}{2}+\frac{c_0\sqrt{d}v_1^2}{2}&-c_0\sqrt{d}v_1+c_0dv_0\\0&0&1&0&0\\0&0&0&1&0\\0&0&0&-\sqrt{d}v_0-v_1&1\end{bmatrix}=\begin{bmatrix}1&0&\frac{c_0\sqrt{d}\left(d^2u_0^2v_0^2-du_0^2v_1^2-du_1^2v_0^2+u_1^2v_1^2\right)}{2}&\frac{c_0\left(\sqrt{d}u_1+du_0\right)\left(dv_0^2-v_1^2\right)}{2}&-c_0\left(\sqrt{d}\left(du_0v_0-u_1v_1\right)-d\left(u_0v_1-u_1v_0\right)\right)\\0&1&-\frac{c_0\left(\sqrt{d}u_1-du_0\right)\left(dv_0^2-v_1^2\right)}{2}&0&0\\0&0&1&0&0\\0&0&0&1&0\\0&0&\sqrt{d}\left(-u_0v_1+u_1v_0\right)-du_0v_0+u_1v_1&0&1\end{bmatrix}$$

$$\alpha=(1,\ -1,\ 0,\ 0,\ 0),\quad \beta=(0,\ 2,\ 0,\ 0,\ 0)$$

$$\left[X_{\alpha}\left(\begin{bmatrix}u_0\\ u_1\end{bmatrix}\right), X_{\beta}\left(v_0\right)\right] = X_{\alpha+\beta}\left(\begin{bmatrix}-u_1v_0\\ u_0v_0\end{bmatrix}\right) X_{2\alpha+\beta}\left(v_0\left(-du_0^2+u_1^2\right)\right)$$

$$\begin{bmatrix}1 & \sqrt{d}u_0+u_1 & 0 & 0 & 0\\ 0 & 1 & 0 & 0 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & 0 & \sqrt{d}u_0-u_1 & 1 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix}\begin{bmatrix}1 & 0 & 0 & 0 & 0\\ 0 & 1 & 0 & -v_0 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix}\begin{bmatrix}1 & -\sqrt{d}u_0-u_1 & 0 & 0 & 0\\ 0 & 1 & 0 & 0 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & 0 & -\sqrt{d}u_0+u_1 & 1 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix}\begin{bmatrix}1 & 0 & 0 & 0 & 0\\ 0 & 1 & 0 & v_0 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix}=\begin{bmatrix}1 & 0 & v_0\left(du_0^2-u_1^2\right) & -v_0\left(\sqrt{d}u_0+u_1\right) & 0\\ 0 & 1 & v_0\left(\sqrt{d}u_0-u_1\right) & 0 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix}$$

$$\alpha=(1,\ -1,\ 0,\ 0,\ 0)\,,\quad \beta=(1,\ 1,\ 0,\ 0,\ 0)$$

$$\left[X_\alpha\left(\begin{bmatrix}u_0\\ u_1\end{bmatrix}\right),X_\beta\left(\begin{bmatrix}v_0\\ v_1\end{bmatrix}\right)\right]=X_{\alpha+\beta}\left(-2du_0v_1-2u_1v_0\right)$$

$$\begin{bmatrix}1 & \sqrt{d}u_0+u_1 & 0 & 0 & 0\\ 0 & 1 & 0 & 0 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & 0 & \sqrt{d}u_0-u_1 & 1 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix}\begin{bmatrix}1 & 0 & 0 & -\sqrt{d}v_1+v_0 & 0\\ 0 & 1 & \sqrt{d}v_1+v_0 & 0 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix}\begin{bmatrix}1 & -\sqrt{d}u_0-u_1 & 0 & 0 & 0\\ 0 & 1 & 0 & 0 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & 0 & -\sqrt{d}u_0+u_1 & 1 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix}\begin{bmatrix}1 & 0 & 0 & \sqrt{d}v_1-v_0 & 0\\ 0 & 1 & -\sqrt{d}v_1-v_0 & 0 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix}=\begin{bmatrix}1 & 0 & 2du_0v_1+2u_1v_0 & 0 & 0\\ 0 & 1 & 0 & 0 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix}$$

$$\alpha=(1,\ 0,\ 0,\ 0,\ 0)\,,\quad \beta=(-1,\ -1,\ 0,\ 0,\ 0)$$

$$\left[X_\alpha\left(\begin{bmatrix}u_0\\ u_1\end{bmatrix}\right),X_\beta\left(\begin{bmatrix}v_0\\ v_1\end{bmatrix}\right)\right]=X_{\alpha+\beta}\left(\begin{bmatrix}u_0v_1-u_1v_0\\ -du_0v_0+u_1v_1\end{bmatrix}\right)X_{2\alpha+\beta}\left(\begin{bmatrix}\frac{c_0v_1\left(-du_0^2+u_1^2\right)}{2}\\ \frac{c_0dv_0\left(du_0^2-u_1^2\right)}{2}\end{bmatrix}\right)X_{2\alpha+2\beta}\left(0\right)$$

$$\begin{bmatrix}1 & 0 & \frac{c_0\sqrt{d}\left(-du_0^2+u_1^2\right)}{2} & 0 & c_0\left(\sqrt{d}u_1-du_0\right)\\ 0 & 1 & 0 & 0 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 1 & 0\\ 0 & 0 & \sqrt{d}u_0+u_1 & 0 & 1\end{bmatrix}\begin{bmatrix}1 & 0 & 0 & 0 & 0\\ 0 & 1 & 0 & 0 & 0\\ 0 & -\sqrt{d}v_0+v_1 & 1 & 0 & 0\\ \sqrt{d}v_0+v_1 & 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix}\begin{bmatrix}1 & 0 & -\frac{c_0d^{\frac{3}{2}}u_0^2}{2}+\frac{c_0\sqrt{d}u_1^2}{2} & 0 & -c_0\sqrt{d}u_1+c_0du_0\\ 0 & 1 & 0 & 0 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 1 & 0\\ 0 & 0 & -\sqrt{d}u_0-u_1 & 0 & 1\end{bmatrix}\begin{bmatrix}1 & 0 & 0 & 0 & 0\\ 0 & 1 & 0 & 0 & 0\\ 0 & \sqrt{d}v_0-v_1 & 1 & 0 & 0\\ -\sqrt{d}v_0-v_1 & 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix}=\begin{bmatrix}1 & -\frac{c_0\left(\sqrt{d}v_1-dv_0\right)\left(du_0^2-u_1^2\right)}{2} & 0 & 0 & 0\\ 0 & 1 & 0 & 0 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & \frac{c_0\sqrt{d}\left(-d^2u_0^2v_0^2+du_0^2v_1^2+du_1^2v_0^2-u_1^2v_1^2\right)}{2} & -\frac{c_0\left(\sqrt{d}v_1+dv_0\right)\left(du_0^2-u_1^2\right)}{2} & 1 & c_0\left(\sqrt{d}\left(du_0v_0-u_1v_1\right)+d\left(u_0v_1-u_1v_0\right)\right)\\ 0 & \sqrt{d}\left(u_0v_1-u_1v_0\right)-du_0v_0+u_1v_1 & 0 & 0 & 1\end{bmatrix}$$

$$\alpha=(1,\ 0,\ 0,\ 0,\ 0)\,,\quad \beta=(-1,\ 1,\ 0,\ 0,\ 0)$$

$$\left[X_\alpha\left(\begin{bmatrix}u_0\\ u_1\end{bmatrix}\right),X_\beta\left(\begin{bmatrix}v_0\\ v_1\end{bmatrix}\right)\right]=X_{\alpha+\beta}\left(\begin{bmatrix}-u_0v_0+u_1v_1\\ du_0v_1-u_1v_0\end{bmatrix}\right)X_{2\alpha+\beta}\left(\begin{bmatrix}\frac{c_0dv_1\left(-du_0^2+u_1^2\right)}{2}\\ \frac{c_0v_0\left(-du_0^2+u_1^2\right)}{2}\end{bmatrix}\right)X_{2\alpha+2\beta}\left(0\right)$$

$$\begin{bmatrix}1 & 0 & \frac{c_0\sqrt{d}\left(-du_0^2+u_1^2\right)}{2} & 0 & c_0\left(\sqrt{d}u_1-du_0\right)\\ 0 & 1 & 0 & 0 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 1 & 0\\ 0 & 0 & \sqrt{d}u_0+u_1 & 0 & 1\end{bmatrix}\begin{bmatrix}1 & 0 & 0 & 0 & 0\\ \sqrt{d}v_1+v_0 & 1 & 0 & 0 & 0\\ 0 & 0 & 1 & \sqrt{d}v_1-v_0 & 0\\ 0 & 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix}\begin{bmatrix}1 & 0 & -\frac{c_0d^{\frac{3}{2}}u_0^2}{2}+\frac{c_0\sqrt{d}u_1^2}{2} & 0 & -c_0\sqrt{d}u_1+c_0du_0\\ 0 & 1 & 0 & 0 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 1 & 0\\ 0 & 0 & -\sqrt{d}u_0-u_1 & 0 & 1\end{bmatrix}\begin{bmatrix}1 & 0 & 0 & 0 & 0\\ -\sqrt{d}v_1-v_0 & 1 & 0 & 0 & 0\\ 0 & 0 & 1 & -\sqrt{d}v_1+v_0 & 0\\ 0 & 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix}=\begin{bmatrix}1 & 0 & 0 & \frac{c_0\left(\sqrt{d}v_0-dv_1\right)\left(du_0^2-u_1^2\right)}{2} & 0\\ 0 & 1 & -\frac{c_0\left(\sqrt{d}v_0+dv_1\right)\left(du_0^2-u_1^2\right)}{2} & \frac{c_0\sqrt{d}\left(d^2u_0^2v_1^2-du_0^2v_0^2-du_1^2v_1^2+u_1^2v_0^2\right)}{2} & c_0\left(\sqrt{d}\left(du_0v_1-u_1v_0\right)+d\left(u_0v_0-u_1v_1\right)\right)\\ 0 & 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 1 & 0\\ 0 & 0 & 0 & \sqrt{d}\left(-u_0v_0+u_1v_1\right)+du_0v_1-u_1v_0 & 1\end{bmatrix}$$

$$\alpha=(1,\ 0,\ 0,\ 0,\ 0)\,,\quad \beta=(0,\ -1,\ 0,\ 0,\ 0)$$

$$\left[X_\alpha\left(\begin{bmatrix}u_0\\ u_1\end{bmatrix}\right),X_\beta\left(\begin{bmatrix}v_0\\ v_1\end{bmatrix}\right)\right]=X_{\alpha+\beta}\left(\begin{bmatrix}c_0\left(-du_0v_0+u_1v_1\right)\\ c_0d\left(-u_0v_1+u_1v_0\right)\end{bmatrix}\right)$$

$$\begin{bmatrix}1 & 0 & \frac{c_0\sqrt{d}\left(-du_0^2+u_1^2\right)}{2} & 0 & c_0\left(\sqrt{d}u_1-du_0\right)\\ 0 & 1 & 0 & 0 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 1 & 0\\ 0 & 0 & \sqrt{d}u_0+u_1 & 0 & 1\end{bmatrix}\begin{bmatrix}1 & 0 & 0 & 0 & 0\\ 0 & 1 & 0 & 0 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & \frac{c_0\sqrt{d}\left(dv_0^2-v_1^2\right)}{2} & 0 & 1 & c_0\left(-\sqrt{d}v_1+dv_0\right)\\ 0 & \sqrt{d}v_0+v_1 & 0 & 0 & 1\end{bmatrix}\begin{bmatrix}1 & 0 & -\frac{c_0d^{\frac{3}{2}}u_0^2}{2}+\frac{c_0\sqrt{d}u_1^2}{2} & 0 & -c_0\sqrt{d}u_1+c_0du_0\\ 0 & 1 & 0 & 0 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 1 & 0\\ 0 & 0 & -\sqrt{d}u_0-u_1 & 0 & 1\end{bmatrix}\begin{bmatrix}1 & 0 & 0 & 0 & 0\\ 0 & 1 & 0 & 0 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & \frac{c_0d^{\frac{3}{2}}v_0^2}{2}-\frac{c_0\sqrt{d}v_1^2}{2} & 0 & 1 & c_0\sqrt{d}v_1-c_0dv_0\\ 0 & -\sqrt{d}v_0-v_1 & 0 & 0 & 1\end{bmatrix}=\begin{bmatrix}1 & -c_0\left(\sqrt{d}\left(du_0v_0-u_1v_1\right)+d\left(u_0v_1-u_1v_0\right)\right) & 0 & 0 & 0\\ 0 & 1 & 0 & 0 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & 0 & -c_0\left(\sqrt{d}\left(du_0v_0-u_1v_1\right)-d\left(u_0v_1-u_1v_0\right)\right) & 1 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix}$$

$$\alpha=(1,\ 0,\ 0,\ 0,\ 0)\,,\quad \beta=(0,\ 1,\ 0,\ 0,\ 0)$$

$$\left[X_\alpha\left(\begin{bmatrix}u_0\\ u_1\end{bmatrix}\right),X_\beta\left(\begin{bmatrix}v_0\\ v_1\end{bmatrix}\right)\right]=X_{\alpha+\beta}\left(\begin{bmatrix}c_0d\left(-u_0v_1+u_1v_0\right)\\ c_0\left(du_0v_0-u_1v_1\right)\end{bmatrix}\right)$$

$$\begin{bmatrix}1 & 0 & \frac{c_0\sqrt{d}\left(-du_0^2+u_1^2\right)}{2} & 0 & c_0\left(\sqrt{d}u_1-du_0\right)\\ 0 & 1 & 0 & 0 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 1 & 0\\ 0 & 0 & \sqrt{d}u_0+u_1 & 0 & 1\end{bmatrix}\begin{bmatrix}1 & 0 & 0 & 0 & 0\\ 0 & 1 & 0 & \frac{c_0\sqrt{d}\left(-dv_0^2+v_1^2\right)}{2} & c_0\left(\sqrt{d}v_1-dv_0\right)\\ 0 & 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 1 & 0\\ 0 & 0 & 0 & \sqrt{d}v_0+v_1 & 1\end{bmatrix}\begin{bmatrix}1 & 0 & -\frac{c_0d^{\frac{3}{2}}u_0^2}{2}+\frac{c_0\sqrt{d}u_1^2}{2} & 0 & -c_0\sqrt{d}u_1+c_0du_0\\ 0 & 1 & 0 & 0 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 1 & 0\\ 0 & 0 & -\sqrt{d}u_0-u_1 & 0 & 1\end{bmatrix}\begin{bmatrix}1 & 0 & 0 & 0 & 0\\ 0 & 1 & 0 & -\frac{c_0d^{\frac{3}{2}}v_0^2}{2}+\frac{c_0\sqrt{d}v_1^2}{2} & -c_0\sqrt{d}v_1+c_0dv_0\\ 0 & 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 1 & 0\\ 0 & 0 & 0 & -\sqrt{d}v_0-v_1 & 1\end{bmatrix}=\begin{bmatrix}1 & 0 & 0 & -c_0\left(\sqrt{d}\left(du_0v_0-u_1v_1\right)+d\left(u_0v_1-u_1v_0\right)\right) & 0\\ 0 & 1 & c_0\left(\sqrt{d}\left(du_0v_0-u_1v_1\right)-d\left(u_0v_1-u_1v_0\right)\right) & 0 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix}$$

$$\alpha=(1,\ 1,\ 0,\ 0,\ 0)\,,\quad \beta=(-2,\ 0,\ 0,\ 0,\ 0)$$

$$\left[X_\alpha\left(\begin{bmatrix}u_0\\ u_1\end{bmatrix}\right),X_\beta\left(v_0\right)\right]=X_{\alpha+\beta}\left(\begin{bmatrix}-u_0v_0\\ -u_1v_0\end{bmatrix}\right)X_{2\alpha+\beta}\left(v_0\left(du_1^2-u_0^2\right)\right)$$

$$\begin{bmatrix}1 & 0 & 0 & -\sqrt{d}u_1+u_0 & 0\\ 0 & 1 & \sqrt{d}u_1+u_0 & 0 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix}\begin{bmatrix}1 & 0 & 0 & 0 & 0\\ 0 & 1 & 0 & 0 & 0\\ -v_0 & 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix}\begin{bmatrix}1 & 0 & 0 & \sqrt{d}u_1-u_0 & 0\\ 0 & 1 & -\sqrt{d}u_1-u_0 & 0 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix}\begin{bmatrix}1 & 0 & 0 & 0 & 0\\ 0 & 1 & 0 & 0 & 0\\ v_0 & 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix}=\begin{bmatrix}1 & 0 & 0 & 0 & 0\\ -v_0\left(\sqrt{d}u_1+u_0\right) & 1 & 0 & v_0\left(-du_1^2+u_0^2\right) & 0\\ 0 & 0 & 1 & v_0\left(-\sqrt{d}u_1+u_0\right) & 0\\ 0 & 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix}$$

$$\alpha=(1,\ 1,\ 0,\ 0,\ 0)\,,\quad \beta=(-1,\ 0,\ 0,\ 0,\ 0)$$

$$\left[X_\alpha\left(\begin{bmatrix}u_0\\ u_1\end{bmatrix}\right),X_\beta\left(\begin{bmatrix}v_0\\ v_1\end{bmatrix}\right)\right]=X_{\alpha+\beta}\left(\begin{bmatrix}-u_0v_1+u_1v_0\\ du_1v_1-u_0v_0\end{bmatrix}\right)X_{\alpha+2\beta}\left(\begin{bmatrix}\frac{c_0du_1\left(-dv_1^2+v_0^2\right)}{2}\\ \frac{c_0u_0\left(-dv_1^2+v_0^2\right)}{2}\end{bmatrix}\right)X_{2\alpha+2\beta}\left(0\right)$$

$$\begin{bmatrix}1 & 0 & 0 & -\sqrt{d}u_1+u_0 & 0\\ 0 & 1 & \sqrt{d}u_1+u_0 & 0 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix}\begin{bmatrix}1 & 0 & 0 & 0 & 0\\ 0 & 1 & 0 & 0 & 0\\ \frac{c_0\sqrt{d}\left(dv_1^2-v_0^2\right)}{2} & 0 & 1 & 0 & c_0\left(-\sqrt{d}v_0+dv_1\right)\\ 0 & 0 & 0 & 1 & 0\\ \sqrt{d}v_1+v_0 & 0 & 0 & 0 & 1\end{bmatrix}\begin{bmatrix}1 & 0 & 0 & \sqrt{d}u_1-u_0 & 0\\ 0 & 1 & -\sqrt{d}u_1-u_0 & 0 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix}\begin{bmatrix}1 & 0 & 0 & 0 & 0\\ 0 & 1 & 0 & 0 & 0\\ \frac{c_0d^{\frac{3}{2}}v_1^2}{2}-\frac{c_0\sqrt{d}v_0^2}{2} & 0 & 1 & 0 & c_0\sqrt{d}v_0-c_0dv_1\\ 0 & 0 & 0 & 1 & 0\\ -\sqrt{d}v_1-v_0 & 0 & 0 & 0 & 1\end{bmatrix}=\begin{bmatrix}1 & 0 & 0 & 0 & 0\\ -\frac{c_0\left(\sqrt{d}u_0+du_1\right)\left(dv_1^2-v_0^2\right)}{2} & 1 & 0 & \frac{c_0\sqrt{d}\left(d^2u_1^2v_1^2-du_0^2v_1^2-du_1^2v_0^2+u_0^2v_0^2\right)}{2} & c_0\left(\sqrt{d}\left(du_1v_1-u_0v_0\right)+d\left(u_0v_1-u_1v_0\right)\right)\\ 0 & 0 & 1 & -\frac{c_0\left(\sqrt{d}u_0-du_1\right)\left(dv_1^2-v_0^2\right)}{2} & 0\\ 0 & 0 & 0 & 1 & 0\\ 0 & 0 & 0 & \sqrt{d}\left(-u_0v_1+u_1v_0\right)+du_1v_1-u_0v_0 & 1\end{bmatrix}$$

$$\alpha=(1,\ 1,\ 0,\ 0,\ 0)\,,\quad \beta=(-1,\ 1,\ 0,\ 0,\ 0)$$

$$\left[X_\alpha\left(\begin{bmatrix}u_0\\ u_1\end{bmatrix}\right),X_\beta\left(\begin{bmatrix}v_0\\ v_1\end{bmatrix}\right)\right]=X_{\alpha+\beta}\left(-2du_1v_1+2u_0v_0\right)$$

$$\begin{bmatrix}1 & 0 & 0 & -\sqrt{d}u_1+u_0 & 0\\ 0 & 1 & \sqrt{d}u_1+u_0 & 0 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix}\begin{bmatrix}1 & 0 & 0 & 0 & 0\\ \sqrt{d}v_1+v_0 & 1 & 0 & 0 & 0\\ 0 & 0 & 1 & \sqrt{d}v_1-v_0 & 0\\ 0 & 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix}\begin{bmatrix}1 & 0 & 0 & \sqrt{d}u_1-u_0 & 0\\ 0 & 1 & -\sqrt{d}u_1-u_0 & 0 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix}\begin{bmatrix}1 & 0 & 0 & 0 & 0\\ -\sqrt{d}v_1-v_0 & 1 & 0 & 0 & 0\\ 0 & 0 & 1 & -\sqrt{d}v_1+v_0 & 0\\ 0 & 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix}=\begin{bmatrix}1 & 0 & 0 & 0 & 0\\ 0 & 1 & 0 & 2du_1v_1-2u_0v_0 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix}$$

$$\alpha=(1,\ 1,\ 0,\ 0,\ 0)\,,\quad \beta=(0,\ -2,\ 0,\ 0,\ 0)$$

$$\left[X_\alpha\left(\begin{bmatrix}u_0\\ u_1\end{bmatrix}\right),X_\beta\left(v_0\right)\right]=X_{\alpha+\beta}\left(\begin{bmatrix}u_1v_0\\ -u_0v_0\end{bmatrix}\right)X_{2\alpha+\beta}\left(v_0\left(du_1^2-u_0^2\right)\right)$$

$$\begin{bmatrix}1 & 0 & 0 & -\sqrt{d}u_1+u_0 & 0\\ 0 & 1 & \sqrt{d}u_1+u_0 & 0 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix}\begin{bmatrix}1 & 0 & 0 & 0 & 0\\ 0 & 1 & 0 & 0 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & -v_0 & 0 & 1 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix}\begin{bmatrix}1 & 0 & 0 & \sqrt{d}u_1-u_0 & 0\\ 0 & 1 & -\sqrt{d}u_1-u_0 & 0 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix}\begin{bmatrix}1 & 0 & 0 & 0 & 0\\ 0 & 1 & 0 & 0 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & v_0 & 0 & 1 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix}=\begin{bmatrix}1 & v_0\left(\sqrt{d}u_1-u_0\right) & v_0\left(-du_1^2+u_0^2\right) & 0 & 0\\ 0 & 1 & 0 & 0 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & 0 & v_0\left(\sqrt{d}u_1+u_0\right) & 1 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix}$$

$$\alpha=(1,\ 1,\ 0,\ 0,\ 0)\,,\quad \beta=(0,\ -1,\ 0,\ 0,\ 0)$$

$$\left[X_\alpha\left(\begin{bmatrix}u_0\\ u_1\end{bmatrix}\right),X_\beta\left(\begin{bmatrix}v_0\\ v_1\end{bmatrix}\right)\right]=X_{\alpha+\beta}\left(\begin{bmatrix}-u_0v_0-u_1v_1\\ -du_1v_0-u_0v_1\end{bmatrix}\right)X_{\alpha+2\beta}\left(\begin{bmatrix}\frac{c_0u_0\left(-dv_0^2+v_1^2\right)}{2}\\ \frac{c_0du_1\left(dv_0^2-v_1^2\right)}{2}\end{bmatrix}\right)X_{2\alpha+2\beta}\left(0\right)$$

$$\begin{bmatrix}1 & 0 & 0 & -\sqrt{d}u_1+u_0 & 0\\ 0 & 1 & \sqrt{d}u_1+u_0 & 0 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix}\begin{bmatrix}1 & 0 & 0 & 0 & 0\\ 0 & 1 & 0 & 0 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & \frac{c_0\sqrt{d}\left(dv_0^2-v_1^2\right)}{2} & 0 & 1 & c_0\left(-\sqrt{d}v_1+dv_0\right)\\ 0 & \sqrt{d}v_0+v_1 & 0 & 0 & 1\end{bmatrix}\begin{bmatrix}1 & 0 & 0 & \sqrt{d}u_1-u_0 & 0\\ 0 & 1 & -\sqrt{d}u_1-u_0 & 0 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix}\begin{bmatrix}1 & 0 & 0 & 0 & 0\\ 0 & 1 & 0 & 0 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & \frac{c_0d^{\frac{3}{2}}v_0^2}{2}-\frac{c_0\sqrt{d}v_1^2}{2} & 0 & 1 & c_0\sqrt{d}v_1-c_0dv_0\\ 0 & -\sqrt{d}v_0-v_1 & 0 & 0 & 1\end{bmatrix}=\begin{bmatrix}1 & -\frac{c_0\left(\sqrt{d}u_0-du_1\right)\left(dv_0^2-v_1^2\right)}{2} & \frac{c_0\sqrt{d}\left(d^2u_1^2v_0^2-du_0^2v_0^2-du_1^2v_1^2+u_0^2v_1^2\right)}{2} & 0 & -c_0\left(\sqrt{d}\left(du_1v_0+u_0v_1\right)-d\left(u_0v_0+u_1v_1\right)\right)\\ 0 & 1 & 0 & 0 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & 0 & -\frac{c_0\left(\sqrt{d}u_0+du_1\right)\left(dv_0^2-v_1^2\right)}{2} & 1 & 0\\ 0 & 0 & -\sqrt{d}\left(u_0v_0+u_1v_1\right)-du_1v_0-u_0v_1 & 0 & 1\end{bmatrix}$$

$$\alpha=(1,\ 1,\ 0,\ 0,\ 0)\,,\quad \beta=(1,\ -1,\ 0,\ 0,\ 0)$$

$$\left[X_\alpha\left(\begin{bmatrix}u_0\\ u_1\end{bmatrix}\right),X_\beta\left(\begin{bmatrix}v_0\\ v_1\end{bmatrix}\right)\right]=X_{\alpha+\beta}\left(2du_1v_0+2u_0v_1\right)$$

$$\begin{bmatrix}1 & 0 & 0 & -\sqrt{d}u_1+u_0 & 0\\ 0 & 1 & \sqrt{d}u_1+u_0 & 0 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix}\begin{bmatrix}1 & \sqrt{d}v_0+v_1 & 0 & 0 & 0\\ 0 & 1 & 0 & 0 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & 0 & \sqrt{d}v_0-v_1 & 1 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix}\begin{bmatrix}1 & 0 & 0 & \sqrt{d}u_1-u_0 & 0\\ 0 & 1 & -\sqrt{d}u_1-u_0 & 0 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix}\begin{bmatrix}1 & -\sqrt{d}v_0-v_1 & 0 & 0 & 0\\ 0 & 1 & 0 & 0 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & 0 & -\sqrt{d}v_0+v_1 & 1 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix}=\begin{bmatrix}1 & 0 & -2du_1v_0-2u_0v_1 & 0 & 0\\ 0 & 1 & 0 & 0 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix}$$

$$\alpha=(2,\ 0,\ 0,\ 0,\ 0)\,,\quad \beta=(-1,\ -1,\ 0,\ 0,\ 0)$$

$$\left[X_\alpha\left(u_0\right),X_\beta\left(\begin{bmatrix}v_0\\ v_1\end{bmatrix}\right)\right]=X_{\alpha+\beta}\left(\begin{bmatrix}u_0v_0\\ -u_0v_1\end{bmatrix}\right)X_{\alpha+2\beta}\left(u_0\left(-dv_0^2+v_1^2\right)\right)$$

$$\begin{bmatrix}1 & 0 & -u_0 & 0 & 0\\ 0 & 1 & 0 & 0 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix}\begin{bmatrix}1 & 0 & 0 & 0 & 0\\ 0 & 1 & 0 & 0 & 0\\ 0 & -\sqrt{d}v_0+v_1 & 1 & 0 & 0\\ \sqrt{d}v_0+v_1 & 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix}\begin{bmatrix}1 & 0 & u_0 & 0 & 0\\ 0 & 1 & 0 & 0 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix}\begin{bmatrix}1 & 0 & 0 & 0 & 0\\ 0 & 1 & 0 & 0 & 0\\ 0 & \sqrt{d}v_0-v_1 & 1 & 0 & 0\\ -\sqrt{d}v_0-v_1 & 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix}=\begin{bmatrix}1 & u_0\left(\sqrt{d}v_0-v_1\right) & 0 & 0 & 0\\ 0 & 1 & 0 & 0 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & u_0\left(dv_0^2-v_1^2\right) & u_0\left(\sqrt{d}v_0+v_1\right) & 1 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix}$$

$$\alpha=(2,\ 0,\ 0,\ 0,\ 0)\,,\quad \beta=(-1,\ 1,\ 0,\ 0,\ 0)$$

$$\left[X_\alpha\left(u_0\right),X_\beta\left(\begin{bmatrix}v_0\\ v_1\end{bmatrix}\right)\right]=X_{\alpha+\beta}\left(\begin{bmatrix}u_0v_0\\ u_0v_1\end{bmatrix}\right)X_{\alpha+2\beta}\left(u_0\left(dv_1^2-v_0^2\right)\right)$$

$$\begin{bmatrix}1 & 0 & -u_0 & 0 & 0\\ 0 & 1 & 0 & 0 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix}\begin{bmatrix}1 & 0 & 0 & 0 & 0\\ \sqrt{d}v_1+v_0 & 1 & 0 & 0 & 0\\ 0 & 0 & 1 & \sqrt{d}v_1-v_0 & 0\\ 0 & 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix}\begin{bmatrix}1 & 0 & u_0 & 0 & 0\\ 0 & 1 & 0 & 0 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix}\begin{bmatrix}1 & 0 & 0 & 0 & 0\\ -\sqrt{d}v_1-v_0 & 1 & 0 & 0 & 0\\ 0 & 0 & 1 & -\sqrt{d}v_1+v_0 & 0\\ 0 & 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix}=\begin{bmatrix}1 & 0 & 0 & u_0\left(-\sqrt{d}v_1+v_0\right) & 0\\ 0 & 1 & u_0\left(\sqrt{d}v_1+v_0\right) & u_0\left(-dv_1^2+v_0^2\right) & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix}$$

6 Weyl group

One of the defining characteristics of a root system Φ is that for any two roots $\alpha, \beta \in \Phi$, the reflection of β across the hyperplane perpendicular to α is another root. The reflection across the hyperplane perpendicular to α is denoted σ_α, and the previous sentence can be rephrased as saying that σ_α is a (bijective) function from Φ to itself. The reflection of β is then denoted $\sigma_\alpha(\beta)$. For each root $\alpha \in \Phi$, there is an element $w_\alpha \in G$ with the following properties:

1. $w_\alpha \in N_G(T)$, i.e. w_α normalizes the torus, i.e. $w_\alpha \cdot t \cdot w_\alpha^{-1} \in T$ for all $t \in T$.
2. $w_\alpha^2 \in Z_G(T)$, i.e. w_α^2 centralizes the torus, i.e. $w_\alpha^2 \cdot t \cdot (w_\alpha^2)^{-1} = t$ for all $t \in T$. Under some assumptions, $T = Z_G(T)$, though this is not true for all of the groups considered in this package.
3. For any root $\beta \in \Phi$, the conjugation of the root subgroup U_β by w_α is the root subgroup $U_{\sigma_\alpha(\beta)}$. That is, for $x \in U_\beta$,
$$w_\alpha \cdot x \cdot w_\alpha^{-1} \in U_{\sigma_\alpha}(\beta)$$
More precisely, given an element $x_\beta(u) \in U_\beta$, there is some function $\varphi_{\alpha\beta}(u)$ so that
$$w_\alpha \cdot x_\beta(u) \cdot w_\alpha^{-1} = x_{\sigma_\alpha(\beta)}\Big(\varphi_{\alpha\beta}(u)\Big)$$

Taken together, properies (1) and (2) above say that viewed as an element of the quotient group $N_G(T)/Z_G(T)$, (the class of) w_α is an element of order 2 (or order 1 but that doesn't happen for other reasons).

6.1 Weyl group elements

The table below lists roots along with an associated Weyl group element.

α	w_α
$(-2,\ 0)$	$\begin{bmatrix} 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$
$(-1,\ -1)$	$\begin{bmatrix} 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$
$(-1,\ 0)$	$\begin{bmatrix} 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$
$(-1,\ 1)$	$\begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$
$(0,\ -2)$	$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$

α	w_α
$(0,\ -1)$	$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$
$(0,\ 1)$	$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$
$(0,\ 2)$	$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$
$(1,\ -1)$	$\begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$
$(1,\ 0)$	$\begin{bmatrix} 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$
$(1,\ 1)$	$\begin{bmatrix} 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$
$(2,\ 0)$	$\begin{bmatrix} 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$

6.2 Weyl group conjugation coefficients

The table below lists roots along with the coefficients obtain in performing conjugation by an associated Weyl element.

α	β	$\gamma = \sigma_\alpha(\beta)$	$\varphi_{\alpha\beta}(u)$
$(-2,\ 0)$	$(-2,\ 0)$	$(2,\ 0)$	$-u_0$
$(-2,\ 0)$	$(-1,\ -1)$	$(1,\ -1)$	$\begin{bmatrix} -u_0 & u_1 \end{bmatrix}$
$(-2,\ 0)$	$(-1,\ 0)$	$(1,\ 0)$	$\begin{bmatrix} -u_1 & -u_0 \end{bmatrix}$
$(-2,\ 0)$	$(-1,\ 1)$	$(1,\ 1)$	$\begin{bmatrix} -u_0 & -u_1 \end{bmatrix}$
$(-2,\ 0)$	$(0,\ -2)$	$(0,\ -2)$	u_0
$(-2,\ 0)$	$(0,\ -1)$	$(0,\ -1)$	$\begin{bmatrix} u_0 & u_1 \end{bmatrix}$
$(-2,\ 0)$	$(0,\ 1)$	$(0,\ 1)$	$\begin{bmatrix} u_0 & u_1 \end{bmatrix}$
$(-2,\ 0)$	$(0,\ 2)$	$(0,\ 2)$	u_0
$(-2,\ 0)$	$(1,\ -1)$	$(-1,\ -1)$	$\begin{bmatrix} u_0 & -u_1 \end{bmatrix}$

α	β	$\gamma = \sigma_\alpha(\beta)$	$\varphi_{\alpha\beta}(u)$
$(-2,\ 0)$	$(1,\ 0)$	$(-1,\ 0)$	$\begin{bmatrix} u_1 & u_0 \end{bmatrix}$
$(-2,\ 0)$	$(1,\ 1)$	$(-1,\ 1)$	$\begin{bmatrix} u_0 & u_1 \end{bmatrix}$
$(-2,\ 0)$	$(2,\ 0)$	$(-2,\ 0)$	$-u_0$
$(-1,\ -1)$	$(-2,\ 0)$	$(0,\ 2)$	$-u_0$
$(-1,\ -1)$	$(-1,\ -1)$	$(1,\ 1)$	$\begin{bmatrix} -u_1 & u_0 \end{bmatrix}$
$(-1,\ -1)$	$(-1,\ 0)$	$(0,\ 1)$	$\begin{bmatrix} -u_1 & -u_0 \end{bmatrix}$
$(-1,\ -1)$	$(-1,\ 1)$	$(-1,\ 1)$	$\begin{bmatrix} -u_0 & u_1 \end{bmatrix}$
$(-1,\ -1)$	$(0,\ -2)$	$(2,\ 0)$	$-u_0$
$(-1,\ -1)$	$(0,\ -1)$	$(1,\ 0)$	$\begin{bmatrix} -u_0 & -u_1 \end{bmatrix}$
$(-1,\ -1)$	$(0,\ 1)$	$(-1,\ 0)$	$\begin{bmatrix} u_1 & u_0 \end{bmatrix}$
$(-1,\ -1)$	$(0,\ 2)$	$(-2,\ 0)$	$-u_0$
$(-1,\ -1)$	$(1,\ -1)$	$(1,\ -1)$	$\begin{bmatrix} u_0 & -u_1 \end{bmatrix}$
$(-1,\ -1)$	$(1,\ 0)$	$(0,\ -1)$	$\begin{bmatrix} u_0 & u_1 \end{bmatrix}$
$(-1,\ -1)$	$(1,\ 1)$	$(-1,\ -1)$	$\begin{bmatrix} u_1 & -u_0 \end{bmatrix}$
$(-1,\ -1)$	$(2,\ 0)$	$(0,\ -2)$	$-u_0$
$(-1,\ 0)$	$(-2,\ 0)$	$(2,\ 0)$	$-u_0$
$(-1,\ 0)$	$(-1,\ -1)$	$(1,\ -1)$	$\begin{bmatrix} -u_0 & u_1 \end{bmatrix}$
$(-1,\ 0)$	$(-1,\ 0)$	$(1,\ 0)$	$\begin{bmatrix} -u_1 & -u_0 \end{bmatrix}$
$(-1,\ 0)$	$(-1,\ 1)$	$(1,\ 1)$	$\begin{bmatrix} -u_0 & -u_1 \end{bmatrix}$
$(-1,\ 0)$	$(0,\ -2)$	$(0,\ -2)$	u_0
$(-1,\ 0)$	$(0,\ -1)$	$(0,\ -1)$	$\begin{bmatrix} u_0 & u_1 \end{bmatrix}$
$(-1,\ 0)$	$(0,\ 1)$	$(0,\ 1)$	$\begin{bmatrix} u_0 & u_1 \end{bmatrix}$
$(-1,\ 0)$	$(0,\ 2)$	$(0,\ 2)$	u_0
$(-1,\ 0)$	$(1,\ -1)$	$(-1,\ -1)$	$\begin{bmatrix} u_0 & -u_1 \end{bmatrix}$
$(-1,\ 0)$	$(1,\ 0)$	$(-1,\ 0)$	$\begin{bmatrix} u_1 & u_0 \end{bmatrix}$
$(-1,\ 0)$	$(1,\ 1)$	$(-1,\ 1)$	$\begin{bmatrix} u_0 & u_1 \end{bmatrix}$
$(-1,\ 0)$	$(2,\ 0)$	$(-2,\ 0)$	$-u_0$
$(-1,\ 1)$	$(-2,\ 0)$	$(0,\ -2)$	u_0
$(-1,\ 1)$	$(-1,\ -1)$	$(-1,\ -1)$	$\begin{bmatrix} -u_0 & u_1 \end{bmatrix}$
$(-1,\ 1)$	$(-1,\ 0)$	$(0,\ -1)$	$\begin{bmatrix} u_1 & u_0 \end{bmatrix}$
$(-1,\ 1)$	$(-1,\ 1)$	$(1,\ -1)$	$\begin{bmatrix} u_1 & u_0 \end{bmatrix}$
$(-1,\ 1)$	$(0,\ -2)$	$(-2,\ 0)$	u_0
$(-1,\ 1)$	$(0,\ -1)$	$(-1,\ 0)$	$\begin{bmatrix} u_1 & u_0 \end{bmatrix}$
$(-1,\ 1)$	$(0,\ 1)$	$(1,\ 0)$	$\begin{bmatrix} u_0 & u_1 \end{bmatrix}$
$(-1,\ 1)$	$(0,\ 2)$	$(2,\ 0)$	u_0
$(-1,\ 1)$	$(1,\ -1)$	$(-1,\ 1)$	$\begin{bmatrix} u_1 & u_0 \end{bmatrix}$
$(-1,\ 1)$	$(1,\ 0)$	$(0,\ 1)$	$\begin{bmatrix} u_0 & u_1 \end{bmatrix}$
$(-1,\ 1)$	$(1,\ 1)$	$(1,\ 1)$	$\begin{bmatrix} u_0 & -u_1 \end{bmatrix}$
$(-1,\ 1)$	$(2,\ 0)$	$(0,\ 2)$	u_0
$(0,\ -2)$	$(-2,\ 0)$	$(-2,\ 0)$	u_0
$(0,\ -2)$	$(-1,\ -1)$	$(-1,\ 1)$	$\begin{bmatrix} u_1 & u_0 \end{bmatrix}$
$(0,\ -2)$	$(-1,\ 0)$	$(-1,\ 0)$	$\begin{bmatrix} u_0 & u_1 \end{bmatrix}$
$(0,\ -2)$	$(-1,\ 1)$	$(-1,\ -1)$	$\begin{bmatrix} -u_1 & -u_0 \end{bmatrix}$
$(0,\ -2)$	$(0,\ -2)$	$(0,\ 2)$	$-u_0$
$(0,\ -2)$	$(0,\ -1)$	$(0,\ 1)$	$\begin{bmatrix} -u_0 & -u_1 \end{bmatrix}$
$(0,\ -2)$	$(0,\ 1)$	$(0,\ -1)$	$\begin{bmatrix} u_0 & u_1 \end{bmatrix}$
$(0,\ -2)$	$(0,\ 2)$	$(0,\ -2)$	$-u_0$
$(0,\ -2)$	$(1,\ -1)$	$(1,\ 1)$	$\begin{bmatrix} -u_1 & u_0 \end{bmatrix}$
$(0,\ -2)$	$(1,\ 0)$	$(1,\ 0)$	$\begin{bmatrix} u_0 & u_1 \end{bmatrix}$
$(0,\ -2)$	$(1,\ 1)$	$(1,\ -1)$	$\begin{bmatrix} -u_1 & u_0 \end{bmatrix}$

α	β	$\gamma = \sigma_\alpha(\beta)$	$\varphi_{\alpha\beta}(u)$
$(0,\ -2)$	$(2,\ 0)$	$(2,\ 0)$	u_0
$(0,\ -1)$	$(-2,\ 0)$	$(-2,\ 0)$	u_0
$(0,\ -1)$	$(-1,\ -1)$	$(-1,\ 1)$	$\begin{bmatrix} u_1 & u_0 \end{bmatrix}$
$(0,\ -1)$	$(-1,\ 0)$	$(-1,\ 0)$	$\begin{bmatrix} u_0 & u_1 \end{bmatrix}$
$(0,\ -1)$	$(-1,\ 1)$	$(-1,\ -1)$	$\begin{bmatrix} -u_1 & -u_0 \end{bmatrix}$
$(0,\ -1)$	$(0,\ -2)$	$(0,\ 2)$	$-u_0$
$(0,\ -1)$	$(0,\ -1)$	$(0,\ 1)$	$\begin{bmatrix} -u_0 & -u_1 \end{bmatrix}$
$(0,\ -1)$	$(0,\ 1)$	$(0,\ -1)$	$\begin{bmatrix} u_0 & u_1 \end{bmatrix}$
$(0,\ -1)$	$(0,\ 2)$	$(0,\ -2)$	$-u_0$
$(0,\ -1)$	$(1,\ -1)$	$(1,\ 1)$	$\begin{bmatrix} -u_1 & u_0 \end{bmatrix}$
$(0,\ -1)$	$(1,\ 0)$	$(1,\ 0)$	$\begin{bmatrix} u_0 & u_1 \end{bmatrix}$
$(0,\ -1)$	$(1,\ 1)$	$(1,\ -1)$	$\begin{bmatrix} -u_1 & u_0 \end{bmatrix}$
$(0,\ -1)$	$(2,\ 0)$	$(2,\ 0)$	u_0
$(0,\ 1)$	$(-2,\ 0)$	$(-2,\ 0)$	u_0
$(0,\ 1)$	$(-1,\ -1)$	$(-1,\ 1)$	$\begin{bmatrix} u_1 & u_0 \end{bmatrix}$
$(0,\ 1)$	$(-1,\ 0)$	$(-1,\ 0)$	$\begin{bmatrix} u_0 & u_1 \end{bmatrix}$
$(0,\ 1)$	$(-1,\ 1)$	$(-1,\ -1)$	$\begin{bmatrix} -u_1 & -u_0 \end{bmatrix}$
$(0,\ 1)$	$(0,\ -2)$	$(0,\ 2)$	$-u_0$
$(0,\ 1)$	$(0,\ -1)$	$(0,\ 1)$	$\begin{bmatrix} -u_0 & -u_1 \end{bmatrix}$
$(0,\ 1)$	$(0,\ 1)$	$(0,\ -1)$	$\begin{bmatrix} u_0 & u_1 \end{bmatrix}$
$(0,\ 1)$	$(0,\ 2)$	$(0,\ -2)$	$-u_0$
$(0,\ 1)$	$(1,\ -1)$	$(1,\ 1)$	$\begin{bmatrix} -u_1 & u_0 \end{bmatrix}$
$(0,\ 1)$	$(1,\ 0)$	$(1,\ 0)$	$\begin{bmatrix} u_0 & u_1 \end{bmatrix}$
$(0,\ 1)$	$(1,\ 1)$	$(1,\ -1)$	$\begin{bmatrix} -u_1 & u_0 \end{bmatrix}$
$(0,\ 1)$	$(2,\ 0)$	$(2,\ 0)$	u_0
$(0,\ 2)$	$(-2,\ 0)$	$(-2,\ 0)$	u_0
$(0,\ 2)$	$(-1,\ -1)$	$(-1,\ 1)$	$\begin{bmatrix} u_1 & u_0 \end{bmatrix}$
$(0,\ 2)$	$(-1,\ 0)$	$(-1,\ 0)$	$\begin{bmatrix} u_0 & u_1 \end{bmatrix}$
$(0,\ 2)$	$(-1,\ 1)$	$(-1,\ -1)$	$\begin{bmatrix} -u_1 & -u_0 \end{bmatrix}$
$(0,\ 2)$	$(0,\ -2)$	$(0,\ 2)$	$-u_0$
$(0,\ 2)$	$(0,\ -1)$	$(0,\ 1)$	$\begin{bmatrix} -u_0 & -u_1 \end{bmatrix}$
$(0,\ 2)$	$(0,\ 1)$	$(0,\ -1)$	$\begin{bmatrix} u_0 & u_1 \end{bmatrix}$
$(0,\ 2)$	$(0,\ 2)$	$(0,\ -2)$	$-u_0$
$(0,\ 2)$	$(1,\ -1)$	$(1,\ 1)$	$\begin{bmatrix} -u_1 & u_0 \end{bmatrix}$
$(0,\ 2)$	$(1,\ 0)$	$(1,\ 0)$	$\begin{bmatrix} u_0 & u_1 \end{bmatrix}$
$(0,\ 2)$	$(1,\ 1)$	$(1,\ -1)$	$\begin{bmatrix} -u_1 & u_0 \end{bmatrix}$
$(0,\ 2)$	$(2,\ 0)$	$(2,\ 0)$	u_0
$(1,\ -1)$	$(-2,\ 0)$	$(0,\ -2)$	u_0
$(1,\ -1)$	$(-1,\ -1)$	$(-1,\ -1)$	$\begin{bmatrix} -u_0 & u_1 \end{bmatrix}$
$(1,\ -1)$	$(-1,\ 0)$	$(0,\ -1)$	$\begin{bmatrix} u_1 & u_0 \end{bmatrix}$
$(1,\ -1)$	$(-1,\ 1)$	$(1,\ -1)$	$\begin{bmatrix} u_1 & u_0 \end{bmatrix}$
$(1,\ -1)$	$(0,\ -2)$	$(-2,\ 0)$	u_0
$(1,\ -1)$	$(0,\ -1)$	$(-1,\ 0)$	$\begin{bmatrix} u_1 & u_0 \end{bmatrix}$
$(1,\ -1)$	$(0,\ 1)$	$(1,\ 0)$	$\begin{bmatrix} u_0 & u_1 \end{bmatrix}$
$(1,\ -1)$	$(0,\ 2)$	$(2,\ 0)$	u_0
$(1,\ -1)$	$(1,\ -1)$	$(-1,\ 1)$	$\begin{bmatrix} u_1 & u_0 \end{bmatrix}$
$(1,\ -1)$	$(1,\ 0)$	$(0,\ 1)$	$\begin{bmatrix} u_0 & u_1 \end{bmatrix}$
$(1,\ -1)$	$(1,\ 1)$	$(1,\ 1)$	$\begin{bmatrix} u_0 & -u_1 \end{bmatrix}$
$(1,\ -1)$	$(2,\ 0)$	$(0,\ 2)$	u_0
$(1,\ 0)$	$(-2,\ 0)$	$(2,\ 0)$	$-u_0$

α	β	$\gamma = \sigma_\alpha(\beta)$	$\varphi_{\alpha\beta}(u)$
$(1,\ 0)$	$(-1,\ -1)$	$(1,\ -1)$	$\begin{bmatrix}-u_0 & u_1\end{bmatrix}$
$(1,\ 0)$	$(-1,\ 0)$	$(1,\ 0)$	$\begin{bmatrix}-u_1 & -u_0\end{bmatrix}$
$(1,\ 0)$	$(-1,\ 1)$	$(1,\ 1)$	$\begin{bmatrix}-u_0 & -u_1\end{bmatrix}$
$(1,\ 0)$	$(0,\ -2)$	$(0,\ -2)$	u_0
$(1,\ 0)$	$(0,\ -1)$	$(0,\ -1)$	$\begin{bmatrix}u_0 & u_1\end{bmatrix}$
$(1,\ 0)$	$(0,\ 1)$	$(0,\ 1)$	$\begin{bmatrix}u_0 & u_1\end{bmatrix}$
$(1,\ 0)$	$(0,\ 2)$	$(0,\ 2)$	u_0
$(1,\ 0)$	$(1,\ -1)$	$(-1,\ -1)$	$\begin{bmatrix}u_0 & -u_1\end{bmatrix}$
$(1,\ 0)$	$(1,\ 0)$	$(-1,\ 0)$	$\begin{bmatrix}u_1 & u_0\end{bmatrix}$
$(1,\ 0)$	$(1,\ 1)$	$(-1,\ 1)$	$\begin{bmatrix}u_0 & u_1\end{bmatrix}$
$(1,\ 0)$	$(2,\ 0)$	$(-2,\ 0)$	$-u_0$
$(1,\ 1)$	$(-2,\ 0)$	$(0,\ 2)$	$-u_0$
$(1,\ 1)$	$(-1,\ -1)$	$(1,\ 1)$	$\begin{bmatrix}-u_1 & u_0\end{bmatrix}$
$(1,\ 1)$	$(-1,\ 0)$	$(0,\ 1)$	$\begin{bmatrix}-u_1 & -u_0\end{bmatrix}$
$(1,\ 1)$	$(-1,\ 1)$	$(-1,\ 1)$	$\begin{bmatrix}-u_0 & u_1\end{bmatrix}$
$(1,\ 1)$	$(0,\ -2)$	$(2,\ 0)$	$-u_0$
$(1,\ 1)$	$(0,\ -1)$	$(1,\ 0)$	$\begin{bmatrix}-u_0 & -u_1\end{bmatrix}$
$(1,\ 1)$	$(0,\ 1)$	$(-1,\ 0)$	$\begin{bmatrix}u_1 & u_0\end{bmatrix}$
$(1,\ 1)$	$(0,\ 2)$	$(-2,\ 0)$	$-u_0$
$(1,\ 1)$	$(1,\ -1)$	$(1,\ -1)$	$\begin{bmatrix}u_0 & -u_1\end{bmatrix}$
$(1,\ 1)$	$(1,\ 0)$	$(0,\ -1)$	$\begin{bmatrix}u_0 & u_1\end{bmatrix}$
$(1,\ 1)$	$(1,\ 1)$	$(-1,\ -1)$	$\begin{bmatrix}u_1 & -u_0\end{bmatrix}$
$(1,\ 1)$	$(2,\ 0)$	$(0,\ -2)$	$-u_0$
$(2,\ 0)$	$(-2,\ 0)$	$(2,\ 0)$	$-u_0$
$(2,\ 0)$	$(-1,\ -1)$	$(1,\ -1)$	$\begin{bmatrix}-u_0 & u_1\end{bmatrix}$
$(2,\ 0)$	$(-1,\ 0)$	$(1,\ 0)$	$\begin{bmatrix}-u_1 & -u_0\end{bmatrix}$
$(2,\ 0)$	$(-1,\ 1)$	$(1,\ 1)$	$\begin{bmatrix}-u_0 & -u_1\end{bmatrix}$
$(2,\ 0)$	$(0,\ -2)$	$(0,\ -2)$	u_0
$(2,\ 0)$	$(0,\ -1)$	$(0,\ -1)$	$\begin{bmatrix}u_0 & u_1\end{bmatrix}$
$(2,\ 0)$	$(0,\ 1)$	$(0,\ 1)$	$\begin{bmatrix}u_0 & u_1\end{bmatrix}$
$(2,\ 0)$	$(0,\ 2)$	$(0,\ 2)$	u_0
$(2,\ 0)$	$(1,\ -1)$	$(-1,\ -1)$	$\begin{bmatrix}u_0 & -u_1\end{bmatrix}$
$(2,\ 0)$	$(1,\ 0)$	$(-1,\ 0)$	$\begin{bmatrix}u_1 & u_0\end{bmatrix}$
$(2,\ 0)$	$(1,\ 1)$	$(-1,\ 1)$	$\begin{bmatrix}u_0 & u_1\end{bmatrix}$
$(2,\ 0)$	$(2,\ 0)$	$(-2,\ 0)$	$-u_0$

6.3 Weyl group conjugation identities

$$\alpha = (-2,\ 0,\ 0,\ 0,\ 0)\,, \beta = (-2,\ 0,\ 0,\ 0,\ 0)\,, \sigma_\alpha(\beta) = (2,\ 0,\ 0,\ 0,\ 0)$$

$$w_\alpha X_\beta\left(u_0\right) w_\alpha^{-1} = X_{\sigma_\alpha(\beta)}\left(-u_0\right)$$

$$\begin{bmatrix} 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ -u_0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 0 & -1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & u_0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\alpha = (-2,\ 0,\ 0,\ 0,\ 0)\,, \beta = (-1,\ -1,\ 0,\ 0,\ 0)\,, \sigma_\alpha(\beta) = (1,\ -1,\ 0,\ 0,\ 0)$$

$$w_\alpha X_\beta\left(\begin{bmatrix} u_0 \\ u_1 \end{bmatrix}\right) w_\alpha^{-1} = X_{\sigma_\alpha(\beta)}\left(\begin{bmatrix} -u_0 \\ u_1 \end{bmatrix}\right)$$

$$\begin{bmatrix} 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & -\sqrt{d}u_0 + u_1 & 1 & 0 & 0 \\ \sqrt{d}u_0 + u_1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 0 & -1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -\sqrt{d}u_0 + u_1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & -\sqrt{d}u_0 - u_1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\alpha = (-2,\ 0,\ 0,\ 0,\ 0)\,, \beta = (-1,\ 0,\ 0,\ 0,\ 0)\,, \sigma_\alpha(\beta) = (1,\ 0,\ 0,\ 0,\ 0)$$

$$w_\alpha X_\beta \left(\begin{bmatrix} u_0 \\ u_1 \end{bmatrix} \right) w_\alpha^{-1} = X_{\sigma_\alpha(\beta)} \left(\begin{bmatrix} -u_1 \\ -u_0 \end{bmatrix} \right)$$

$$\begin{bmatrix} 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ \frac{c_0\sqrt{d}\left(du_1^2 - u_0^2\right)}{2} & 0 & 1 & 0 & c_0\left(-\sqrt{d}u_0 + du_1\right) \\ 0 & 0 & 0 & 1 & 0 \\ \sqrt{d}u_1 + u_0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 0 & -1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & \frac{c_0\sqrt{d}\left(-du_1^2 + u_0^2\right)}{2} & 0 & c_0\left(-\sqrt{d}u_0 + du_1\right) \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & -\sqrt{d}u_1 - u_0 & 0 & 1 \end{bmatrix}$$

$$\alpha = (-2,\ 0,\ 0,\ 0,\ 0)\,, \beta = (-1,\ 1,\ 0,\ 0,\ 0)\,, \sigma_\alpha(\beta) = (1,\ 1,\ 0,\ 0,\ 0)$$

$$w_\alpha X_\beta \left(\begin{bmatrix} u_0 \\ u_1 \end{bmatrix} \right) w_\alpha^{-1} = X_{\sigma_\alpha(\beta)} \left(\begin{bmatrix} -u_0 \\ -u_1 \end{bmatrix} \right)$$

$$\begin{bmatrix} 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ \sqrt{d}u_1 + u_0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & \sqrt{d}u_1 - u_0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 0 & -1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & \sqrt{d}u_1 - u_0 & 0 \\ 0 & 1 & -\sqrt{d}u_1 - u_0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\alpha = (-2,\ 0,\ 0,\ 0,\ 0)\,, \beta = (0,\ -2,\ 0,\ 0,\ 0)\,, \sigma_\alpha(\beta) = (0,\ -2,\ 0,\ 0,\ 0)$$

$$w_\alpha X_\beta \left(u_0 \right) w_\alpha^{-1} = X_{\sigma_\alpha(\beta)} \left(u_0 \right)$$

$$\begin{bmatrix} 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & -u_0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 0 & -1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & -u_0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\alpha = (-2,\ 0,\ 0,\ 0,\ 0)\,, \beta = (0,\ -1,\ 0,\ 0,\ 0)\,, \sigma_\alpha(\beta) = (0,\ -1,\ 0,\ 0,\ 0)$$

$$w_\alpha X_\beta \left(\begin{bmatrix} u_0 \\ u_1 \end{bmatrix} \right) w_\alpha^{-1} = X_{\sigma_\alpha(\beta)} \left(\begin{bmatrix} u_0 \\ u_1 \end{bmatrix} \right)$$

$$\begin{bmatrix} 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & \frac{c_0\sqrt{d}\left(du_0^2 - u_1^2\right)}{2} & 0 & 1 & c_0\left(-\sqrt{d}u_1 + du_0\right) \\ 0 & \sqrt{d}u_0 + u_1 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 0 & -1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & \frac{c_0\sqrt{d}\left(du_0^2 - u_1^2\right)}{2} & 0 & 1 & c_0\left(-\sqrt{d}u_1 + du_0\right) \\ 0 & \sqrt{d}u_0 + u_1 & 0 & 0 & 1 \end{bmatrix}$$

$$\alpha = (-2,\ 0,\ 0,\ 0,\ 0)\,, \beta = (0,\ 1,\ 0,\ 0,\ 0)\,, \sigma_\alpha(\beta) = (0,\ 1,\ 0,\ 0,\ 0)$$

$$w_\alpha X_\beta \left(\begin{bmatrix} u_0 \\ u_1 \end{bmatrix} \right) w_\alpha^{-1} = X_{\sigma_\alpha(\beta)} \left(\begin{bmatrix} u_0 \\ u_1 \end{bmatrix} \right)$$

$$\begin{bmatrix} 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & \frac{c_0\sqrt{d}\left(-du_0^2 + u_1^2\right)}{2} & c_0\left(\sqrt{d}u_1 - du_0\right) \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & \sqrt{d}u_0 + u_1 & 1 \end{bmatrix} \begin{bmatrix} 0 & 0 & -1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & \frac{c_0\sqrt{d}\left(-du_0^2 + u_1^2\right)}{2} & c_0\left(\sqrt{d}u_1 - du_0\right) \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & \sqrt{d}u_0 + u_1 & 1 \end{bmatrix}$$

$$\alpha = (-2,\ 0,\ 0,\ 0,\ 0)\,, \beta = (0,\ 2,\ 0,\ 0,\ 0)\,, \sigma_\alpha(\beta) = (0,\ 2,\ 0,\ 0,\ 0)$$

$$w_\alpha X_\beta \left(u_0 \right) w_\alpha^{-1} = X_{\sigma_\alpha(\beta)} \left(u_0 \right)$$

$$
\begin{bmatrix} 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}
\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & -u_0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}
\begin{bmatrix} 0 & 0 & -1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}
=
\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & -u_0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}
$$

$$\alpha = (-2,\ 0,\ 0,\ 0,\ 0)\,, \beta = (1,\ -1,\ 0,\ 0,\ 0)\,, \sigma_\alpha(\beta) = (-1,\ -1,\ 0,\ 0,\ 0)$$

$$w_\alpha X_\beta \left(\begin{bmatrix} u_0 \\ u_1 \end{bmatrix} \right) w_\alpha^{-1} = X_{\sigma_\alpha(\beta)} \left(\begin{bmatrix} u_0 \\ -u_1 \end{bmatrix} \right)$$

$$
\begin{bmatrix} 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}
\begin{bmatrix} 1 & \sqrt{d}u_0 + u_1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & \sqrt{d}u_0 - u_1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}
\begin{bmatrix} 0 & 0 & -1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}
=
\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & -\sqrt{d}u_0 - u_1 & 1 & 0 & 0 \\ \sqrt{d}u_0 - u_1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}
$$

$$\alpha = (-2,\ 0,\ 0,\ 0,\ 0)\,, \beta = (1,\ 0,\ 0,\ 0,\ 0)\,, \sigma_\alpha(\beta) = (-1,\ 0,\ 0,\ 0,\ 0)$$

$$w_\alpha X_\beta \left(\begin{bmatrix} u_0 \\ u_1 \end{bmatrix} \right) w_\alpha^{-1} = X_{\sigma_\alpha(\beta)} \left(\begin{bmatrix} u_1 \\ u_0 \end{bmatrix} \right)$$

$$
\begin{bmatrix} 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}
\begin{bmatrix} 1 & 0 & \frac{c_0\sqrt{d}\left(-du_0^2+u_1^2\right)}{2} & 0 & c_0\left(\sqrt{d}u_1 - du_0\right) \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & \sqrt{d}u_0 + u_1 & 0 & 1 \end{bmatrix}
\begin{bmatrix} 0 & 0 & -1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}
=
\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ \frac{c_0\sqrt{d}\left(du_0^2-u_1^2\right)}{2} & 0 & 1 & 0 & c_0\left(-\sqrt{d}u_1 + du_0\right) \\ 0 & 0 & 0 & 1 & 0 \\ \sqrt{d}u_0 + u_1 & 0 & 0 & 0 & 1 \end{bmatrix}
$$

$$\alpha = (-2,\ 0,\ 0,\ 0,\ 0)\,, \beta = (1,\ 1,\ 0,\ 0,\ 0)\,, \sigma_\alpha(\beta) = (-1,\ 1,\ 0,\ 0,\ 0)$$

$$w_\alpha X_\beta \left(\begin{bmatrix} u_0 \\ u_1 \end{bmatrix} \right) w_\alpha^{-1} = X_{\sigma_\alpha(\beta)} \left(\begin{bmatrix} u_0 \\ u_1 \end{bmatrix} \right)$$

$$
\begin{bmatrix} 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}
\begin{bmatrix} 1 & 0 & 0 & -\sqrt{d}u_1 + u_0 & 0 \\ 0 & 1 & \sqrt{d}u_1 + u_0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}
\begin{bmatrix} 0 & 0 & -1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}
=
\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ \sqrt{d}u_1 + u_0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & \sqrt{d}u_1 - u_0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}
$$

$$\alpha = (-2,\ 0,\ 0,\ 0,\ 0)\,, \beta = (2,\ 0,\ 0,\ 0,\ 0)\,, \sigma_\alpha(\beta) = (-2,\ 0,\ 0,\ 0,\ 0)$$

$$w_\alpha X_\beta \left(u_0 \right) w_\alpha^{-1} = X_{\sigma_\alpha(\beta)} \left(-u_0 \right)$$

$$
\begin{bmatrix} 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}
\begin{bmatrix} 1 & 0 & -u_0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}
\begin{bmatrix} 0 & 0 & -1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}
=
\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ u_0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}
$$

$$\alpha = (-1,\ -1,\ 0,\ 0,\ 0)\,, \beta = (-2,\ 0,\ 0,\ 0,\ 0)\,, \sigma_\alpha(\beta) = (0,\ 2,\ 0,\ 0,\ 0)$$

$$w_\alpha X_\beta \left(u_0 \right) w_\alpha^{-1} = X_{\sigma_\alpha(\beta)} \left(-u_0 \right)$$

$$
\begin{bmatrix} 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}
\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ -u_0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}
\begin{bmatrix} 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & -1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}
=
\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & u_0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}
$$

$$\alpha = (-1,\ -1,\ 0,\ 0,\ 0)\,, \beta = (-1,\ -1,\ 0,\ 0,\ 0)\,, \sigma_\alpha(\beta) = (1,\ 1,\ 0,\ 0,\ 0)$$

$$w_\alpha X_\beta \left(\begin{bmatrix} u_0 \\ u_1 \end{bmatrix} \right) w_\alpha^{-1} = X_{\sigma_\alpha(\beta)} \left(\begin{bmatrix} -u_1 \\ u_0 \end{bmatrix} \right)$$

$$\begin{bmatrix} 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & -\sqrt{d}u_0 + u_1 & 1 & 0 & 0 \\ \sqrt{d}u_0 + u_1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & -1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & -\sqrt{d}u_0 - u_1 & 0 \\ 0 & 1 & \sqrt{d}u_0 - u_1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\alpha = (-1,\ -1,\ 0,\ 0,\ 0)\,, \beta = (-1,\ 0,\ 0,\ 0,\ 0)\,, \sigma_\alpha(\beta) = (0,\ 1,\ 0,\ 0,\ 0)$$

$$w_\alpha X_\beta \left(\begin{bmatrix} u_0 \\ u_1 \end{bmatrix} \right) w_\alpha^{-1} = X_{\sigma_\alpha(\beta)} \left(\begin{bmatrix} -u_1 \\ -u_0 \end{bmatrix} \right)$$

$$\begin{bmatrix} 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ \frac{c_0\sqrt{d}\left(du_1^2 - u_0^2\right)}{2} & 0 & 1 & 0 & c_0\left(-\sqrt{d}u_0 + du_1\right) \\ 0 & 0 & 0 & 1 & 0 \\ \sqrt{d}u_1 + u_0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & -1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & \frac{c_0\sqrt{d}\left(-du_1^2 + u_0^2\right)}{2} & c_0\left(-\sqrt{d}u_0 + du_1\right) \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -\sqrt{d}u_1 - u_0 & 1 \end{bmatrix}$$

$$\alpha = (-1,\ -1,\ 0,\ 0,\ 0)\,, \beta = (-1,\ 1,\ 0,\ 0,\ 0)\,, \sigma_\alpha(\beta) = (-1,\ 1,\ 0,\ 0,\ 0)$$

$$w_\alpha X_\beta \left(\begin{bmatrix} u_0 \\ u_1 \end{bmatrix} \right) w_\alpha^{-1} = X_{\sigma_\alpha(\beta)} \left(\begin{bmatrix} -u_0 \\ u_1 \end{bmatrix} \right)$$

$$\begin{bmatrix} 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ \sqrt{d}u_1 + u_0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & \sqrt{d}u_1 - u_0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & -1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ \sqrt{d}u_1 - u_0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & \sqrt{d}u_1 + u_0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\alpha = (-1,\ -1,\ 0,\ 0,\ 0)\,, \beta = (0,\ -2,\ 0,\ 0,\ 0)\,, \sigma_\alpha(\beta) = (2,\ 0,\ 0,\ 0,\ 0)$$

$$w_\alpha X_\beta \left(u_0 \right) w_\alpha^{-1} = X_{\sigma_\alpha(\beta)} \left(-u_0 \right)$$

$$\begin{bmatrix} 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & -u_0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & -1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & u_0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\alpha = (-1,\ -1,\ 0,\ 0,\ 0)\,, \beta = (0,\ -1,\ 0,\ 0,\ 0)\,, \sigma_\alpha(\beta) = (1,\ 0,\ 0,\ 0,\ 0)$$

$$w_\alpha X_\beta \left(\begin{bmatrix} u_0 \\ u_1 \end{bmatrix} \right) w_\alpha^{-1} = X_{\sigma_\alpha(\beta)} \left(\begin{bmatrix} -u_0 \\ -u_1 \end{bmatrix} \right)$$

$$\begin{bmatrix} 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & \frac{c_0\sqrt{d}\left(du_0^2 - u_1^2\right)}{2} & 0 & 1 & c_0\left(-\sqrt{d}u_1 + du_0\right) \\ 0 & \sqrt{d}u_0 + u_1 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & -1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & \frac{c_0\sqrt{d}\left(-du_0^2 + u_1^2\right)}{2} & 0 & c_0\left(-\sqrt{d}u_1 + du_0\right) \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & -\sqrt{d}u_0 - u_1 & 0 & 1 \end{bmatrix}$$

$$\alpha = (-1,\ -1,\ 0,\ 0,\ 0)\,, \beta = (0,\ 1,\ 0,\ 0,\ 0)\,, \sigma_\alpha(\beta) = (-1,\ 0,\ 0,\ 0,\ 0)$$

$$w_\alpha X_\beta \left(\begin{bmatrix} u_0 \\ u_1 \end{bmatrix} \right) w_\alpha^{-1} = X_{\sigma_\alpha(\beta)} \left(\begin{bmatrix} u_1 \\ u_0 \end{bmatrix} \right)$$

$$\begin{bmatrix} 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & \frac{c_0\sqrt{d}\left(-du_0^2 + u_1^2\right)}{2} & c_0\left(\sqrt{d}u_1 - du_0\right) \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & \sqrt{d}u_0 + u_1 & 1 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & -1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ \frac{c_0\sqrt{d}\left(du_0^2 - u_1^2\right)}{2} & 0 & 1 & 0 & c_0\left(-\sqrt{d}u_1 + du_0\right) \\ 0 & 0 & 0 & 1 & 0 \\ \sqrt{d}u_0 + u_1 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\alpha = (-1,\ -1,\ 0,\ 0,\ 0)\,, \beta = (0,\ 2,\ 0,\ 0,\ 0)\,, \sigma_\alpha(\beta) = (-2,\ 0,\ 0,\ 0,\ 0)$$

$$w_\alpha X_\beta \left(u_0 \right) w_\alpha^{-1} = X_{\sigma_\alpha(\beta)} \left(-u_0 \right)$$

$$\begin{bmatrix}0 & 0 & 0 & 1 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & -1 & 0 & 0 & 0\\ -1 & 0 & 0 & 0 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix}\begin{bmatrix}1 & 0 & 0 & 0 & 0\\ 0 & 1 & 0 & -u_0 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix}\begin{bmatrix}0 & 0 & 0 & -1 & 0\\ 0 & 0 & -1 & 0 & 0\\ 0 & 1 & 0 & 0 & 0\\ 1 & 0 & 0 & 0 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix} = \begin{bmatrix}1 & 0 & 0 & 0 & 0\\ 0 & 1 & 0 & 0 & 0\\ u_0 & 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix}$$

$$\alpha = (-1,\ -1,\ 0,\ 0,\ 0)\,, \beta = (1,\ -1,\ 0,\ 0,\ 0)\,, \sigma_\alpha(\beta) = (1,\ -1,\ 0,\ 0,\ 0)$$

$$w_\alpha X_\beta\left(\begin{bmatrix}u_0\\ u_1\end{bmatrix}\right)w_\alpha^{-1} = X_{\sigma_\alpha(\beta)}\left(\begin{bmatrix}u_0\\ -u_1\end{bmatrix}\right)$$

$$\begin{bmatrix}0 & 0 & 0 & 1 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & -1 & 0 & 0 & 0\\ -1 & 0 & 0 & 0 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix}\begin{bmatrix}1 & \sqrt{d}u_0 + u_1 & 0 & 0 & 0\\ 0 & 1 & 0 & 0 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & 0 & \sqrt{d}u_0 - u_1 & 1 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix}\begin{bmatrix}0 & 0 & 0 & -1 & 0\\ 0 & 0 & -1 & 0 & 0\\ 0 & 1 & 0 & 0 & 0\\ 1 & 0 & 0 & 0 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix} = \begin{bmatrix}1 & \sqrt{d}u_0 - u_1 & 0 & 0 & 0\\ 0 & 1 & 0 & 0 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & 0 & \sqrt{d}u_0 + u_1 & 1 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix}$$

$$\alpha = (-1,\ -1,\ 0,\ 0,\ 0)\,, \beta = (1,\ 0,\ 0,\ 0,\ 0)\,, \sigma_\alpha(\beta) = (0,\ -1,\ 0,\ 0,\ 0)$$

$$w_\alpha X_\beta\left(\begin{bmatrix}u_0\\ u_1\end{bmatrix}\right)w_\alpha^{-1} = X_{\sigma_\alpha(\beta)}\left(\begin{bmatrix}u_0\\ u_1\end{bmatrix}\right)$$

$$\begin{bmatrix}0 & 0 & 0 & 1 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & -1 & 0 & 0 & 0\\ -1 & 0 & 0 & 0 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix}\begin{bmatrix}1 & 0 & \frac{c_0\sqrt{d}\left(-du_0^2+u_1^2\right)}{2} & 0 & c_0\left(\sqrt{d}u_1 - du_0\right)\\ 0 & 1 & 0 & 0 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 1 & 0\\ 0 & 0 & \sqrt{d}u_0 + u_1 & 0 & 1\end{bmatrix}\begin{bmatrix}0 & 0 & 0 & -1 & 0\\ 0 & 0 & -1 & 0 & 0\\ 0 & 1 & 0 & 0 & 0\\ 1 & 0 & 0 & 0 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix} = \begin{bmatrix}1 & 0 & 0 & 0 & 0\\ 0 & 1 & 0 & 0 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & \frac{c_0\sqrt{d}\left(du_0^2-u_1^2\right)}{2} & 0 & 1 & c_0\left(-\sqrt{d}u_1 + du_0\right)\\ 0 & \sqrt{d}u_0 + u_1 & 0 & 0 & 1\end{bmatrix}$$

$$\alpha = (-1,\ -1,\ 0,\ 0,\ 0)\,, \beta = (1,\ 1,\ 0,\ 0,\ 0)\,, \sigma_\alpha(\beta) = (-1,\ -1,\ 0,\ 0,\ 0)$$

$$w_\alpha X_\beta\left(\begin{bmatrix}u_0\\ u_1\end{bmatrix}\right)w_\alpha^{-1} = X_{\sigma_\alpha(\beta)}\left(\begin{bmatrix}u_1\\ -u_0\end{bmatrix}\right)$$

$$\begin{bmatrix}0 & 0 & 0 & 1 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & -1 & 0 & 0 & 0\\ -1 & 0 & 0 & 0 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix}\begin{bmatrix}1 & 0 & 0 & -\sqrt{d}u_1 + u_0 & 0\\ 0 & 1 & \sqrt{d}u_1 + u_0 & 0 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix}\begin{bmatrix}0 & 0 & 0 & -1 & 0\\ 0 & 0 & -1 & 0 & 0\\ 0 & 1 & 0 & 0 & 0\\ 1 & 0 & 0 & 0 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix} = \begin{bmatrix}1 & 0 & 0 & 0 & 0\\ 0 & 1 & 0 & 0 & 0\\ 0 & -\sqrt{d}u_1 - u_0 & 1 & 0 & 0\\ \sqrt{d}u_1 - u_0 & 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix}$$

$$\alpha = (-1,\ -1,\ 0,\ 0,\ 0)\,, \beta = (2,\ 0,\ 0,\ 0,\ 0)\,, \sigma_\alpha(\beta) = (0,\ -2,\ 0,\ 0,\ 0)$$

$$w_\alpha X_\beta\left(u_0\right)w_\alpha^{-1} = X_{\sigma_\alpha(\beta)}\left(-u_0\right)$$

$$\begin{bmatrix}0 & 0 & 0 & 1 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & -1 & 0 & 0 & 0\\ -1 & 0 & 0 & 0 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix}\begin{bmatrix}1 & 0 & -u_0 & 0 & 0\\ 0 & 1 & 0 & 0 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix}\begin{bmatrix}0 & 0 & 0 & -1 & 0\\ 0 & 0 & -1 & 0 & 0\\ 0 & 1 & 0 & 0 & 0\\ 1 & 0 & 0 & 0 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix} = \begin{bmatrix}1 & 0 & 0 & 0 & 0\\ 0 & 1 & 0 & 0 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & u_0 & 0 & 1 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix}$$

$$\alpha = (-1,\ 1,\ 0,\ 0,\ 0)\,, \beta = (-2,\ 0,\ 0,\ 0,\ 0)\,, \sigma_\alpha(\beta) = (0,\ -2,\ 0,\ 0,\ 0)$$

$$w_\alpha X_\beta\left(u_0\right)w_\alpha^{-1} = X_{\sigma_\alpha(\beta)}\left(u_0\right)$$

$$\begin{bmatrix}0 & 1 & 0 & 0 & 0\\ 1 & 0 & 0 & 0 & 0\\ 0 & 0 & 0 & 1 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix}\begin{bmatrix}1 & 0 & 0 & 0 & 0\\ 0 & 1 & 0 & 0 & 0\\ -u_0 & 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix}\begin{bmatrix}0 & 1 & 0 & 0 & 0\\ 1 & 0 & 0 & 0 & 0\\ 0 & 0 & 0 & 1 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix} = \begin{bmatrix}1 & 0 & 0 & 0 & 0\\ 0 & 1 & 0 & 0 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & -u_0 & 0 & 1 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix}$$

$$\alpha = (-1,\ 1,\ 0,\ 0,\ 0)\,, \beta = (-1,\ -1,\ 0,\ 0,\ 0)\,, \sigma_\alpha(\beta) = (-1,\ -1,\ 0,\ 0,\ 0)$$

$$w_\alpha X_\beta\left(\begin{bmatrix}u_0\\ u_1\end{bmatrix}\right)w_\alpha^{-1} = X_{\sigma_\alpha(\beta)}\left(\begin{bmatrix}-u_0\\ u_1\end{bmatrix}\right)$$

$$\begin{bmatrix}0 & 1 & 0 & 0 & 0\\ 1 & 0 & 0 & 0 & 0\\ 0 & 0 & 0 & 1 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix}\begin{bmatrix}1 & 0 & 0 & 0 & 0\\ 0 & 1 & 0 & 0 & 0\\ 0 & -\sqrt{d}u_0 + u_1 & 1 & 0 & 0\\ \sqrt{d}u_0 + u_1 & 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix}\begin{bmatrix}0 & 1 & 0 & 0 & 0\\ 1 & 0 & 0 & 0 & 0\\ 0 & 0 & 0 & 1 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix} = \begin{bmatrix}1 & 0 & 0 & 0 & 0\\ 0 & 1 & 0 & 0 & 0\\ 0 & \sqrt{d}u_0 + u_1 & 1 & 0 & 0\\ -\sqrt{d}u_0 + u_1 & 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix}$$

$$\alpha = (-1,\ 1,\ 0,\ 0,\ 0)\,, \beta = (-1,\ 0,\ 0,\ 0,\ 0)\,, \sigma_\alpha(\beta) = (0,\ -1,\ 0,\ 0,\ 0)$$

$$w_\alpha X_\beta \left(\begin{bmatrix}u_0\\ u_1\end{bmatrix}\right) w_\alpha^{-1} = X_{\sigma_\alpha(\beta)} \left(\begin{bmatrix}u_1\\ u_0\end{bmatrix}\right)$$

$$\begin{bmatrix}0 & 1 & 0 & 0 & 0\\ 1 & 0 & 0 & 0 & 0\\ 0 & 0 & 0 & 1 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix}\begin{bmatrix}1 & 0 & 0 & 0 & 0\\ 0 & 1 & 0 & 0 & 0\\ \frac{c_0\sqrt{d}\left(du_1^2 - u_0^2\right)}{2} & 0 & 1 & 0 & c_0\left(-\sqrt{d}u_0 + du_1\right)\\ 0 & 0 & 0 & 1 & 0\\ \sqrt{d}u_1 + u_0 & 0 & 0 & 0 & 1\end{bmatrix}\begin{bmatrix}0 & 1 & 0 & 0 & 0\\ 1 & 0 & 0 & 0 & 0\\ 0 & 0 & 0 & 1 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix} = \begin{bmatrix}1 & 0 & 0 & 0 & 0\\ 0 & 1 & 0 & 0 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & \frac{c_0\sqrt{d}\left(du_1^2 - u_0^2\right)}{2} & 0 & 1 & c_0\left(-\sqrt{d}u_0 + du_1\right)\\ 0 & \sqrt{d}u_1 + u_0 & 0 & 0 & 1\end{bmatrix}$$

$$\alpha = (-1,\ 1,\ 0,\ 0,\ 0)\,, \beta = (-1,\ 1,\ 0,\ 0,\ 0)\,, \sigma_\alpha(\beta) = (1,\ -1,\ 0,\ 0,\ 0)$$

$$w_\alpha X_\beta \left(\begin{bmatrix}u_0\\ u_1\end{bmatrix}\right) w_\alpha^{-1} = X_{\sigma_\alpha(\beta)} \left(\begin{bmatrix}u_1\\ u_0\end{bmatrix}\right)$$

$$\begin{bmatrix}0 & 1 & 0 & 0 & 0\\ 1 & 0 & 0 & 0 & 0\\ 0 & 0 & 0 & 1 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix}\begin{bmatrix}1 & 0 & 0 & 0 & 0\\ \sqrt{d}u_1 + u_0 & 1 & 0 & 0 & 0\\ 0 & 0 & 1 & \sqrt{d}u_1 - u_0 & 0\\ 0 & 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix}\begin{bmatrix}0 & 1 & 0 & 0 & 0\\ 1 & 0 & 0 & 0 & 0\\ 0 & 0 & 0 & 1 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix} = \begin{bmatrix}1 & \sqrt{d}u_1 + u_0 & 0 & 0 & 0\\ 0 & 1 & 0 & 0 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & 0 & \sqrt{d}u_1 - u_0 & 1 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix}$$

$$\alpha = (-1,\ 1,\ 0,\ 0,\ 0)\,, \beta = (0,\ -2,\ 0,\ 0,\ 0)\,, \sigma_\alpha(\beta) = (-2,\ 0,\ 0,\ 0,\ 0)$$

$$w_\alpha X_\beta \left(u_0\right) w_\alpha^{-1} = X_{\sigma_\alpha(\beta)} \left(u_0\right)$$

$$\begin{bmatrix}0 & 1 & 0 & 0 & 0\\ 1 & 0 & 0 & 0 & 0\\ 0 & 0 & 0 & 1 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix}\begin{bmatrix}1 & 0 & 0 & 0 & 0\\ 0 & 1 & 0 & 0 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & -u_0 & 0 & 1 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix}\begin{bmatrix}0 & 1 & 0 & 0 & 0\\ 1 & 0 & 0 & 0 & 0\\ 0 & 0 & 0 & 1 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix} = \begin{bmatrix}1 & 0 & 0 & 0 & 0\\ 0 & 1 & 0 & 0 & 0\\ -u_0 & 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix}$$

$$\alpha = (-1,\ 1,\ 0,\ 0,\ 0)\,, \beta = (0,\ -1,\ 0,\ 0,\ 0)\,, \sigma_\alpha(\beta) = (-1,\ 0,\ 0,\ 0,\ 0)$$

$$w_\alpha X_\beta \left(\begin{bmatrix}u_0\\ u_1\end{bmatrix}\right) w_\alpha^{-1} = X_{\sigma_\alpha(\beta)} \left(\begin{bmatrix}u_1\\ u_0\end{bmatrix}\right)$$

$$\begin{bmatrix}0 & 1 & 0 & 0 & 0\\ 1 & 0 & 0 & 0 & 0\\ 0 & 0 & 0 & 1 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix}\begin{bmatrix}1 & 0 & 0 & 0 & 0\\ 0 & 1 & 0 & 0 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & \frac{c_0\sqrt{d}\left(du_0^2 - u_1^2\right)}{2} & 0 & 1 & c_0\left(-\sqrt{d}u_1 + du_0\right)\\ 0 & \sqrt{d}u_0 + u_1 & 0 & 0 & 1\end{bmatrix}\begin{bmatrix}0 & 1 & 0 & 0 & 0\\ 1 & 0 & 0 & 0 & 0\\ 0 & 0 & 0 & 1 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix} = \begin{bmatrix}1 & 0 & 0 & 0 & 0\\ 0 & 1 & 0 & 0 & 0\\ \frac{c_0\sqrt{d}\left(du_0^2 - u_1^2\right)}{2} & 0 & 1 & 0 & c_0\left(-\sqrt{d}u_1 + du_0\right)\\ 0 & 0 & 0 & 1 & 0\\ \sqrt{d}u_0 + u_1 & 0 & 0 & 0 & 1\end{bmatrix}$$

$$\alpha = (-1,\ 1,\ 0,\ 0,\ 0)\,, \beta = (0,\ 1,\ 0,\ 0,\ 0)\,, \sigma_\alpha(\beta) = (1,\ 0,\ 0,\ 0,\ 0)$$

$$w_\alpha X_\beta \left(\begin{bmatrix}u_0\\ u_1\end{bmatrix}\right) w_\alpha^{-1} = X_{\sigma_\alpha(\beta)} \left(\begin{bmatrix}u_0\\ u_1\end{bmatrix}\right)$$

$$\begin{bmatrix}0 & 1 & 0 & 0 & 0\\ 1 & 0 & 0 & 0 & 0\\ 0 & 0 & 0 & 1 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix}\begin{bmatrix}1 & 0 & 0 & 0 & 0\\ 0 & 1 & 0 & \frac{c_0\sqrt{d}\left(-du_0^2 + u_1^2\right)}{2} & c_0\left(\sqrt{d}u_1 - du_0\right)\\ 0 & 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 1 & 0\\ 0 & 0 & 0 & \sqrt{d}u_0 + u_1 & 1\end{bmatrix}\begin{bmatrix}0 & 1 & 0 & 0 & 0\\ 1 & 0 & 0 & 0 & 0\\ 0 & 0 & 0 & 1 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 0 & 1\end{bmatrix} = \begin{bmatrix}1 & 0 & \frac{c_0\sqrt{d}\left(-du_0^2 + u_1^2\right)}{2} & 0 & c_0\left(\sqrt{d}u_1 - du_0\right)\\ 0 & 1 & 0 & 0 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 1 & 0\\ 0 & 0 & \sqrt{d}u_0 + u_1 & 0 & 1\end{bmatrix}$$

$$\alpha = (-1,\ 1,\ 0,\ 0,\ 0)\,, \beta = (0,\ 2,\ 0,\ 0,\ 0)\,, \sigma_\alpha(\beta) = (2,\ 0,\ 0,\ 0,\ 0)$$

$$w_\alpha X_\beta\left(u_0\right) w_\alpha^{-1} = X_{\sigma_\alpha(\beta)}\left(u_0\right)$$

$$\begin{bmatrix}0&1&0&0&0\\1&0&0&0&0\\0&0&0&1&0\\0&0&1&0&0\\0&0&0&0&1\end{bmatrix}\begin{bmatrix}1&0&0&0&0\\0&1&0&-u_0&0\\0&0&1&0&0\\0&0&0&1&0\\0&0&0&0&1\end{bmatrix}\begin{bmatrix}0&1&0&0&0\\1&0&0&0&0\\0&0&0&1&0\\0&0&1&0&0\\0&0&0&0&1\end{bmatrix} = \begin{bmatrix}1&0&-u_0&0&0\\0&1&0&0&0\\0&0&1&0&0\\0&0&0&1&0\\0&0&0&0&1\end{bmatrix}$$

$$\alpha = (-1,\ 1,\ 0,\ 0,\ 0)\,, \beta = (1,\ -1,\ 0,\ 0,\ 0)\,, \sigma_\alpha(\beta) = (-1,\ 1,\ 0,\ 0,\ 0)$$

$$w_\alpha X_\beta\left(\begin{bmatrix}u_0\\u_1\end{bmatrix}\right) w_\alpha^{-1} = X_{\sigma_\alpha(\beta)}\left(\begin{bmatrix}u_1\\u_0\end{bmatrix}\right)$$

$$\begin{bmatrix}0&1&0&0&0\\1&0&0&0&0\\0&0&0&1&0\\0&0&1&0&0\\0&0&0&0&1\end{bmatrix}\begin{bmatrix}1&\sqrt{d}u_0+u_1&0&0&0\\0&1&0&0&0\\0&0&1&0&0\\0&0&\sqrt{d}u_0-u_1&1&0\\0&0&0&0&1\end{bmatrix}\begin{bmatrix}0&1&0&0&0\\1&0&0&0&0\\0&0&0&1&0\\0&0&1&0&0\\0&0&0&0&1\end{bmatrix} = \begin{bmatrix}1&0&0&0&0\\\sqrt{d}u_0+u_1&1&0&0&0\\0&0&1&\sqrt{d}u_0-u_1&0\\0&0&0&1&0\\0&0&0&0&1\end{bmatrix}$$

$$\alpha = (-1,\ 1,\ 0,\ 0,\ 0)\,, \beta = (1,\ 0,\ 0,\ 0,\ 0)\,, \sigma_\alpha(\beta) = (0,\ 1,\ 0,\ 0,\ 0)$$

$$w_\alpha X_\beta\left(\begin{bmatrix}u_0\\u_1\end{bmatrix}\right) w_\alpha^{-1} = X_{\sigma_\alpha(\beta)}\left(\begin{bmatrix}u_0\\u_1\end{bmatrix}\right)$$

$$\begin{bmatrix}0&1&0&0&0\\1&0&0&0&0\\0&0&0&1&0\\0&0&1&0&0\\0&0&0&0&1\end{bmatrix}\begin{bmatrix}1&0&\frac{c_0\sqrt{d}\left(-du_0^2+u_1^2\right)}{2}&0&c_0\left(\sqrt{d}u_1-du_0\right)\\0&1&0&0&0\\0&0&1&0&0\\0&0&0&1&0\\0&0&\sqrt{d}u_0+u_1&0&1\end{bmatrix}\begin{bmatrix}0&1&0&0&0\\1&0&0&0&0\\0&0&0&1&0\\0&0&1&0&0\\0&0&0&0&1\end{bmatrix} = \begin{bmatrix}1&0&0&0&0\\0&1&0&\frac{c_0\sqrt{d}\left(-du_0^2+u_1^2\right)}{2}&c_0\left(\sqrt{d}u_1-du_0\right)\\0&0&1&0&0\\0&0&0&1&0\\0&0&0&\sqrt{d}u_0+u_1&1\end{bmatrix}$$

$$\alpha = (-1,\ 1,\ 0,\ 0,\ 0)\,, \beta = (1,\ 1,\ 0,\ 0,\ 0)\,, \sigma_\alpha(\beta) = (1,\ 1,\ 0,\ 0,\ 0)$$

$$w_\alpha X_\beta\left(\begin{bmatrix}u_0\\u_1\end{bmatrix}\right) w_\alpha^{-1} = X_{\sigma_\alpha(\beta)}\left(\begin{bmatrix}u_0\\-u_1\end{bmatrix}\right)$$

$$\begin{bmatrix}0&1&0&0&0\\1&0&0&0&0\\0&0&0&1&0\\0&0&1&0&0\\0&0&0&0&1\end{bmatrix}\begin{bmatrix}1&0&0&-\sqrt{d}u_1+u_0&0\\0&1&\sqrt{d}u_1+u_0&0&0\\0&0&1&0&0\\0&0&0&1&0\\0&0&0&0&1\end{bmatrix}\begin{bmatrix}0&1&0&0&0\\1&0&0&0&0\\0&0&0&1&0\\0&0&1&0&0\\0&0&0&0&1\end{bmatrix} = \begin{bmatrix}1&0&0&\sqrt{d}u_1+u_0&0\\0&1&-\sqrt{d}u_1+u_0&0&0\\0&0&1&0&0\\0&0&0&1&0\\0&0&0&0&1\end{bmatrix}$$

$$\alpha = (-1,\ 1,\ 0,\ 0,\ 0)\,, \beta = (2,\ 0,\ 0,\ 0,\ 0)\,, \sigma_\alpha(\beta) = (0,\ 2,\ 0,\ 0,\ 0)$$

$$w_\alpha X_\beta\left(u_0\right) w_\alpha^{-1} = X_{\sigma_\alpha(\beta)}\left(u_0\right)$$

$$\begin{bmatrix}0&1&0&0&0\\1&0&0&0&0\\0&0&0&1&0\\0&0&1&0&0\\0&0&0&0&1\end{bmatrix}\begin{bmatrix}1&0&-u_0&0&0\\0&1&0&0&0\\0&0&1&0&0\\0&0&0&1&0\\0&0&0&0&1\end{bmatrix}\begin{bmatrix}0&1&0&0&0\\1&0&0&0&0\\0&0&0&1&0\\0&0&1&0&0\\0&0&0&0&1\end{bmatrix} = \begin{bmatrix}1&0&0&0&0\\0&1&0&-u_0&0\\0&0&1&0&0\\0&0&0&1&0\\0&0&0&0&1\end{bmatrix}$$

$$\alpha = (0,\ -2,\ 0,\ 0,\ 0)\,, \beta = (-2,\ 0,\ 0,\ 0,\ 0)\,, \sigma_\alpha(\beta) = (-2,\ 0,\ 0,\ 0,\ 0)$$

$$w_\alpha X_\beta\left(u_0\right) w_\alpha^{-1} = X_{\sigma_\alpha(\beta)}\left(u_0\right)$$

$$\begin{bmatrix}1&0&0&0&0\\0&0&0&1&0\\0&0&1&0&0\\0&-1&0&0&0\\0&0&0&0&1\end{bmatrix}\begin{bmatrix}1&0&0&0&0\\0&1&0&0&0\\-u_0&0&1&0&0\\0&0&0&1&0\\0&0&0&0&1\end{bmatrix}\begin{bmatrix}1&0&0&0&0\\0&0&0&-1&0\\0&0&1&0&0\\0&1&0&0&0\\0&0&0&0&1\end{bmatrix} = \begin{bmatrix}1&0&0&0&0\\0&1&0&0&0\\-u_0&0&1&0&0\\0&0&0&1&0\\0&0&0&0&1\end{bmatrix}$$

$$\alpha = (0,\ -2,\ 0,\ 0,\ 0)\,, \beta = (-1,\ -1,\ 0,\ 0,\ 0)\,, \sigma_\alpha(\beta) = (-1,\ 1,\ 0,\ 0,\ 0)$$

$$w_\alpha X_\beta\left(\begin{bmatrix}u_0\\u_1\end{bmatrix}\right)w_\alpha^{-1}=X_{\sigma_\alpha(\beta)}\left(\begin{bmatrix}u_1\\u_0\end{bmatrix}\right)$$

$$\begin{bmatrix}1&0&0&0&0\\0&0&0&1&0\\0&0&1&0&0\\0&-1&0&0&0\\0&0&0&0&1\end{bmatrix}\begin{bmatrix}1&0&0&0&0\\0&1&0&0&0\\0&-\sqrt{d}u_0+u_1&1&0&0\\\sqrt{d}u_0+u_1&0&0&1&0\\0&0&0&0&1\end{bmatrix}\begin{bmatrix}1&0&0&0&0\\0&0&0&-1&0\\0&0&1&0&0\\0&1&0&0&0\\0&0&0&0&1\end{bmatrix}=\begin{bmatrix}1&0&0&0&0\\\sqrt{d}u_0+u_1&1&0&0&0\\0&0&1&\sqrt{d}u_0-u_1&0\\0&0&0&1&0\\0&0&0&0&1\end{bmatrix}$$

$$\alpha=(0,\ -2,\ 0,\ 0,\ 0)\,,\beta=(-1,\ 0,\ 0,\ 0,\ 0)\,,\sigma_\alpha(\beta)=(-1,\ 0,\ 0,\ 0,\ 0)$$

$$w_\alpha X_\beta\left(\begin{bmatrix}u_0\\u_1\end{bmatrix}\right)w_\alpha^{-1}=X_{\sigma_\alpha(\beta)}\left(\begin{bmatrix}u_0\\u_1\end{bmatrix}\right)$$

$$\begin{bmatrix}1&0&0&0&0\\0&0&0&1&0\\0&0&1&0&0\\0&-1&0&0&0\\0&0&0&0&1\end{bmatrix}\begin{bmatrix}1&0&0&0&0\\0&1&0&0&0\\\frac{c_0\sqrt{d}\left(du_1^2-u_0^2\right)}{2}&0&1&0&c_0\left(-\sqrt{d}u_0+du_1\right)\\0&0&0&1&0\\\sqrt{d}u_1+u_0&0&0&0&1\end{bmatrix}\begin{bmatrix}1&0&0&0&0\\0&0&0&-1&0\\0&0&1&0&0\\0&1&0&0&0\\0&0&0&0&1\end{bmatrix}=\begin{bmatrix}1&0&0&0&0\\0&1&0&0&0\\\frac{c_0\sqrt{d}\left(du_1^2-u_0^2\right)}{2}&0&1&0&c_0\left(-\sqrt{d}u_0+du_1\right)\\0&0&0&1&0\\\sqrt{d}u_1+u_0&0&0&0&1\end{bmatrix}$$

$$\alpha=(0,\ -2,\ 0,\ 0,\ 0)\,,\beta=(-1,\ 1,\ 0,\ 0,\ 0)\,,\sigma_\alpha(\beta)=(-1,\ -1,\ 0,\ 0,\ 0)$$

$$w_\alpha X_\beta\left(\begin{bmatrix}u_0\\u_1\end{bmatrix}\right)w_\alpha^{-1}=X_{\sigma_\alpha(\beta)}\left(\begin{bmatrix}-u_1\\-u_0\end{bmatrix}\right)$$

$$\begin{bmatrix}1&0&0&0&0\\0&0&0&1&0\\0&0&1&0&0\\0&-1&0&0&0\\0&0&0&0&1\end{bmatrix}\begin{bmatrix}1&0&0&0&0\\\sqrt{d}u_1+u_0&1&0&0&0\\0&0&1&\sqrt{d}u_1-u_0&0\\0&0&0&1&0\\0&0&0&0&1\end{bmatrix}\begin{bmatrix}1&0&0&0&0\\0&0&0&-1&0\\0&0&1&0&0\\0&1&0&0&0\\0&0&0&0&1\end{bmatrix}=\begin{bmatrix}1&0&0&0&0\\0&1&0&0&0\\0&\sqrt{d}u_1-u_0&1&0&0\\-\sqrt{d}u_1-u_0&0&0&1&0\\0&0&0&0&1\end{bmatrix}$$

$$\alpha=(0,\ -2,\ 0,\ 0,\ 0)\,,\beta=(0,\ -2,\ 0,\ 0,\ 0)\,,\sigma_\alpha(\beta)=(0,\ 2,\ 0,\ 0,\ 0)$$

$$w_\alpha X_\beta\left(u_0\right)w_\alpha^{-1}=X_{\sigma_\alpha(\beta)}\left(-u_0\right)$$

$$\begin{bmatrix}1&0&0&0&0\\0&0&0&1&0\\0&0&1&0&0\\0&-1&0&0&0\\0&0&0&0&1\end{bmatrix}\begin{bmatrix}1&0&0&0&0\\0&1&0&0&0\\0&0&1&0&0\\0&-u_0&0&1&0\\0&0&0&0&1\end{bmatrix}\begin{bmatrix}1&0&0&0&0\\0&0&0&-1&0\\0&0&1&0&0\\0&1&0&0&0\\0&0&0&0&1\end{bmatrix}=\begin{bmatrix}1&0&0&0&0\\0&1&0&u_0&0\\0&0&1&0&0\\0&0&0&1&0\\0&0&0&0&1\end{bmatrix}$$

$$\alpha=(0,\ -2,\ 0,\ 0,\ 0)\,,\beta=(0,\ -1,\ 0,\ 0,\ 0)\,,\sigma_\alpha(\beta)=(0,\ 1,\ 0,\ 0,\ 0)$$

$$w_\alpha X_\beta\left(\begin{bmatrix}u_0\\u_1\end{bmatrix}\right)w_\alpha^{-1}=X_{\sigma_\alpha(\beta)}\left(\begin{bmatrix}-u_0\\-u_1\end{bmatrix}\right)$$

$$\begin{bmatrix}1&0&0&0&0\\0&0&0&1&0\\0&0&1&0&0\\0&-1&0&0&0\\0&0&0&0&1\end{bmatrix}\begin{bmatrix}1&0&0&0&0\\0&1&0&0&0\\0&0&1&0&0\\0&\frac{c_0\sqrt{d}\left(du_0^2-u_1^2\right)}{2}&0&1&c_0\left(-\sqrt{d}u_1+du_0\right)\\0&\sqrt{d}u_0+u_1&0&0&1\end{bmatrix}\begin{bmatrix}1&0&0&0&0\\0&0&0&-1&0\\0&0&1&0&0\\0&1&0&0&0\\0&0&0&0&1\end{bmatrix}=\begin{bmatrix}1&0&0&0&0\\0&1&0&\frac{c_0\sqrt{d}\left(-du_0^2+u_1^2\right)}{2}&c_0\left(-\sqrt{d}u_1+du_0\right)\\0&0&1&0&0\\0&0&0&1&0\\0&0&0&-\sqrt{d}u_0-u_1&1\end{bmatrix}$$

$$\alpha=(0,\ -2,\ 0,\ 0,\ 0)\,,\beta=(0,\ 1,\ 0,\ 0,\ 0)\,,\sigma_\alpha(\beta)=(0,\ -1,\ 0,\ 0,\ 0)$$

$$w_\alpha X_\beta\left(\begin{bmatrix}u_0\\u_1\end{bmatrix}\right)w_\alpha^{-1}=X_{\sigma_\alpha(\beta)}\left(\begin{bmatrix}u_0\\u_1\end{bmatrix}\right)$$

$$\begin{bmatrix}1&0&0&0&0\\0&0&0&1&0\\0&0&1&0&0\\0&-1&0&0&0\\0&0&0&0&1\end{bmatrix}\begin{bmatrix}1&0&0&0&0\\0&1&0&\frac{c_0\sqrt{d}\left(-du_0^2+u_1^2\right)}{2}&c_0\left(\sqrt{d}u_1-du_0\right)\\0&0&1&0&0\\0&0&0&1&0\\0&0&0&\sqrt{d}u_0+u_1&1\end{bmatrix}\begin{bmatrix}1&0&0&0&0\\0&0&0&-1&0\\0&0&1&0&0\\0&1&0&0&0\\0&0&0&0&1\end{bmatrix}=\begin{bmatrix}1&0&0&0&0\\0&1&0&0&0\\0&0&1&0&0\\0&\frac{c_0\sqrt{d}\left(du_0^2-u_1^2\right)}{2}&0&1&c_0\left(-\sqrt{d}u_1+du_0\right)\\0&\sqrt{d}u_0+u_1&0&0&1\end{bmatrix}$$

$$\alpha = (0,\ -2,\ 0,\ 0,\ 0)\,, \beta = (0,\ 2,\ 0,\ 0,\ 0)\,, \sigma_\alpha(\beta) = (0,\ -2,\ 0,\ 0,\ 0)$$

$$w_\alpha X_\beta\,(u_0)\,w_\alpha^{-1} = X_{\sigma_\alpha(\beta)}\,(-u_0)$$

$$\begin{bmatrix}1&0&0&0&0\\0&0&0&1&0\\0&0&1&0&0\\0&-1&0&0&0\\0&0&0&0&1\end{bmatrix}\begin{bmatrix}1&0&0&0&0\\0&1&0&-u_0&0\\0&0&1&0&0\\0&0&0&1&0\\0&0&0&0&1\end{bmatrix}\begin{bmatrix}1&0&0&0&0\\0&0&0&-1&0\\0&0&1&0&0\\0&1&0&0&0\\0&0&0&0&1\end{bmatrix} = \begin{bmatrix}1&0&0&0&0\\0&1&0&0&0\\0&0&1&0&0\\0&u_0&0&1&0\\0&0&0&0&1\end{bmatrix}$$

$$\alpha = (0,\ -2,\ 0,\ 0,\ 0)\,, \beta = (1,\ -1,\ 0,\ 0,\ 0)\,, \sigma_\alpha(\beta) = (1,\ 1,\ 0,\ 0,\ 0)$$

$$w_\alpha X_\beta\left(\begin{bmatrix}u_0\\u_1\end{bmatrix}\right)w_\alpha^{-1} = X_{\sigma_\alpha(\beta)}\left(\begin{bmatrix}-u_1\\u_0\end{bmatrix}\right)$$

$$\begin{bmatrix}1&0&0&0&0\\0&0&0&1&0\\0&0&1&0&0\\0&-1&0&0&0\\0&0&0&0&1\end{bmatrix}\begin{bmatrix}1&\sqrt{d}u_0+u_1&0&0&0\\0&1&0&0&0\\0&0&1&0&0\\0&0&\sqrt{d}u_0-u_1&1&0\\0&0&0&0&1\end{bmatrix}\begin{bmatrix}1&0&0&0&0\\0&0&0&-1&0\\0&0&1&0&0\\0&1&0&0&0\\0&0&0&0&1\end{bmatrix} = \begin{bmatrix}1&0&0&-\sqrt{d}u_0-u_1&0\\0&1&\sqrt{d}u_0-u_1&0&0\\0&0&1&0&0\\0&0&0&1&0\\0&0&0&0&1\end{bmatrix}$$

$$\alpha = (0,\ -2,\ 0,\ 0,\ 0)\,, \beta = (1,\ 0,\ 0,\ 0,\ 0)\,, \sigma_\alpha(\beta) = (1,\ 0,\ 0,\ 0,\ 0)$$

$$w_\alpha X_\beta\left(\begin{bmatrix}u_0\\u_1\end{bmatrix}\right)w_\alpha^{-1} = X_{\sigma_\alpha(\beta)}\left(\begin{bmatrix}u_0\\u_1\end{bmatrix}\right)$$

$$\begin{bmatrix}1&0&0&0&0\\0&0&0&1&0\\0&0&1&0&0\\0&-1&0&0&0\\0&0&0&0&1\end{bmatrix}\begin{bmatrix}1&0&\frac{c_0\sqrt{d}\left(-du_0^2+u_1^2\right)}{2}&0&c_0\left(\sqrt{d}u_1-du_0\right)\\0&1&0&0&0\\0&0&1&0&0\\0&0&0&1&0\\0&0&\sqrt{d}u_0+u_1&0&1\end{bmatrix}\begin{bmatrix}1&0&0&0&0\\0&0&0&-1&0\\0&0&1&0&0\\0&1&0&0&0\\0&0&0&0&1\end{bmatrix} = \begin{bmatrix}1&0&\frac{c_0\sqrt{d}\left(-du_0^2+u_1^2\right)}{2}&0&c_0\left(\sqrt{d}u_1-du_0\right)\\0&1&0&0&0\\0&0&1&0&0\\0&0&0&1&0\\0&0&\sqrt{d}u_0+u_1&0&1\end{bmatrix}$$

$$\alpha = (0,\ -2,\ 0,\ 0,\ 0)\,, \beta = (1,\ 1,\ 0,\ 0,\ 0)\,, \sigma_\alpha(\beta) = (1,\ -1,\ 0,\ 0,\ 0)$$

$$w_\alpha X_\beta\left(\begin{bmatrix}u_0\\u_1\end{bmatrix}\right)w_\alpha^{-1} = X_{\sigma_\alpha(\beta)}\left(\begin{bmatrix}-u_1\\u_0\end{bmatrix}\right)$$

$$\begin{bmatrix}1&0&0&0&0\\0&0&0&1&0\\0&0&1&0&0\\0&-1&0&0&0\\0&0&0&0&1\end{bmatrix}\begin{bmatrix}1&0&0&-\sqrt{d}u_1+u_0&0\\0&1&\sqrt{d}u_1+u_0&0&0\\0&0&1&0&0\\0&0&0&1&0\\0&0&0&0&1\end{bmatrix}\begin{bmatrix}1&0&0&0&0\\0&0&0&-1&0\\0&0&1&0&0\\0&1&0&0&0\\0&0&0&0&1\end{bmatrix} = \begin{bmatrix}1&-\sqrt{d}u_1+u_0&0&0&0\\0&1&0&0&0\\0&0&1&0&0\\0&0&-\sqrt{d}u_1-u_0&1&0\\0&0&0&0&1\end{bmatrix}$$

$$\alpha = (0,\ -2,\ 0,\ 0,\ 0)\,, \beta = (2,\ 0,\ 0,\ 0,\ 0)\,, \sigma_\alpha(\beta) = (2,\ 0,\ 0,\ 0,\ 0)$$

$$w_\alpha X_\beta\,(u_0)\,w_\alpha^{-1} = X_{\sigma_\alpha(\beta)}\,(u_0)$$

$$\begin{bmatrix}1&0&0&0&0\\0&0&0&1&0\\0&0&1&0&0\\0&-1&0&0&0\\0&0&0&0&1\end{bmatrix}\begin{bmatrix}1&0&-u_0&0&0\\0&1&0&0&0\\0&0&1&0&0\\0&0&0&1&0\\0&0&0&0&1\end{bmatrix}\begin{bmatrix}1&0&0&0&0\\0&0&0&-1&0\\0&0&1&0&0\\0&1&0&0&0\\0&0&0&0&1\end{bmatrix} = \begin{bmatrix}1&0&-u_0&0&0\\0&1&0&0&0\\0&0&1&0&0\\0&0&0&1&0\\0&0&0&0&1\end{bmatrix}$$